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## DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

# COURSE MATERIAL

**Course Name with Code** : **24AD301 / FOUNDATIONS OF DATA SCIENCE**

**Course Faculty** : **A. SANGEETHA**

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# UNIT I
# INTRODUCTION

**Need for data science - benefits and uses - facets of data - data science process - setting the research goal - retrieving data - cleaning, integrating, and transforming data - exploratory data analysis - build the models - presenting and building applications - Frequency distributions - Outliers - relative frequency distributions - cumulative frequency distributions - frequency distributions for nominal data - interpreting distributions.**

## 1.1 INTRODUCTION

"An automated, machine-based way to extract meaningful insights from big amounts of data. Or a field of study dedicated to collecting, storing, analyzing, and extracting information about people, machines, and the wider world"

Data science combines three areas of expertise: business knowledge, statistical analysis, and computer science. A skilful data scientist uses their business knowledge to understand a problem, applies statistical techniques to collect data and model solutions, and writes programs to run their analysis and generate results.

### 1.1.1 Data Science Definition

Data Science refers to the multi-disciplinary field that extracts knowledge and insights from data using processes, algorithms, and systems based in the scientific method.

Data science is the field of study that combines domain expertise, programming skills, and knowledge of mathematics and statistics to extract meaningful insights from data.

### 1.1.2 Why do we Need Data Science?

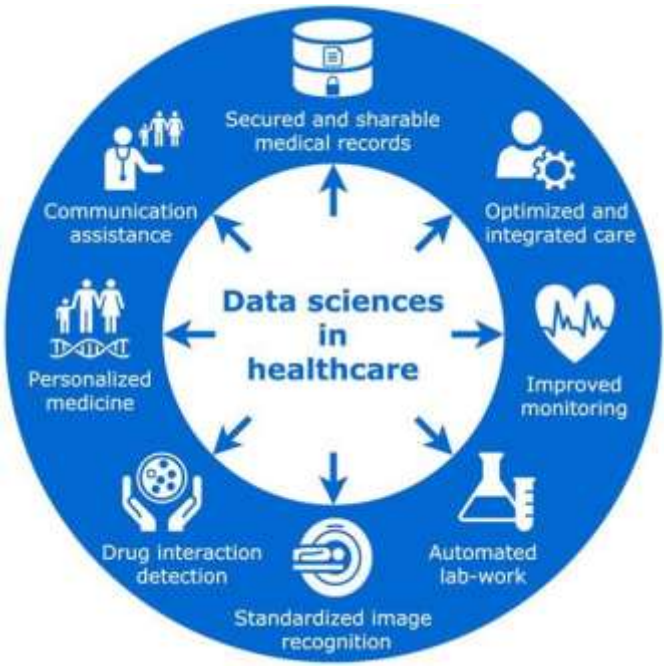
From business to the health industry, science to our everyday lives, marketing to research, in fact, for everything in a fraternity, data is required to thrust the movement forward. Computer science and information technology have taken over our lives, and it is advancing with each passing day with such velocity and variety that the operational techniques used a few years back have now become obsolete.

The same is the case with challenges and problems. The problems and concerns of the past for a specific theme, illness, or shortfall may not be the same today as they have advanced in terms of complexity.

### 1.1.3 Data Science for Business:

Data science is changing the course of the action plan of business procedures and plans. It is solving intricate problems with the help of technology. Data science provides the inside knowledge, which is derived from the big data after processes of extraction and the information. This information is sometimes collected from the ongoing sources within the system, and most of the time, it is mined from external sources.

### 1.1.4 Data Science for Health Care:




The life-threatening health issues are required to be dealt with utter care and diligence to target the actual cause and find a cure with few side effects. The data congregating in clinics and hospitals is huge. The clinical history of a patient and the medicinal treatment given is computerized and kept as a digitalized record, which makes it easier for the

practitioner and doctors to detect the complex diseases at an early stage and understand its complexity.

### 1.1.5 Data Science in Education:

A large amount of student data is being gathered through online forums, websites, and online learning portals, which display their choices concerning selection in the field of study, preferred universities, personal interests, occupational choices, and career. The fact and figures obtained through this data help design the guide and roadmap suggested for higher studies.

### 1.1.6 Data Science in E-Commerce:

Data Science applications help enterprises for making predictions about **profit**, **loss**, and **sales**.

Companies also use Data Science for influencing customers. So that the customers buy their products by using the customers' data for evaluating their needs and interests.

## 1.2 BENEFITS AND USES OF DATA SCIENCE

Data science are used almost everywhere in both commercial and non commercial settings. Commercial companies in almost every industry use data science and big data to gain insights into their customers, processes, staff, completion, and products. Many companies use data science to offer customers a better user experience, as well as to cross-sell, up-sell, and personalize their offerings.

Human resource professionals use people analytics and text mining to screen candidates, monitor the mood of employees, and study informal networks among co-workers. Financial institutions use data science to predict stock markets, determine the risk of lending money, and learn how to attract new clients for their services.

Governmental organizations are also aware of data's value. Many governmental organizations not only rely on internal data scientists to discover valuable information, but also share their data with the public. You can use this data to gain insights or build data-driven applications. Data.gov is but one example; it's the home of the US Government's open data. A data scientist in a governmental organization gets to work on diverse projects such as detecting fraud and other criminal activity or optimizing project funding.

### 1.2.1 Benefits of Data Science:

- ✓ Allowing Management to take better business decisions based on the data.
- ✓ Defining better goals by directing actions based on data trends
- ✓ Helps in improving the relevance of the company's product in the market with proper analysis of the current market and customer purchase trends.
- ✓ It helps in hiring a smooth and less tiring process for the organization.
- ✓ It can help in the expansion of business by analyzing the customer trends and usage of a product.
- ✓ As the applications of Data Science are versatile as it is being used by E-Commerce companies, Medical and Pharma industries, Web-based businesses, Financial and Banking sector, Consulting companies, to give some examples.

In other cases, the benefits include

- Reduced fraud
- More effective risk management
- More profitable financial trading
- Increased manufacturing uptime
- Better supply chain performance
- Stronger cyber security protections and improved patient outcomes.
- Improves Business Predictions
- Helps in Sales and Marketing.
- Automatic Recruitment Process.

### 1.2.2 Data science applications and uses

Common applications that data scientists engage in include predictive modeling, pattern recognition, anomaly detection, classification, categorization and sentiment analysis, as well as development of technologies such as recommendation engines, personalization systems and artificial intelligence (AI) tools like chat bots and autonomous vehicles and machines.

Those applications drive a wide variety of use cases in organizations, including the following:

- customer analytics
- fraud detection
- risk management
- stock trading
- targeted advertising
- website personalization
- customer service
- predictive maintenance
- logistics and supply chain management
- image recognition
- speech recognition
- natural language processing
- cyber security
- medical diagnosis

## 1.3 FACETS OF DATA

In data science you'll come across many different types of data, and each of them tends to require different tools and techniques.

The main categories of data are these:

1. Structured
2. Unstructured
3. Natural language
4. Machine-generated
5. Graph-based
6. Audio, video, and images
7. Streaming

**1. Structured data**

`**Structured data** is when data is in a standardized format, has a well-defined structure, complies to a data model, follows a persistent order, and is easily accessed by humans and programs. This data type is generally stored in a database.

**2. Unstructured data**

Unstructured or qualitative data — is just the opposite. It doesn't fit nicely into a spreadsheet or database. It can be textual or non-textual. It can be human- or machine-generated.

Examples of unstructured data include:

**Media:** Audio and video files, images

**Text files:** Word docs, PowerPoint presentations, email, chat logs

**Email:** There's some internal metadata structure, so it's sometimes called semi-structured, but the message field is unstructured and difficult to analyze with traditional tools.

**Social Media:** Data from social networking sites like Facebook, Twitter and LinkedIn

**Mobile data:** Text messages, locations

**Communications:** Chat, call recordings

**Figure 1.3.2 Email is simultaneously an example of unstructured data and natural language data.**

### 3. Natural language

Natural language is a special type of unstructured data; it's challenging to process because it requires knowledge of specific data science techniques and linguistics.

The natural language processing community has had success in entity recognition, topic recognition, summarization, text completion, and sentiment analysis, but models trained in one domain don't generalize well to other domains. Even state-of-the-art techniques aren't able to decipher the meaning of every piece of text. This shouldn't be a surprise though: humans struggle with natural language as well. It's ambiguous by nature.

### 3. Machine-generated data

Machine-generated data is information that's automatically created by a computer, process, application, or other machine without human intervention. Machine-generated data is becoming a major data resource and will continue to do so. Wikibon has forecast that the market value of the industrial Internet (a term coined by Frost & Sullivan to refer to the integration of complex physical machinery with networked sensors and software) will be approximately $540 billion in 2020. IDC (International Data Corporation) has estimated there will be 26 times more connected things than people in 2020. This network is commonly referred to as the internet of things.

The analysis of machine data relies on highly scalable tools, due to its high volume and speed. Examples of machine data are web server logs, call detail records, network event logs, and telemetry (figure 1.3.2).

```
CSIPERF:TXCOMMIT;313236
2014-11-28 11:36:13, Info                  CSI    00000153 Creating NT transaction (seq
69), objectname [6]"(null)"
2014-11-28 11:36:13, Info                  CSI    00000154 Created NT transaction (seq 69)
result 0x00000000, handle @0x4e54
2014-11-28 11:36:13, Info                  CSI    00000155@2014/11/28:10:36:13.471
Beginning NT transaction commit...
2014-11-28 11:36:13, Info                  CSI    00000156@2014/11/28:10:36:13.705 CSI perf
trace:
CSIPERF:TXCOMMIT;273983
2014-11-28 11:36:13, Info                  CSI    00000157 Creating NT transaction (seq
70), objectname [6]"(null)"
2014-11-28 11:36:13, Info                  CSI    00000158 Created NT transaction (seq 70)
result 0x00000000, handle @0x4e5c
2014-11-28 11:36:13, Info                  CSI    00000159@2014/11/28:10:36:13.764
Beginning NT transaction commit...
2014-11-28 11:36:14, Info                  CSI    0000015a@2014/11/28:10:36:14.094 CSI perf
trace:
CSIPERF:TXCOMMIT;386259
2014-11-28 11:36:14, Info                  CSI    0000015b Creating NT transaction (seq
71), objectname [6]"(null)"
2014-11-28 11:36:14, Info                  CSI    0000015c Created NT transaction (seq 71)
result 0x00000000, handle @0x4e5c
2014-11-28 11:36:14, Info                  CSI    0000015d@2014/11/28:10:36:14.106
Beginning NT transaction commit...
2014-11-28 11:36:14, Info                  CSI    0000015e@2014/11/28:10:36:14.428 CSI perf
trace:
CSIPERF:TXCOMMIT;375581
```

**Figure 1.3.3 Example of machine-generated data**

### 4. Graph-based or network data

"Graph data" can be a confusing term because any data can be shown in a graph. "Graph" in this case points to mathematical graph theory. In graph theory, a graph is a mathematical structure to model pair-wise relationships between objects.

Graph or network data is, in short, data that focuses on the relationship or adjacency of

objects.

The graph structures use nodes, edges, and properties to represent and store graphical data. Graph-based data is a natural way to represent social networks, and its structure allows you to calculate specific metrics such as the influence of a person and the shortest path between two people.

Examples of graph-based data can be found on many social media websites (figure 3.4).

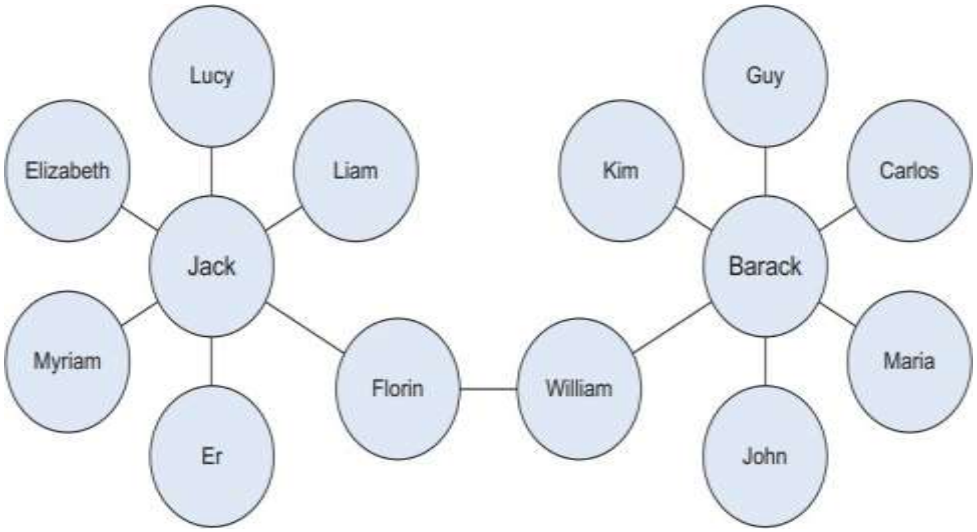



**Figure 1.3.4 Friends in a social network are an example of graph-based data.**

Graph databases are used to store graph-based data and are queried with specialized query languages such as SPARQL. Graph data poses its challenges, but for a computer interpreting additive and image data, it can be even more difficult.

### 5. Audio, image, and video

Audio, image, and video are data types that pose specific challenges to a data scientist. Tasks that are trivial for humans, such as recognizing objects in pictures, turn out to be challenging for computers. MLBAM (Major League Baseball Advanced Media) announced in 2014 that they'll increase video capture to approximately 7 TB per game for the purpose of live, in-game analytics.

### 6. Streaming data

While streaming data can take almost any of the previous forms, it has an extra property. The data flows into the system when an event happens instead of being loaded into a data store in a batch. Although this isn't really a different type of data, we treat it here as such because you need to adapt your process to deal with this type of information. Examples

are the “What’s trending” on Twitter, live sporting or music events, and the stock market.

## 1.4 THE DATA SCIENCE PROCESS

The data science process typically consists of six steps, as you can see in the mind map in figure 4.1.

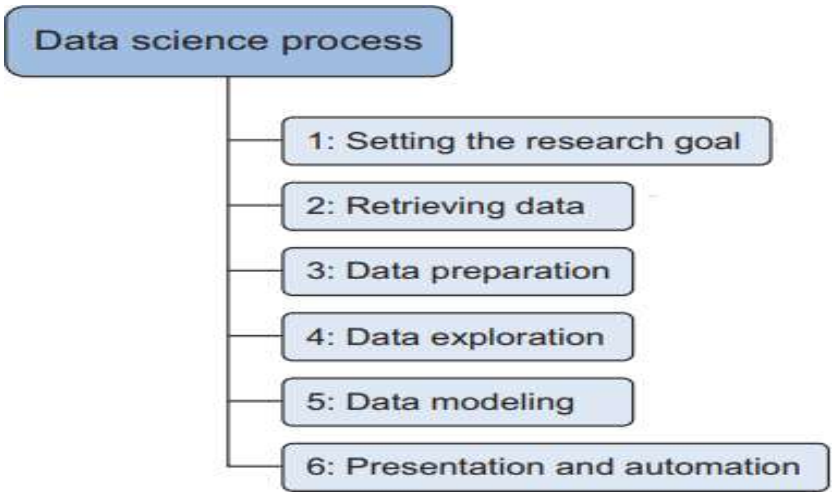



**Figure 1.4.1 The data science process**

### 1.4.1 Overview of the data science process

Following a structured approach to data science helps you to maximize your chances of success in a data science project at the lowest cost. It also makes it possible to take up a project as a team, with each team member focusing on what they do best. Take care, however: this approach may not be suitable for every type of project or be the only way to do good data science.

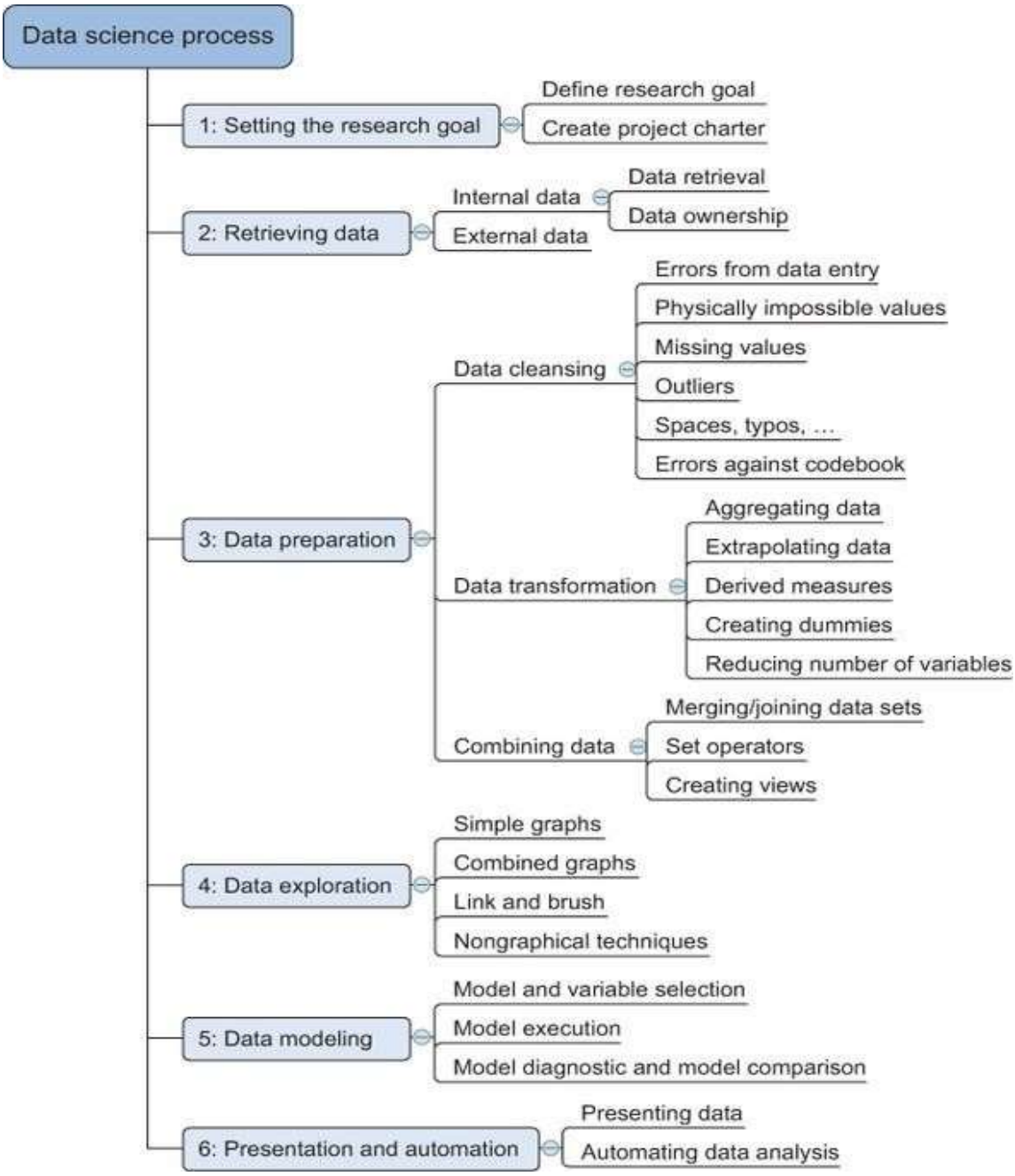



**Figure 1.4.2 The six steps of the data science process**

Figure 1.4.2 summarizes the data science process and shows the main steps and actions you'll take during a project.

- The first step of this process is **setting a research goal**. The main purpose here is making sure all the stakeholders understand the what, how, and why of the project. In every serious project this will result in a project charter.

- The second phase is **data retrieval**. You want to have data available for analysis, so this step includes finding suitable data and getting access to the data from the data owner. The result is data in its raw form, which probably needs polishing and transformation before it becomes usable.

- Now that you have the raw data, it's time to **prepare** it. This includes transforming the data from a raw form into data that's directly usable in your models. To achieve this, you'll detect and correct different kinds of errors in the data, combine data from different data sources, and transform it. If you have successfully completed this step, you can progress to data visualization and modeling.

- The fourth step is **data exploration**. The goal of this step is to gain a deep understanding of the data. You'll look for patterns, correlations, and deviations based on visual and descriptive techniques. The insights you gain from this phase will enable you to start modeling.

- Finally, we get to the **model building** (often referred to as "data modeling" throughout this book). It is now that you attempt to gain the insights or make the predictions stated in your project charter. Now is the time to bring out the heavy guns, but remember research has taught us that often (but not always) a combination of simple models tends to outperform one complicated model. If you've done this phase right, you're almost done.

- The last step of the data science model **is presenting your results and automating the analysis, if needed**. One goal of a project is to change a process and/or make better decisions. You may still need to convince the business that your findings will indeed change the business process as expected. This is where you can shine in your influencer role. The importance of this step is more apparent in projects on a strategic and tactical level. Certain projects require you to perform the business process over and over again, so automating the project will save time.

1. **Setting the research goals and creating a project charter**

A project starts by understanding the what, the why, and the how of your project. What does the company expect you to do? And why does management place such a value on your research? Is it part of a bigger strategic picture or a "lone wolf" project originating from an opportunity someone detected? Answering these three questions (what, why, how) is the goal of the first phase, so that everybody knows what to do and can agree on the best course of action.

Data science is mostly applied in the context of an organization. When the business asks you to perform a data science project, you'll first prepare a project charter. This charter contains information such as what you're going to research, how the company benefits from that, what data and resources you need, a timetable, and deliverables.

- ✓ Spend time understanding the goals and context of your research
- ✓ Create a project charter

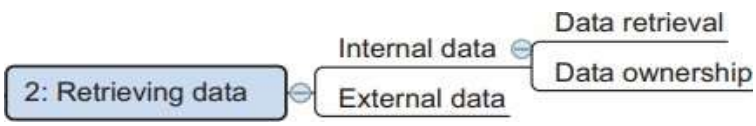
A project charter requires teamwork, and your input covers at least the following:

- A clear research goal
- The project mission and context
- How you're going to perform your analysis
- What resources you expect to use
- Proof that it's an achievable project, or proof of concepts
- Deliverables and a measure of success

A timeline your client can use this information to make an estimation of the project costs and the data and people required for your project to become a success.

2. **Retrieving data**

The second step is to collect data. You've stated in the project charter which data you need and where you can find it. In this step you ensure that you can use the data in your program, which means checking the existence of, quality, and access to the data. Data can also be delivered by third-party companies and takes many forms ranging from Excel spreadsheets to different types of databases.

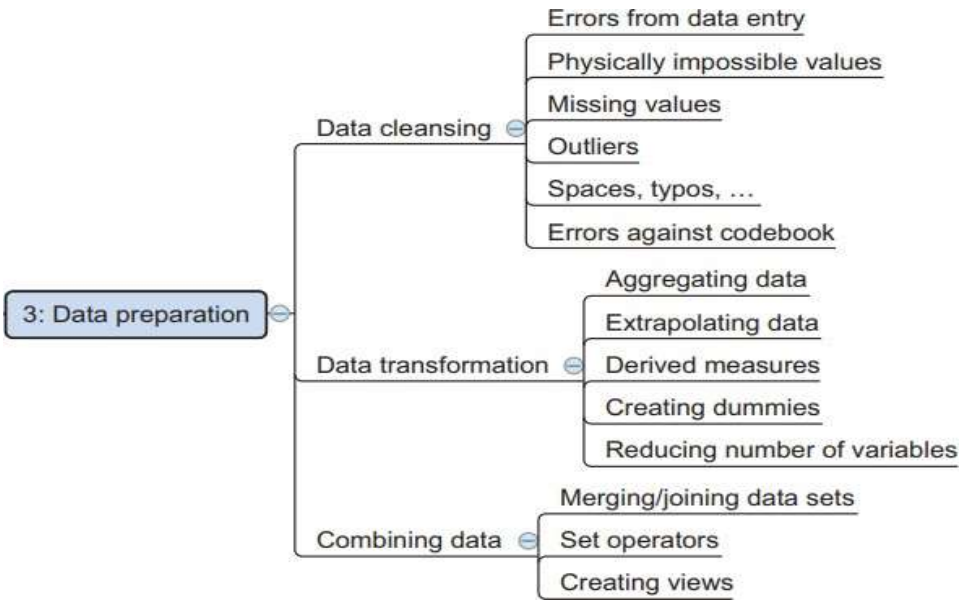


Data can be stored in many forms, ranging from simple text files to tables in a database. The objective now is acquiring all the data you need. This may be difficult, and even if you succeed, data is often like a diamond in the rough: it needs polishing to be of any use to you.

- ✓ Start with data stored within the company
- ✓ Don't be afraid to shop around
- ✓ Do data quality checks now to prevent problems later

### 3. Data preparation

Data collection is an error-prone process; in this phase you enhance the quality of the data and prepare it for use in subsequent steps.



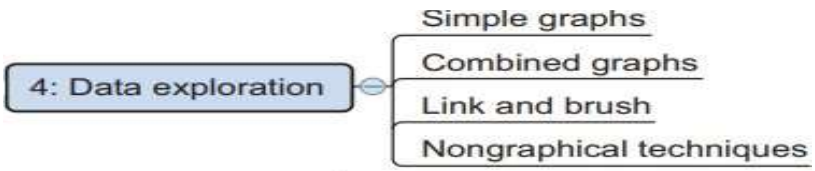

This phase consists of three subphases:

- **data cleansing** removes false values from a data source and inconsistencies across data sources
- **data transformation** ensures that the data is in a suitable format for use in your models.
- **data integration** enriches data sources by combining information from multiple
- data sources

### 4. Data exploration

Data exploration is concerned with building a deeper understanding of your data. You try

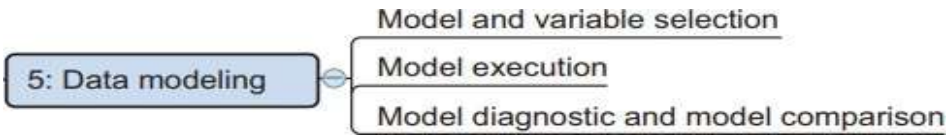
to understand how variables interact with each other, the distribution of the data, and whether there are outliers. To achieve this you mainly use descriptive statistics, visual techniques, and simple modeling. This step often goes by the abbreviation EDA, for Exploratory Data Analysis.




**5. Data modeling or model building**

In this phase you use models, domain knowledge, and insights about the data you found in the previous steps to answer the research question. You select a technique from the fields of statistics, machine learning, operations research, and so on. Building a model is an iterative process that involves selecting the variables for the model, executing the model, and model diagnostics.

With clean data in place and a good understanding of the content, you're ready to build models with the goal of making better predictions, classifying objects, or gaining an understanding of the system that you're modeling.




Building a model is an iterative process. The way you build your model depends on whether you go with classic statistics or the somewhat more recent machine learning school, and the type of technique you want to use.

Either way, most models consist of the following main steps:

1. Selection of a modeling technique and variables to enter in the model
2. Execution of the model
3. Diagnosis and model comparison

6. **Presentation and automation**

Finally, you present the results to your business. These results can take many forms, ranging from presentations to research reports. Sometimes you'll need to automate the

execution of the process because the business will want to use the insights you gained in another project or enable an operational process to use the outcome from your model.

After you've successfully analyzed the data and built a well-performing model, you're ready to present your findings to the world. This is an exciting part; all your hours of hard work have paid off and you can explain what you found to the stakeholders.

Summary of data science process consists of six steps:

1. Setting the research goal—Defining the what, the why, and the how of your project in a project charter.

2. Retrieving data—Finding and getting access to data needed in your project. This data is either found within the company or retrieved from a third party.

3. Data preparation—Checking and remediating data errors, enriching the data with data from other data sources, and transforming it into a suitable format for your models.

4. Data exploration—Diving deeper into your data using descriptive statistics and visual techniques.

5. Data modeling—Using machine learning and statistical techniques to achieve your project goal.

6. Presentation and automation—Presenting your results to the stakeholders and industrializing your analysis process for repetitive reuse and integration with other tools.

**1.5 FREQUENCY DISTRIBUTIONS (TABLES)**

Before jumping to frequency distribution, let us first understand what frequency is. Frequency refers to a measure of how often something has happened. The frequency of any observation tells you the repeated number of times a specific observation occurs in the observed data. Tables can show both qualitative and quantitative variables; qualitative variables are also known as categorical and represent different non-measurable categories like eye colour, brands, etc., while quantitative variables are numeric.

### 1.5.1 WHAT IS FREQUENCY DISTRIBUTION?

Frequency distribution is used to organize the collected data in table form. The data could be marks scored by students, temperatures of different towns, points scored in a volleyball match, etc. After data collection, we have to show data in a meaningful manner for better understanding. Organize the data in such a way that all its features are summarized in a table. This is known as frequency distribution.

There are two types of frequency distributions -grouped and ungrouped. Their usage depends on the data on which we are working. Their analysis is a really important part of probability and statistics.

Let's consider an example to understand this better. The following are the scores of 10 students in the G.K. quiz released by Mr. Chris 15, 17, 20, 15, 20, 17, 17, 14, 14, 20. Let's represent this data in frequency distribution and find out the number of students who got the same marks.

| Quiz Marks | No. of Students |
|---|---|
| 15 | 2 |
| 17 | 3 |
| 20 | 3 |
| 14 | 2 |

### 1.5.2 FREQUENCY DISTRIBUTIONS FOR QUANTITATIVE DATA

A frequency distribution is a collection of observations produced by sorting oobservations into classes and showing their frequency (f ) of occurrence in each class. Table 1 shows one way to organize the weights of the male statistics students listed in Table 1.1. First, arrange a column of consecutive numbers, beginning with the lightest weight (133) at the bottom and ending with the heaviest weight (245) at the top. (Because of the extreme length of this column, many intermediate numbers have been omitted in Table 2.1, a procedure never followed in practice.) Then place a short vertical stroke or tally next to a number each time its value appears in the original set of data; once this process has been completed, substitute for each tally count (not shown in Table 2.1) a number indicating the frequency ( f ) of occurrence of each weight.

| FREQUENCY DISTRIBUTION (UNGROUPED DATA) | |
|---|---|
| **WEIGHT** | ***f*** |
| 245 | 1 |
| 244 | 0 |
| 243 | 0 |
| 242 | 0 |
| * | |
| * | |
| * | |
| 161 | 0 |
| 160 | 4 |
| 159 | 1 |
| 158 | 2 |
| 157 | 3 |
| * | |
| * | |
| * | |
| 136 | 0 |
| 135 | 2 |
| 134 | 0 |
| 133 | 1 |
| Total | 53 |

***Frequency Distribution***
*A collection of observations produced by sorting observations into classes and showing their frequency (f) of occurrence in each class.*

***Frequency Distribution for Ungrouped Data***
*A frequency distribution produced whenever observations are sorted into classes of* single *values.*

**Table 1**

When observations are sorted into classes of single values, as in Table 1, the result is referred to as a frequency distribution for ungrouped data.

Table 2 shows another way to organize the weights in Table 1 according to their frequency of occurrence. When observations are sorted into classes of more than one value, as in Table 2, the result is referred to as a frequency distribution for grouped data. Let's look at the general structure of this frequency distribution. Data are grouped into class intervals with 10 possible values each. The bottom class includes the smallest observation (133), and the top class includes the largest observation (245).

The distance between bottom and top is occupied by an orderly series of classes. The frequency ( f ) column shows the frequency of observations in each class and, at the bottom, the total number of observations in all classes. Let's summarize the more important properties of the distribution of weights in Table 2. Although ranging from the 130s to the 240s, the weights peak in the 150s, with a progressively decreasing but relatively heavy concentration in the 160s and 170s. Furthermore, the distribution of weights is not balanced about its peak, but tilted in the direction of the heavier weights.

**FREQUENCY DISTRIBUTION (GROUPED DATA)**

| WEIGHT | *f* |
|---|---|
| 240–249 | 1 |
| 230–239 | 0 |
| 220–229 | 3 |
| 210–219 | 0 |
| 200–209 | 2 |
| 190–199 | 4 |
| 180–189 | 3 |
| 170–179 | 7 |
| 160–169 | 12 |
| 150–159 | 17 |
| 140–149 | 1 |
| 130–139 | 3 |
| Total | 53 |

***Frequency Distribution for Grouped Data***

*A frequency distribution produced whenever observations are sorted into classes of more than one value.*

**Table 2**

### 1.5.3 GUIDELINES FOR FREQUENCY DISTRIBUTIONS

The "Guidelines for Frequency Distributions" box lists seven rules for producing a well-constructed frequency distribution. The first three rules are essential and should not be violated.

**How Many Classes?**

The seventh guideline requires a few more comments. The use of too many classes—as in Table 3, in which the weights are grouped into 24 classes, each with an interval of 5—tends to defeat the purpose of a frequency distribution, namely, to provide a reasonably concise description of data. On the other hand, the use of too few classes—as in Table 4, in which the weights are grouped into three classes, each with an interval of 50—can mask important data patterns such as the high density of weights in the 150s and 160s.

**Table 3**
**FREQUENCY DISTRIBUTION WITH TOO MANY INTERVALS**

| WEIGHT | f |
|---|---|
| 245–249 | 1 |
| 240–244 | 0 |
| 235–239 | 0 |
| 230–234 | 0 |
| 225–229 | 2 |
| 220–224 | 1 |
| 215–219 | 0 |
| 210–214 | 0 |
| 205–209 | 2 |
| 200–204 | 0 |
| 195–199 | 0 |
| 190–194 | 4 |
| 185–189 | 1 |
| 180–184 | 2 |
| 175–179 | 2 |
| 170–174 | 5 |
| 165–169 | 7 |
| 160–164 | 5 |
| 155–159 | 9 |
| 150–154 | 8 |
| 145–149 | 1 |
| 140–144 | 0 |
| 135–139 | 2 |
| 130–134 | 1 |
| Total | 53 |

**Table 4**
**FREQUENCY DISTRIBUTION WITH TOO FEW INTERVALS**

| WEIGHT | f |
|---|---|
| 200–249 | 6 |
| 150–199 | 43 |
| 100–149 | 4 |
| Total | 53 |

**GUIDELINES FOR FREQUENCY DISTRIBUTIONS**

**Essential**

1. ***Each observation should be included in one, and only one, class.***
   **Example:** 130–139, 140–149, 150–159, etc. It would be incorrect to use 130–140, 140–150, 150–160, etc., in which, because the boundaries of classes overlap, an observation of 140 (or 150) could be assigned to either of two classes.
2. ***List all classes, even those with zero frequencies.***
   **Example:** Listed in Table 2.2 is the class 210–219 and its frequency of zero. It would be incorrect to skip this class because of its zero frequency.
3. ***All classes should have equal intervals.***
   **Example:** 130–139, 140–149, 150–159, etc. It would be incorrect to use 130–139, 140–159, etc., in which the second class interval (140–159) is twice as wide as the first class interval (130–139).

**Optional**

4. ***All classes should have both an upper boundary and a lower boundary.***
   **Example:** 240–249. Less preferred would be 240–above, in which no maximum value can be assigned to observations in this class. (Nevertheless, this type of open-ended class is employed as a space-saving device when many different tables must be listed, as in the *Statistical Abstract of the United States*. An open-ended class appears in the table "Two Age Distributions" in Review Question 2.17 at the end of this chapter.)
5. ***Select the class interval from convenient numbers, such as 1, 2, 3, . . . 10, particularly 5 and 10 or multiples of 5 and 10.***
   **Example:** 130–139, 140–149, in which the class interval of 10 is a convenient number. Less preferred would be 130–142, 143–155, etc., in which the class interval of 13 is not a convenient number.
6. ***The lower boundary of each class interval should be a multiple of the class interval.***
   **Example:** 130–139, 140–149, in which the lower boundaries of 130, 140, are multiples of 10, the class interval. Less preferred would be 135–144, 145–154, etc., in which the lower boundaries of 135 and 145 are not multiples of 10, the class interval.

### When There Are Either Many or Few Observations

There is nothing sacred about 10, the recommended number of classes. When describing large sets of data, you might aim for considerably more than 10 classes in order to portray some of the more fine-grained data patterns that otherwise could vanish. On the other hand, when describing small batches of data, you might aim for fewer than 10 classes in order to spotlight data regularities that otherwise could be blurred. It is best, therefore, to think of 10, the recommended number of classes, as a rough rule of thumb to be applied with discretion.

### Gaps between Classes

In well-constructed frequency tables, the gaps between classes, such as between 149 and

150 in Table 2, show clearly that each observation or score has been assigned to one, and only one, class. The size of the gap should always equal one unit of measurement; that is, it should always equal the smallest possible difference between scores within a particular set of data. Since the gap is never bigger than one unit of measurement, no score can fall into the gap. In the present case, in which the weights are reported to the nearest pound, one pound is the unit of measurement, and therefore, the gap between classes equals one pound. These gaps would not be appropriate if the weights had been reported to the nearest tenth of a pound. In this case, one-tenth of a pound is the unit of measurement, and therefore, the gap should equal one-tenth of a pound.

### Real Limits of Class Intervals

Gaps cannot be ignored when you are determining the actual width of any class interval. The real limits are located at the midpoint of the gap between adjacent tabled boundaries; that is, one-half of one unit of measurement below the lower tabled boundary and one-half of one unit of measurement above the upper tabled boundary.

For example, the real limits for 140–149 in Table 2 are 139.5 (140 minus one-half of the unit of measurement of 1) and 149.5 (149 plus one-half of the unit of measurement of 1), and the actual width of the class interval would be 10 (from 149.5 139.5 = 10). If weights had been reported to the nearest tenth of a pound, the real limits for 140.0–149.9 would be 139.95 (140.0 minus one-half of the unit of measurement of .1) and 149.95 (149.9 plus one-half of one unit of measurement of .1), and the actual width of the class interval still would be 10 (from 149.95 139.95 = 10).

## 1.6 OUTLIERS

Be prepared to deal occasionally with the appearance of one or more very extreme scores, or outliers. A GPA of 0.06, an IQ of 170, summer wages of $62,000—each requires special attention because of its potential impact on a summary of the data.

### Check for Accuracy

Whenever you encounter an outrageously extreme value, such as a GPA of 0.06, attempt to verify its accuracy. For instance, was a respectable GPA of 3.06 recorded erroneously as 0.06? If the outlier survives an accuracy check, it should be treated as a legitimate score.

**Might Exclude from Summaries**

You might choose to segregate (but not to suppress!) an outlier from any summary of the data. For example, you might relegate it to a footnote instead of using excessively wide class intervals in order to include it in a frequency distribution. Or you might use various numerical summaries, such as the median and interquartile range, that ignore extreme scores, including outliers.

**Might Enhance Understanding**

Insofar as a valid outlier can be viewed as the product of special circumstances, it might help you to understand the data. For example, you might understand better why crime rates differ among communities by studying the special circumstances that produce a community with an extremely low (or high) crime rate, or why learning rates differ among third graders by studying a third grader who learns very rapidly (or very slowly).

| SUMMER INCOME | AGE | FAMILY SIZE | GPA |
|---|---|---|---|
| $6,450 | 20 | 2 | 2.30 |
| $4,820 | 19 | 4 | 4.00 |
| $5,650 | 61 | 3 | 3.56 |
| $1,720 | 32 | 6 | 2.89 |
| $600 | 19 | 18 | 2.15 |
| $0 | 22 | 2 | 3.01 |
| $3,482 | 23 | 6 | 3.09 |
| $25,700 | 27 | 3 | 3.50 |
| $8,548 | 21 | 4 | 3.20 |

## 1.7 RELATIVE FREQUENCY DISTRIBUTIONS

An important variation of the frequency distribution is the relative frequency distribution

**Relative frequency distributions show the frequency of each class as a part or fraction of the total frequency for the entire distribution.**

This type of distribution allows us to focus on the relative concentration of observations among different classes within the same distribution. In the case of the weight data in Table 2, it permits us to see that the 160s account for about one-fourth (12/53 = 23, or 23%) of all observations. This type of distribution is especially helpful when you must compare two or more distributions based on different total numbers of observations.

**Constructing Relative Frequency Distributions**

To convert a frequency distribution into a relative frequency distribution, divide the frequency for each class by the total frequency for the entire distribution. Table 5 illustrates a relative frequency distribution based on the weight distribution of Table 2.

**Table 5**
**RELATIVE FREQUENCY DISTRIBUTION**

| WEIGHT | *f* | RELATIVE *f* |
|---|---|---|
| 240–249 | 1 | .02 |
| 230–239 | 0 | .00 |
| 220–229 | 3 | .06 |
| 210–219 | 0 | .00 |
| 200–209 | 2 | .04 |
| 190–199 | 4 | .08 |
| 180–189 | 3 | .06 |
| 170–179 | 7 | .13 |
| 160–169 | 12 | .23 |
| 150–159 | 17 | .32 |
| 140–149 | 1 | .02 |
| 130–139 | 3 | .06 |
| Total | 53 | 1.02* |

** The sum does not equal 1.00 because of rounding-off errors.*

The conversion to proportions is straightforward. For instance, to obtain the proportion of .06 for the class 130–139, divide the frequency of 3 for that class by the total frequency of 53. Repeat this process until a proportion has been calculated for each class.

## 1.8 CUMULATIVE FREQUENCY DISTRIBUTIONS

**Cumulative frequency distributions show the total number of observations in each class and in all lower-ranked classes.**

This type of distribution can be used effectively with sets of scores, such as test scores for intellectual or academic aptitude, when relative standing within the distribution assumes primary importance. Under these circumstances, cumulative frequencies are usually converted, in turn, to cumulative percentages. Cumulative percentages are often

referred to as percentile ranks.

**Table 6**
**CUMULATIVE FREQUENCY DISTRIBUTION**

| WEIGHT | *f* | CUMULATIVE *f* | CUMULATIVE PERCENT |
|---|---|---|---|
| 240–249 | 1 | 53 | 100 |
| 230–239 | 0 | 52 | 98 |
| 220–229 | 3 | 52 | 98 |
| 210–219 | 0 | 49 | 92 |
| 200–209 | 2 | 49 | 92 |
| 190–199 | 4 | 47 | 89 |
| 180–189 | 3 | 43 | 81 |
| 170–179 | 7 | 40 | 75 |
| 160–169 | 12 | 33 | 62 |
| 150–159 | 17 | 21 | 40 |
| 140–149 | 1 | 4 | 8 |
| 130–139 | 3 | 3 | 6 |
| Total | 53 | | |

Cumulative Percentages As has been suggested, if relative standing within a distribution is particularly important, then cumulative frequencies are converted to cumulative percentages. A glance at Table 6 reveals that 75 percent of all weights are the same as or lighter than the weights between 170 and 179 lbs. To obtain this cumulative percentage (75%), the cumulative frequency of 40 for the class 170–179 should be divided by the total frequency of 53 for the entire distribution.

| GRE | (a) CUMULATIVE *f* | (b) CUMULATIVE PERCENT(%) |
|---|---|---|
| 725–749 | 200 | 100 |
| 700–724 | 199 | 100 |
| 675–699 | 196 | 98 |
| 650–674 | 182 | 91 |
| 625–649 | 152 | 76 |
| 600–624 | 118 | 59 |
| 575–599 | 76 | 38 |
| 550–574 | 46 | 23 |
| 525–549 | 19 | 10 |
| 500–524 | 6 | 3 |
| 475–499 | 2 | 1 |

**Percentile Ranks**

When used to describe the relative position of any score within its parent distribution, cumulative percentages are referred to as percentile ranks. The percentile rank of a score indicates the percentage of scores in the entire distribution with similar or smaller values than that score. Thus a weight has a percentile rank of 80 if equal or lighter weights constitute 80 percent of the entire distribution.

**Approximate Percentile Ranks (from Grouped Data)**

The assignment of exact percentile ranks requires that cumulative percentages be obtained from frequency distributions for ungrouped data. If we have access only to a frequency distribution for grouped data, as in Table 6, cumulative percentages can be used to assign approximate percentile ranks. In Table 6, for example, any weight in the class 170–179 could be assigned an approximate percentile rank of 75, since 75 is the cumulative percent for this class.

## 1.9 FREQUENCY DISTRIBUTIONS FOR QUALITATIVE (NOMINAL) DATA

When, among a set of observations, any single observation is a word, letter, or numerical code, the data are qualitative. Frequency distributions for qualitative data are easy to construct. Simply determine the frequency with which observations occupy each class, and report these frequencies as shown in Table 7 for the Facebook profile survey. This frequency distribution reveals that Yes replies are approximately twice as prevalent as No replies.

**Table 7**
**FACEBOOK PROFILE SURVEY**

| *Response* | *f* |
|---|---|
| Yes | 56 |
| No | 27 |
| Total | 83 |

**Ordered Qualitative**

Data It's totally arbitrary whether Yes is listed above or below No in Table 7. When, however, qualitative data have an ordinal level of measurement because observations can be ordered from least to most, that order should be preserved in the frequency table, as illustrated in Table 8, in which military ranks are listed in descending order from general to lieutenant.

**Table 8**
**RANKS OF OFFICERS IN THE U.S. ARMY (PROJECTED 2016)**

| RANK | *f* | PROPORTION | CUMULATIVE PERCENT |
|---|---|---|---|
| General | 311 | .004* | 100.0 |
| Colonel | 13,156 | .167 | 99.6 |
| Major | 16,108 | .204 | 82.9 |
| Captain | 29,169 | .370 | 62.5 |
| Lieutenant | 20,083 | .255 | 25.5 |
| Total | 78,827 | | |

**To avoid a value of .00 for General, proportions are carried three places to the right of the decimal point.*
*Source: http://www.statista.com/statistics*

### Relative and Cumulative Distributions for Qualitative Data

Frequency distributions for qualitative variables can always be converted into relative frequency distributions, as illustrated in Table 8. Furthermore, if measurement is ordinal because observations can be ordered from least to most, cumulative frequencies (and cumulative percentages) can be used. As illustrated in Table 8, it's appropriate to claim, for example, that a captain has an approximate percentile rank of 63 among officers since 62.5 (or 63) is the cumulative percent for this class.

## 1.10 INTERPRETING DISTRIBUTIONS CONSTRUCTED BY OTHERS

When inspecting a distribution for the first time, train yourself to look at the entire table, not just the distribution. Read the title, column headings, and any footnotes. Where do the data come from? Is a source cited? Next, focus on the form of the frequency distribution. Is it well constructed? For quantitative data, does the total number of classes seem to avoid either over- or under-summarizing the data? After these preliminaries, inspect the content of the frequency distribution. What is the approximate range? Does it seem reasonable? (Otherwise, you might be misinterpreting the distribution or the distribution might contain one or more outliers that require special attention.) As best you can, disregard the inevitable irregularities that accompany a frequency distribution and focus on its overall appearance or shape.

# UNIT II
# DESCRIBING DATA

Types of Data - Types of Variables -Describing Data with Tables and Graphs -Describing Data with Averages - Describing variability - range - variance - standard deviation - degrees of freedom - interquartile range - variability for qualitative and ranked data- Normal distribution and Standard (Z) Scores..

## 2.1 THREE TYPES OF DATA

- **Qualitative data** consist of words (Yes or No), letters (Y or N), or numerical codes (0 or 1) that represent a class or category.
- **Ranked data** consist of numbers (1st, 2nd, . . . 40th place) that represent relative standing within a group.
- **Quantitative data** consist of numbers (weights of 238, 170, . . . 185 lbs) that represent an amount or a count. To determine the type of data, focus on a single observation in any collection of observations

## 2.2 TYPES OF VARIABLES

A **variable** is a characteristic or property that can take on different values.

- The weights can be described not only as quantitative data but also as observations for a quantitative variable, since the various weights take on different numerical values.
- By the same token, the replies can be described as observations for a qualitative variable, since the replies to the Facebook profile question take on different values of either Yes or No.
- Given this perspective, any single observation can be described as a constant, since it takes on only one value.

### 2.2.1 Discrete and Continuous Variables

Quantitative variables can be further distinguished as **discrete or continuous.** A **discrete variable** consists of isolated numbers separated by gaps. Discrete variables can only assume specific values that you cannot subdivide. Typically, you count discrete values, and the results are integers.

Examples

- Counts- such as the number of children in a family. (1, 2, 3, etc., but never 1.5)
- These variables cannot have fractional or decimal values. You can have 20 or 21 cats, but not 20.5
- The number of heads in a sequence of coin tosses.
- The result of rolling a die.
- The number of patients in a hospital.
- The population of a country.

While discrete variables have no decimal places, the average of these values can be fractional. For example, families can have only a discrete number of children: 1, 2, 3, etc. However, the average number of children per family can be 2.2.

A **continuous variable** consists of numbers whose values, at least in theory, have no restrictions.

### 2.2.3 Independent and Dependent Variables Independent Variables

In an experiment, an **independent variable** is the treatment manipulated by the investigator.

- Independent variables (IVs) are the ones that you include in the model to explain or predict changes in the dependent variable.
- Independent indicates that they stand alone and other variables in the model do not influence them.
- Independent variables are also known as predictors, factors, treatment variables, explanatory variables, input variables, x-variables, and right-hand variables—because they appear on the right side of the equals sign in a regression equation.
- It is a variable that stands alone and isn't changed by the other variables you are trying to measure.

**Dependent Variable**

*When a variable is believed to have been influenced by the independent variable, it is called a* **dependent variable.** In an experimental setting, the dependent variable is measured, counted, or recorded by the investigator.

**Confounding Variable**

*An uncontrolled variable that compromises the interpretation of a study is known as a* **confounding variable.** Sometimes a confounding variable occurs because it's impossible to assign subjects randomly to different conditions.

## 2.3 Describing Data with Tables and Graphs

### Frequency Distributions for Quantitative Data

- A frequency distribution is a collection of observations produced by sorting observations into classes and showing their frequency (f) of occurrence in each class.
- When observations are sorted into classes of single values, as in Table 2.1, the result is referred to as a frequency distribution for ungrouped data.
- The frequency distribution shown in Table 2.1 is only partially displayed because there are more than 100 possible values between the largest and smallest observations. Frequency distribution table is much more informative if possible observed values is less then 20. If more entry is observed then grouped data is used.

**Grouped Data**

According to their frequency of occurrence. When observations are sorted into classes of more than one value result is referred to as a frequency for grouped data. (Shown in table 2.2)

**Table 2.2 FREQUENCY DISTRIBUTION (GROUPED DATA)**

| WEIGHT | f |
|---|---|
| 240–249 | 1 |
| 230–239 | 0 |
| 220–229 | 3 |
| 210–219 | 0 |
| 200–209 | 2 |
| 190–199 | 4 |
| 180–189 | 3 |
| 170–179 | 7 |
| 160–169 | 12 |
| 150–159 | 17 |
| 140–149 | 1 |
| 130–139 | 3 |
| Total | 53 |

- The general structure of this frequency distribution is the data's are grouped into class intervals with 10 possible values each.
- The frequency ( f ) column shows the frequency of observations in each class and, at the bottom, the total number of observations in all classes.

**Table 2.3 FREQUENCY DISTRIBUTION WITH TOO MANY INTERVALS**

| WEIGHT | f |
|---|---|
| 245–249 | 1 |
| 240–244 | 0 |
| 235–239 | 0 |
| 230–234 | 0 |
| 225–229 | 2 |
| 220–224 | 1 |
| 215–219 | 0 |
| 210–214 | 0 |
| 205–209 | 2 |
| 200–204 | 0 |
| 195–199 | 0 |
| 190–194 | 4 |
| 185–189 | 1 |
| 180–184 | 2 |
| 175–179 | 2 |
| 170–174 | 5 |
| 165–169 | 7 |
| 160–164 | 5 |
| 155–159 | 9 |
| 150–154 | 8 |
| 145–149 | 1 |
| 140–144 | 0 |
| 135–139 | 2 |
| 130–134 | 1 |
| Total | 53 |

**Table 2.4 FREQUENCY DISTRIBUTION WITH TOO FEW INTERVALS**

| WEIGHT | f |
|---|---|
| 200–249 | 6 |
| 150–199 | 43 |
| 100–149 | 4 |
| Total | 53 |

**GUIDELINES FOR FREQUENCY DISTRIBUTIONS**

**Essential**

1. ***Each observation should be included in one, and only one, class.***
   **Example:** 130–139, 140–149, 150–159, etc. It would be incorrect to use 130–140, 140–150, 150–160, etc., in which, because the boundaries of classes overlap, an observation of 140 (or 150) could be assigned to either of two classes.
2. ***List all classes, even those with zero frequencies.***
   **Example:** Listed in Table 2.2 is the class 210–219 and its frequency of zero. It would be incorrect to skip this class because of its zero frequency.
3. ***All classes should have equal intervals.***
   **Example:** 130–139, 140–149, 150–159, etc. It would be incorrect to use 130–139, 140–159, etc., in which the second class interval (140–159) is twice as wide as the first class interval (130–139).

**Optional**

4. ***All classes should have both an upper boundary and a lower boundary.***
   **Example:** 240–249. Less preferred would be 240–above, in which no maximum value can be assigned to observations in this class. (Nevertheless, this type of open-ended class is employed as a space-saving device when many different tables must be listed, as in the *Statistical Abstract of the United States*. An open-ended class appears in the table "Two Age Distributions" in Review Question 2.17 at the end of this chapter.)
5. ***Select the class interval from convenient numbers, such as 1, 2, 3, . . . 10, particularly 5 and 10 or multiples of 5 and 10.***
   **Example:** 130–139, 140–149, in which the class interval of 10 is a convenient number. Less preferred would be 130–142, 143–155, etc., in which the class interval of 13 is not a convenient number.
6. ***The lower boundary of each class interval should be a multiple of the class interval.***
   **Example:** 130–139, 140–149, in which the lower boundaries of 130, 140, are multiples of 10, the class interval. Less preferred would be 135–144, 145–154, etc., in which the lower boundaries of 135 and 145 are not multiples of 10, the class interval.
7. ***Aim for a total of approximately 10 classes.***
   **Example:** The distribution in Table 2.2 uses 12 classes. Less preferred would be the distributions in Tables 2.3 and 2.4. The distribution in Table 2.3 has too many classes (24), whereas the distribution in Table 2.4 has too few classes (3).

### 2.3.1 OUTLIERS

An outlier is an extremely high or extremely low data point relative to the nearest data point and the rest of the neighboring co-existing values in a data graph or dataset you're

working with.

### 2.3.2 RELATIVE FREQUENCY DISTRIBUTIONS

Relative frequency distributions show the frequency of each class as a part or fraction of the total frequency for the entire distribution.

This type of distribution is especially helpful when you must compare two or more distributions based on different total numbers of observations.

The conversion to relative frequencies allows a direct comparison of the shapes of two distributions without adjust other observations.

Table 2.5
RELATIVE FREQUENCY DISTRIBUTION

| WEIGHT | f | RELATIVE f |
|---|---|---|
| 240–249 | 1 | .02 |
| 230–239 | 0 | .00 |
| 220–229 | 3 | .06 |
| 210–219 | 0 | .00 |
| 200–209 | 2 | .04 |
| 190–199 | 4 | .08 |
| 180–189 | 3 | .06 |
| 170–179 | 7 | .13 |
| 160–169 | 12 | .23 |
| 150–159 | 17 | .32 |
| 140–149 | 1 | .02 |
| 130–139 | 3 | .06 |
| Total | 53 | 1.02* |

*The sum does not equal 1.00 because of rounding-off errors.

**Constructing Relative Frequency Distributions**

To convert a frequency distribution into a relative frequency distribution, divide the frequency for each class by the total f requency for the entire distribution.

Table 2.5 illustrates a relative frequency distribution based on the weight distribution of Table 2.2.

**Percentages or Proportions**

Some people prefer to deal with percentages rather than proportions because percentages usually lack decimal points.

A proportion always varies between 0 and 1, whereas a percentage always varies between 0 percent and 100 percent.

### 2.3.3 CUMULATIVE FREQUENCY DISTRIBUTIONS

Cumulative frequency distributions show the total number of observations in each class and in all lower- ranked classes.

Cumulative frequencies are usually converted, in turn, to cumulative percentages. Cumulative percentages are often referred to as percentile ranks.

**Constructing Cumulative Frequency Distributions**

To convert a frequency distribution into a cumulative frequency distribution, add to the frequency of each class the sum of the frequencies of all classes ranked below it.

**Table 2.6**
**CUMULATIVE FREQUENCY DISTRIBUTION**

| WEIGHT | f | CUMULATIVE f | CUMULATIVE PERCENT |
|---|---|---|---|
| 240–249 | 1 | 53 | 100 |
| 230–239 | 0 | 52 | 98 |
| 220–229 | 3 | 52 | 98 |
| 210–219 | 0 | 49 | 92 |
| 200–209 | 2 | 49 | 92 |
| 190–199 | 4 | 47 | 89 |
| 180–189 | 3 | 43 | 81 |
| 170–179 | 7 | 40 | 75 |
| 160–169 | 12 | 33 | 62 |
| 150–159 | 17 | 21 | 40 |
| 140–149 | 1 | 4 | 8 |
| 130–139 | 3 | 3 | 6 |
| Total | 53 | | |

**Cumulative Percentages**

As has been suggested, if relative standing within a distribution is particularly important, then cumulative frequencies are converted to cumulative percentages

**Percentile Ranks**

When used to describe the relative position of any score within its parent distribution, cumulative percentages are referred to as percentile ranks.

### 2.3.4 FREQUENCY DISTRIBUTIONS FOR QUALITATIVE (NOMINAL) DATA

Frequency distributions for qualitative data are easy to construct. Simply determine the frequency with which observations occupy Each class, and report these frequencies as shown in Table 2.7 for the Face book profile survey

**Table 2.7**
**FACEBOOK PROFILE SURVEY**

| *Response* | *f* |
|---|---|
| Yes | 56 |
| No | 27 |
| Total | 83 |

Qualitative data have an ordinal level of measurement because Observations can be ordered from least to most, that order should be preserved in the frequency table

**Relative and Cumulative Distributions for Qualitative Data**

Frequency distributions for qualitative variables can always be converted into relative frequency distributions.

if measurement is ordinal because observations can be ordered from least to most, cumulative frequencies (and cumulative percentages) can be used.

**Table 2.8**
**RANKS OF OFFICERS IN THE U.S. ARMY (PROJECTED 2016)**

| RANK | f | PROPORTION | CUMULATIVE PERCENT |
|---|---|---|---|
| General | 311 | .004* | 100.0 |
| Colonel | 13,156 | .167 | 99.6 |
| Major | 16,108 | .204 | 82.9 |
| Captain | 29,169 | .370 | 62.5 |
| Lieutenant | 20,083 | .255 | 25.5 |
| Total | 78,827 | | |

**To avoid a value of .00 for General, proportions are carried three places to the right of the decimal point.*

## 2.4 GRAPHS

Data can be described clearly and concisely with the aid of a well-constructed frequency distribution. And data can often be described even more vividly by converting frequency distributions into graphs.

### 2.4.1 GRAPHS FOR QUANTITATIVE DATA

**Histograms**

A bar-type graph for quantitative data. The common boundaries between adjacent bars emphasize the continuity of the data, as with continuous variables.

A histogram is a display of statistical information that uses rectangles to show the frequency of data items in successive numerical intervals of equal size.

**Important features of histograms**

- Equal units along the horizontal axis (the X axis, or abscissa) reflect the various class intervals of the frequency distribution.
- Equal units along the vertical axis (the Y axis, or ordinate) reflect increases in frequency. (The units along the vertical axis do not have to be the same width as those along the horizontal axis.)
- The intersection of the two axes defines the origin at which both numerical scales equal 0.
- Numerical scales always increase from left to right along the horizontal axis and from bottom to top along the vertical axis
- The body of the histogram consists of a series of bars whose heights reflect the frequencies for the various classes.
- The adjacent bars in histograms have common boundaries that emphasize the continuity of quantitative data for continuous variables.
- The introduction of gaps between adjacent bars would suggest an artificial disruption in the data more appropriate for discrete quantitative variables or for qualitative variables.

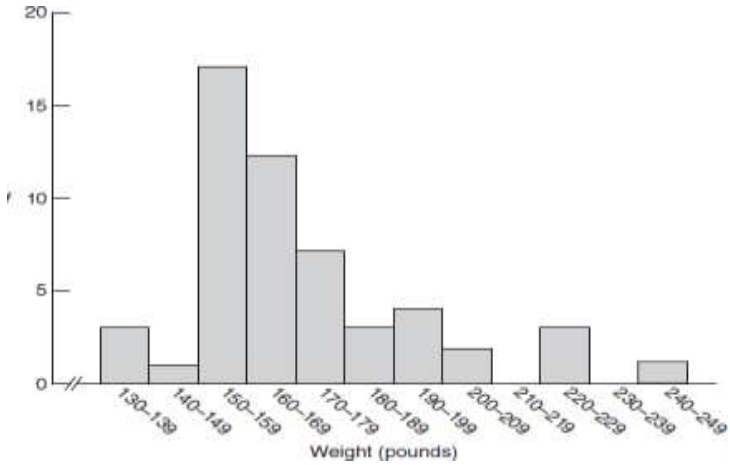



**Figure:2.4 Histogram**

### 2.4.2 Frequency Polygon

An important variation on a histogram is the frequency polygon, or line graph. Frequency polygons may be constructed directly from frequency distributions.

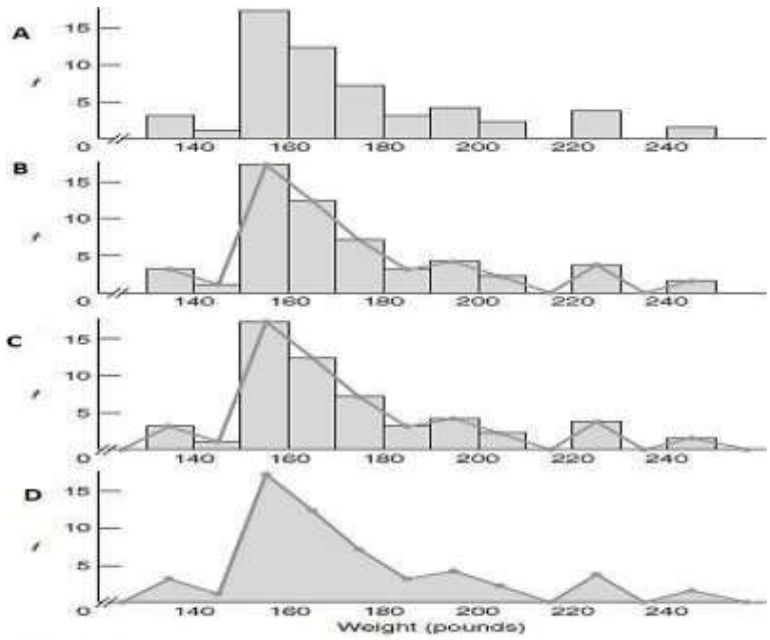



**FIGURE**
*Transition from histogram to frequency polygon.*

**Step-by-step transformation of a histogram into a frequency polygon**

A. This panel shows the histogram for the weight distribution.
B. Place dots at the midpoints of each bar top or, in the absence of bar tops, at midpoints for classes on the horizontal axis, and connect them with straight lines.
C. Anchor the frequency polygon to the horizontal axis. First, extend the upper tail to the

midpoint of the first unoccupied class on the upper flank of the histogram. Then extend the lower tail to the midpoint of the first unoccupied class on the lower flank of the histogram. Now all of the area under the frequency polygon is enclosed completely.

D. Finally, erase all of the histogram bars, leaving only the frequency polygon.

For example

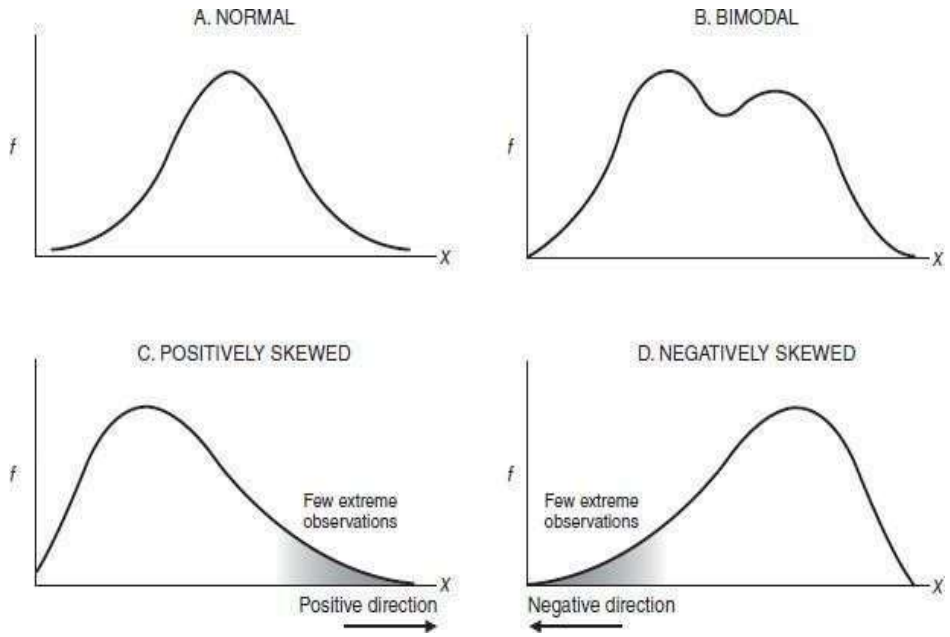
Enter each raw score into the stem and leaf display. As suggested by the shaded coding in Table 2.9, the first raw score of 160 reappears as a leaf of 0 on a stem of 16. The next raw score of 193 reappears as a leaf of 3 on a stem of 19, and the third raw score of 226 reappears as a leaf of 6 on a stem of 22, and so on, until each raw score reappears as a leaf on its appropriate stem.

**Table 2.9**
**CONSTRUCTING STEM AND LEAF DISPLAY FROM WEIGHTS OF MALE STATISTICS STUDENTS**

| RAW SCORES | | | | | STEM AND LEAF DISPLAY |
|---|---|---|---|---|---|
| 160 | 165 | 135 | 175 | | |
| 193 | 168 | 245 | 165 | 13 | 3 5 5 |
| 226 | 169 | 170 | 185 | 14 | 5 |
| 152 | 160 | 156 | 154 | 15 | 2 7 1 7 8 0 2 0 2 6 9 8 2 6 4 7 6 |
| 180 | 170 | 160 | 179 | 16 | 0 3 5 8 9 0 0 0 6 5 5 5 |
| 205 | 150 | 225 | 165 | 17 | 2 0 0 0 2 5 9 |
| 163 | 152 | 190 | 206 | 18 | 0 0 5 |
| 157 | 160 | 159 | 165 | 19 | 3 0 0 0 |
| 151 | 190 | 172 | 157 | 20 | 5 6 |
| 157 | 150 | 190 | 156 | 21 | |
| 220 | 133 | 166 | 135 | 22 | 6 0 5 |
| 145 | 180 | 158 | | 23 | |
| 158 | 152 | 152 | | 24 | 5 |
| 172 | 170 | 156 | | | |

## 2.5 TYPICAL SHAPES

Whether expressed as a histogram, a frequency polygon, or a stem and leaf display, an important characteristic of a frequency distribution is its shape. Below figure shows some of the more typical shapes for smoothed frequency polygons (which ignore the inevitable irregularities of real data).

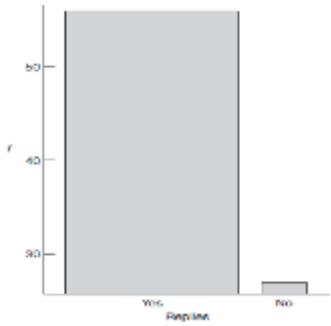



## 2.6 A GRAPH FOR QUALITATIVE (NOMINAL) DATA

- As with histograms, equal segments along the horizontal axis are allocated to the different words or classes that appear in the frequency distribution for qualitative data. Likewise, equal segments along the vertical axis reflect increases in frequency. The body of the bar graph consists of a series of bars whose heights reflect the frequencies for the various words or classes.
- A person's answer to the question "Do you have a Facebook profile?" is either Yes or No, not some impossible intermediate value, such as 40 percent Yes and 60 percent No.
- Gaps are placed between adjacent bars of bar graphs to emphasize the discontinuous nature of qualitative data.

## 2.7 MISLEADING GRAPHS

Graphs can be constructed in an unscrupulous manner to support a particular point of view. Popular sayings says, including "Numbers don't lie, but statisticians do" and "There are three kinds of lies— lies, damned lies, and statistics."




**CONSTRUCTING GRAPHS**

1. ***Decide on the appropriate type of graph,*** recalling that histograms and frequency polygons are appropriate for quantitative data, while bar graphs are appropriate for qualitative data and also are sometimes used with discrete quantitative data.
2. ***Draw the horizontal axis, then the vertical axis,*** remembering that the vertical axis should be about as tall as the horizontal axis is wide.
3. ***Identify the string of class intervals that eventually will be superimposed on the horizontal axis.*** For qualitative data or ungrouped quantitative data, this is easy—just use the classes suggested by the data. For grouped quantitative data, proceed as if you were creating a set of class intervals for a frequency distribution. (See the box "Constructing Frequency Distributions" on page 27.)
4. ***Superimpose the string of class intervals (with gaps for bar graphs) along the entire length of the horizontal axis.*** For histograms and frequency polygons, be prepared for some trial and error—use a pencil! Do not use a string of empty class intervals to bridge a sizable gap between the origin of 0 and the smallest class interval. Instead, use wiggly lines to signal a break in scale, then begin with the smallest class interval. Also, do not clutter the horizontal scale with excessive numbers—use just a few convenient numbers.
5. ***Along the entire length of the vertical axis, superimpose a progression of convenient numbers,*** beginning at the bottom with 0 and ending at the top with a number as large as or slightly larger than the maximum observed frequency. If there is a considerable gap between the origin of 0 and the smallest observed frequency, use wiggly lines to signal a break in scale.
6. Using the scaled axes, ***construct bars (or dots and lines) to reflect the frequency of observations within each class interval.*** For frequency polygons, dots should be located above the midpoints of class intervals, and both tails of the graph should be anchored to the horizontal axis, as described under "Frequency Polygons" in Section 2.8.
7. ***Supply labels for both axes and a title (or even an explanatory sentence) for the graph.***

Unlike the independent variable, the dependent variable isn't manipulated by the investigator. Instead, it represents an outcome: the data produced by the experiment.

**Confounding Variable**

An uncontrolled variable that compromises the interpretation of a study is known as a **confounding variable.** Sometimes a confounding variable occurs because it's impossible to assign subjects randomly to different conditions.

## 2.8 DESCRIBING DATA WITH AVERAGES

Long-term investors in the stock market are able to ignore, with only an occasional sleepless night, daily fluctuations in their stocks by remembering that, **on average**, the annual growth rate of stocks during the past 50 years has exceeded by several percentage points that of more conservative investments in bonds. You might stop smoking because, **on average**, nonsmokers can expect to live longer than heavy smokers (by as much as 10

years, according to some researchers). You might strengthen your resolve to graduate from college upon hearing that, **on average**, the lifetime earnings of college graduates are almost double those of high school graduates.

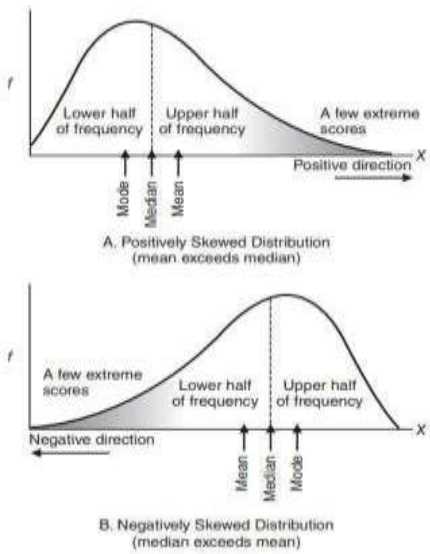
Each of these has its special uses, but the mean is the most important average in both descriptive and inferential statistics.

### 2.8.1 MODE

**The mode reflects the value of the most frequently occurring score.**

Table 1 shows the number of years served by 20 recent U.S. presidents, beginning with Benjamin Harrison (4 years) and ending with Bill Clinton (8 years). Four years is the modal term, since the greatest number of presidents, 7, served this term. Note that the mode equals 4 years, the value of the most frequently occurring term, not 7, the frequency with which that term occurred.

### 2.8.2 MEDIAN




The median reflects the middle value when observations are ordered from least to most.

#### Finding the Median

Table 2 shows how to find the median for two different sets of scores. The numbers in shaded squares cross-reference instructions in the top panel with examples in the bottom panel. Study Table 2 before reading on.

**Table 2**
**FINDING THE MEDIAN**

**A. INSTRUCTIONS**

1. Order scores from least to most.
2. Find the middle position by adding one to the total number of scores and dividing by 2.
3. *If the middle position is a whole number*, as in the left-hand panel below, use this number to *count* into the set of ordered scores.
4. The value of the median equals the value of the score located at the middle position.
5. *If the middle position is not a whole number*, as in the right-hand panel below, use the two nearest whole numbers to *count* into the set of ordered scores.
6. The value of the median equals the value midway between those of the two middlemost scores; to find the midway value, add the two given values and divide by 2.

| **B. EXAMPLES** | |
|---|---|
| Set of five scores: | Set of six scores: |
| 2, 8, 2, 7, 6 | 3, 8, 9, 3, 1, 8 |
| 1 2, 2, 6, 7, 8 | 1 1, 3, 3, 8, 8, 9 |
| 2 $\frac{5+1}{2}=3$ | 2 $\frac{6+1}{2}=3.5$ |
| 2, 2, 6, 7, 8 ↑ | |
| 3 1, 2, 3 | |
| 4 median = 6 | 1, 3, 3, 8, 8, 9 ↑ ↑ |
| | 5 1, 2, 3, 4 |
| | 6 median $=\frac{3+8}{2}=5.5$ |

*Strictly speaking, the median always has a percentile rank of exactly 50 only insofar as interpolation procedures, not discussed in this book, identify the value of the median with a single point along the numerical scale for the data.

Then, following the instructions in Table 2, verify that the median term for the 20 presidents equals 4.5 years, since 4.5 is the value midway between the values (4 and 5) of the two middlemost (10th- and 11th-ranked) terms in Table 3.

**Table 3**
**TERMS IN YEARS OF 20 RECENT U.S. PRESIDENTS**

| ARRANGED BY LENGTH | DEVIATION FROM MEAN | SUM OF DEVIATIONS |
|---|---|---|
| 12 | 6.40 | |
| 8 | 2.40 | |
| 8 | 2.40 | |
| 8 | 2.40 | 21.6 |
| 8 | 2.40 | |
| 8 | 2.40 | |
| 8 | 2.40 | |
| 6 | 0.40 | |
| 6 | 0.40 | |
| (mean = 5.60) | | 0 |
| 5 | –0.60 | |
| 4 | –1.60 | |
| 4 | –1.60 | |
| 4 | –1.60 | |
| 4 | –1.60 | |
| 4 | –1.60 | –21.6 |
| 4 | –1.60 | |
| 4 | –1.60 | |
| 3 | –2.60 | |
| 2 | –3.60 | |
| 2 | –3.60 | |

### 2.8.3 MEAN

The mean is the most common average, one you have doubtless calculated many times.

The mean is found by adding all scores and then dividing by the number of scores. That is, To find the mean term for the 20 presidents, add all 20 terms in Table 1 (4 + . . . + 4 + 8) to obtain a sum of 112 years, and then divide this sum by 20, the number of presidents, to obtain a mean of 5.60 years. There is no requirement that presidential terms be ranked before calculating the mean. Even when large sets of unorganized data are involved, the calculation of the mean is usually straightforward, particularly with the aid of a calculator or computer.

**Sample or Population?**

Statisticians distinguish between two types of means—the population mean and the sample mean—depending on whether the data are viewed as a **population** (a complete set of scores) or as a **sample** (a subset of scores).

**Formula for Sample Mean**

It's usually more efficient to substitute symbols for words in statistical formulas, including the word formula given above for the mean.

**SAMPLE MEAN**

$$\bar{X} = \frac{\sum X}{n} \qquad (1)$$

When symbols are used, X designates the sample mean, and the formula becomes and reads: "X-bar equals the sum of the variable X divided by the sample size n." [Note that the uppercase Greek letter sigma (Σ) is read as the sum of, not as sigma. To avoid confusion, read only the lowercase Greek letter sigma (σ) as sigma since it has an entirely different meaning in statistics.]

In **Formula (1)**, the variable X can be replaced, in turn, by each of the 20 presidential terms in **Table 1**, beginning with 4 and ending with 8. The symbol Σ, the uppercase Greek letter sigma, specifies that all scores represented by the variable X be added (4 +. . . + 4 + 8) to find the sum of 112. (Notice that this sum contains the values of all scores including duplications.) Then divide this sum by n, the sample size—20 in the present example—to obtain the mean presidential term of 5.60 years.

**Formula for Population Mean**

The formula for the population mean differs from that for the sample mean only because of a change in some symbols. In statistics, Greek symbols usually describe population characteristics, such as the population mean, while English letters usually describe sample characteristics, such as the sample mean. The **population mean** is represented by μ (pronounced "mu"), the lowercase Greek letter m for mean, where the uppercase letter N refers to the **population size**. Otherwise, the calculations are the same as those for the sample mean.

**Mean as Balance Point**

**POPULATION MEAN**

$$\mu = \frac{\sum X}{N} \qquad (2)$$

Imagine that the histogram for the terms of the 20 presidents has been

constructed out of some rigid material such as wood. Also imagine that, while using only one finger placed under its base, you wish to lift the histogram without disturbing its horizontal balance. To accomplish this, your finger should be at 5.60, the value of the mean, shown as a dot. If your finger were to the right of thispoint, the entire histogram would seesaw down to the left; if your finger were to the left of this point, the histogram would seesaw down to the right.

### If Distribution Is Skewed

When extreme scores cause a distribution to be skewed, as for the infant death rates for selected countries listed in **Table 4**, the values of the three averages can differ appreciably. The modal infant death rate of 4 describes the most typical rate (since it occurs most frequently, five times, in **Table 4**).

### Interpreting Differences between Mean and Median

Ideally, when a distribution is skewed, report both the mean and the median. Appreciable differences between the values of the mean and median signal the presence of a skewed distribution.

**Table 4**
**INFANT DEATH RATES FOR SELECTED COUNTRIES (2012)**

| COUNTRY | INFANT DEATH RATE* |
|---|---|
| Sierra Leone | 182 |
| Pakistan | 86 |
| Ghana | 72 |
| India | 56 |
| South Africa | 45 |
| Cambodia | 40 |
| Mexico | 16 |
| China | 14 |
| Brazil | 14 |
| United States | 7 |
| Cuba | 6 |
| United Kingdom | 6 |
| Netherlands | 4 |
| Israel | 4 |
| France | 4 |
| Denmark | 4 |
| Germany | 4 |
| Japan | 3 |
| Sweden | 3 |

**Rates per 1000 live births.*
*Source: 2014 World Development Indicators.*

**Figure 1.15.2** summarizes the relationship between the various averages and the two types of skewed distributions (shown as smoothed curves).

### Special Status of the Mean

As has been seen, the mean sometimes fails to describe the typical or middle-ranked value of a distribution. Therefore, it should be used in conjunction with another average, such as the median. In the long run, however, the mean is the single most

preferred average for quantitative data. After this chapter, it will be used almost exclusively. The mean serves as a key component in an important statistical measure, the standard deviation. In inferential statistics , it emerges as a well-documented measure to be used when generalizing beyond actual scores in surveys and experiments.

**Using the Word Average**

Strictly speaking, an average can refer to the mode, median, or mean—or even to some more exotic average, such as the geometric mean or the harmonic mean. Conventional usage prescribes that average usually signifies mean, and this connotation is often reinforced by the context. For instance, grade point average is virtually synonymous with mean grade point.

## 2.9 AVERAGES FOR QUALITATIVE AND RANKED DATA

### Mode Always Appropriate for Qualitative Data

So far, we have been talking about quantitative data for which, in principle, all three averages can be used. But when the data are qualitative, your choice among averages is restricted. The mode always can be used with qualitative data. For instance, Yes qualifies as the modal or most typical response for the Facebook profile.

### Median Sometimes Appropriate

The median can be used whenever it is possible to order qualitative data from least to most because the level of measurement is ordinal. It's easiest to determine the median class for ordered qualitative data by using relative frequencies, as in **Table 5**. (Otherwise, first convert regular frequencies to relative frequencies.)

**Table 5**
**FINDING THE MEDIAN FOR ORDERED QUALITATIVE DATA: RANKS OF OFFICERS IN THE U.S. ARMY (PROJECTED 2016)**

| RANK | % | CUMULATIVE % |
|---|---|---|
| General | 0.4 | |
| Colonel | 16.7 | |
| Major | 20.4 | |
| Captain | 37.0 | 25.5+37.0 = 62.5 |
| Lieutenant | 25.5 | 25.5 |
| | 100.0 | |

*Source: http://www.Statista.com/statistics*

One word of caution when you are finding the median for ordered qualitative data. Avoid a common error that identifies the median simply with the middle or two middlemost classes, such as "between captain and major," without regard to the

cumulative relative frequencies and the location of the 50th percentile. In other words, do not treat the various classes as though they have the same frequencies when they actually have different frequencies.

**Inappropriate Averages**

It would not be appropriate to report a median for unordered qualitative data with nominal measurement, such as the ancestries of Americans. Nor would it be appropriate to report a mean for any qualitative data, such as the ranks of officers in the U.S. Army. After all, words cannot be added and then divided, as required by the formula for the mean.

| | | | | |
|---|---|---|---|---|
| DB | DB | C | LH | DB |
| C | SP | LH | DB | O |
| O | SP | C | DB | LH |
| DB | C | DB | O | DB |

Find the mode and, if possible, the median.

**Averages for Ranked Data**

When the data consist of a series of ranks, with its ordinal level of measurement, the median rank always can be obtained. It's simply the middlemost or average of the two middlemost ranks. For example, imagine that **Table 1** displays not weights for the 53 male statistics students, but only ranks for their weights, beginning with rank 1 for the lightest man (133 lbs) and ending with rank 53 for the heaviest man (245 lbs). Recalling how to find the median when there is an even number of scores, as described in **Table 2** assign the average of the two middlemost ranks (26th and 27th), that is, 26.5, as the median rank.

## 2.10 DESCRIBING VARIABILITY

**Preview:**

Averages are important, but they tell only part of the story. Most of us would refuse to forge a swift-flowing stream knowing only that the water depth averages 5 feet. Statistics flourishes because we live in a world of variability; no two people are identical, and a few are really far out. When summarizing a set of data, we specify not only measures of central tendency, such as the mean, but also measures of variability, that is, measures of the amount by which scores are dispersed or scattered in a distribution.

Variability in statistics refers to the difference being exhibited by data points within a data set, as related to each other or as related to the mean. This can be expressed through the range, variance or standard deviation of a data set.

2.10.1.1 **Variability** refers to the divergence of data from its mean value, and is commonly used in the statistical and financial sectors.

2.10.1.2 **Variability** in statistics is the degree to which data in a set varies, or how much difference there is in a single set of data. The variability definition also refers to the consistency of the pattern in a set of data.

2.10.1.3 **Variability** in finance is most commonly applied to variability of returns, wherein investors prefer investments that have higher return with less variability.

2.10.1.4 **Variability** is used to standardize the returns obtained on an investment and provides a point of comparison for additional analysis.

**INTUITIVE APPROACH**

You probably already possess an intuitive feel for differences in variability. In **Figure 2.10.1**, each of the three frequency distributions consists of seven scores with the same mean (10) but with different variabilities. (Ignore the numbers in boxes; their significance will be explained later.)

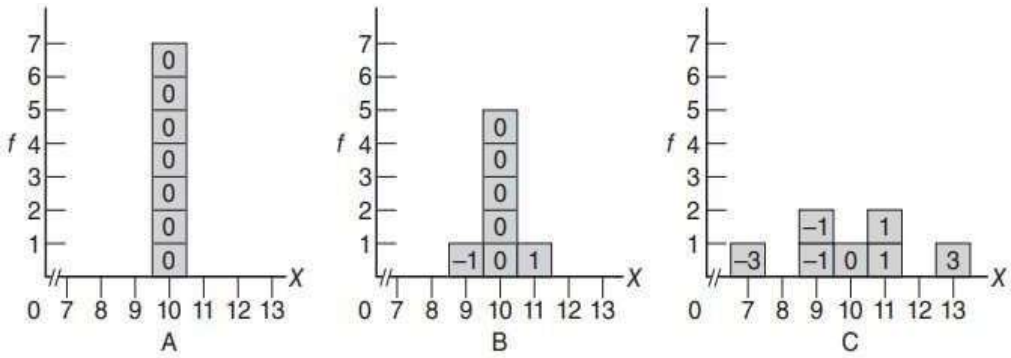



**Figure 2.10**

**Importance of Variability**

Variability assumes a key role in an analysis of research results. For example, a researcher might ask: Does fitness training improve, on average, the scores of depressed patients on a mental-wellness test?

To illustrate the importance of variability, **Figure 2.1.2** shows the outcomes for two fictitious experiments, each with the same mean difference of 2, but with the two groups in experiment B having less variability than the two groups in experiment C. Notice that groups B and C in **Figure 2.1.2** are the same as their counterparts in **Figure 2.1.1**.

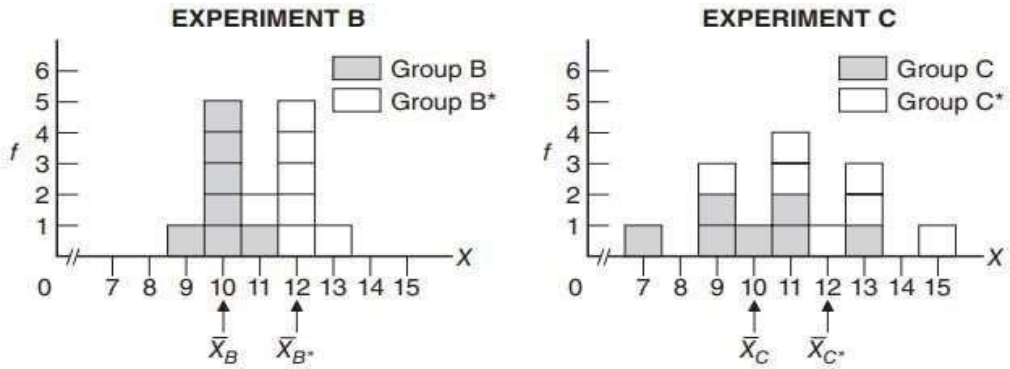



**Figure 2.10.1 Two experiments with the same mean difference but dissimilar variabilities.**

Before reading on, decide which mean difference of 2 in **Figure 2.10.2** is more apparent. The mean difference for experiment B should seem more apparent because of the smaller variabilities within both groups B and B*. Just as it's easier to hear a phone message when static is reduced, it's easier to see a difference between group means when variabilities within groups are reduced.

## 2.11 RANGE

**The range is the difference between the largest and smallest scores.**

Exact measures of variability not only aid communication but also are essential tools in statistics. One such measure is the range.

In Figure 2.1.2, distribution A, the least variable, has the smallest range of 0 (from 10 to 10); distribution B, the moderately variable, has an intermediate range of 2 (from 11 to 9); and distribution C, the most variable, has the largest range of 6 (from 13 to 7), in agreement with our intuitive judgments about differences in variability. The range is a handy measure of variability that can readily be calculated and understood.

| Dataset 1 | Dataset 2 |
|---|---|
| 20 | 11 |
| 21 | 16 |
| 22 | 19 |
| 25 | 23 |
| 26 | 25 |
| 29 | 32 |
| 33 | 39 |
| 34 | 46 |
| 38 | 52 |

While the range is easy to understand, it is based on only the two most extreme values in the dataset, which makes it very susceptible to outliers. If one of those numbers is unusually high or low, it affects the entire range even if it is atypical.

**Shortcomings of Range**

The range has several shortcomings. First, since its value depends on only two scores—the largest and the smallest—it fails to use the information provided by the remaining

scores. Furthermore, the value of the range tends to increase with increases in the total number of scores.

of the sample is below and 50% is above. In other words the median is at the 50th percentile. Quartiles can also be defined. The 1st quartile is at the 25th percentile. The 2nd quartile is at the 50th percentile. The 3rd quartile is at the 75th percentile. And, the 4th quartile is at the 100th percentile.

Since most distributions are spread more widely in their extremities than their middle, the IQR tends to be less than half the size of the range.

**CALCULATION OF THE IQR**

**A. INSTRUCTIONS**

1. Order scores from least to most.
2. To determine how far to penetrate the set of ordered scores, begin at either end, then add 1 to the total number of scores and divide by 4. If necessary, round the result to the nearest whole number.
3. Beginning with the largest score, count the requisite number of steps (calculated in step 2) into the ordered scores to find the location of the third quartile.
4. The third quartile equals the value of the score at this location.
5. Beginning with the smallest score, again count the requisite number of steps into the ordered scores to find the location of the first quartile.
6. The first quartile equals the value of the score at this location.
7. The IQR equals the third quartile minus the first quartile.

**B. EXAMPLE**

1. 7, 9, 9, 10, 11, 11, 13
2. $(7 + 1)/4 = 2$
3. 7, 9, 9, 10, 11, 11, 13

   2 1
4. third quartile = 11
5. 7, 9, 9, 10, 11, 11, 13

   1 2
6. first quartile = 9
7. IQR = 11 − 9 = 2

The calculation of the IQR is relatively straightforward, as you can see by studying Table. This table shows that the IQR equals 2 for distribution C (7, 9, 9, 10, 11,11, 13) shown in Figure 2.1.1.

**Not Sensitive to Extreme Scores**

A key property of the IQR is its resistance to the distorting effect of extreme scores, or outliers. For example, if the smallest score (7) in distribution C of Figure

## 2.12 VARIANCE

Variance is the average squared difference of the values from the mean. Unlike the previous measures of variability, the variance includes all values in the calculation by comparing each value to the mean. To calculate this statistic, you calculate a set of squared differences between the data points and the mean, sum them, and then divide by the number of observations. Hence, it's the average squared difference.

**Sample variance**

To use a sample to estimate the variance for a population, use the following formula. Using the previous equation with sample data tends to underestimate the variability. Because it's usually impossible to measure an entire population, statisticians use the equation for sample variances much more frequently.

$$s^2 = \frac{\sum (X - M)^2}{N - 1}$$

In the equation, $s^2$ is the sample variance, and M is the sample mean. N-1 in the denominator corrects for the tendency of a sample to underestimate the population variance.

**Mean of the Squared Deviations**

Before calculating the variance (a type of mean), negative signs must be eliminated from deviation scores. Squaring each deviation—that is, multiplying each deviation by itself—generates a set of squared deviation scores, all of which are positive

**Weakness of Variance**

In the case of the weights of 53 male students in Table, it is useful to know that the mean for the distribution of weights equals 169.51 pounds, but it is confusing to know that, because of the squared deviations, the variance for the same distribution equals 544.29squared pounds. What, you might reasonably ask, are squared pounds?

| QUANTITATIVE DATA: WEIGHTS (IN POUNDS) OF MALE STATISTICS STUDENTS | | | | | | | |
|---|---|---|---|---|---|---|---|
| 160 | 168 | 133 | 170 | 150 | 165 | 158 | 165 |
| 193 | 169 | 245 | 160 | 152 | 190 | 179 | 157 |
| 226 | 160 | 170 | 180 | 150 | 156 | 190 | 156 |
| 157 | 163 | 152 | 158 | 225 | 135 | 165 | 135 |
| 180 | 172 | 160 | 170 | 145 | 185 | 152 | |
| 205 | 151 | 220 | 166 | 152 | 159 | 156 | |
| 165 | 157 | 190 | 206 | 172 | 175 | 154 | |

## 2.13 STANDARD DEVIATION

To rid ourselves of these mind-boggling units of measurement, simply take the square root of the variance. This produces a new measure, known as the standard deviation, that describes variability in the original units of measurement. For example, the standard deviation for the distribution of weights equals the square root of 544.29 squared pounds, that is, 23.33 pounds.

The variance often assumes a special role in more advanced statistical work, including that described in Chapters 16, 17, and 18. Otherwise, because of its unintelligible u nits of measurement, the variance serves mainly as a stepping stone, only a square root away from a more preferred measure of variability, the standard deviation,

$$standard\ deviation = \sqrt{variance}$$

the square root of the mean of all squared deviations from the mean, that is,

### Standard Deviation: An Interpretation

You might find it helpful to think of the standard deviation as a rough measure of the average (or standard) amount by which scores deviate on either side of their mean.

For distribution C in **Figure 2.1.1**, the square root of the variance of 3.14 yields a standard deviation of 1.77. Given this perspective, a standard deviation of 1.77 is a rough measure of the average amount by which the seven scores in distribution C (7, 9, 9, 10, 11, 11, 13) deviate on either side of their mean of 10.

+ 2s, another person's weight as two-thirds of one standard deviation below the mean, X – 2⁄3s, and so on.

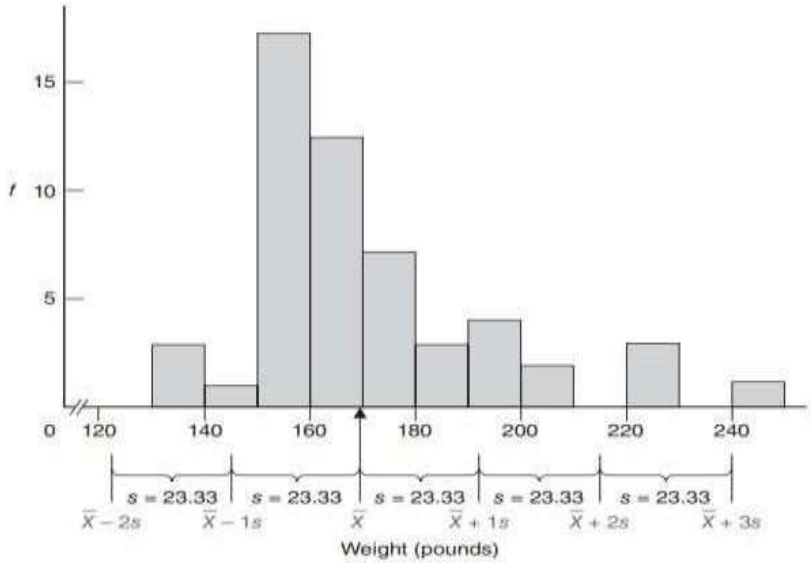



**Figure 2.13 Weight distribution with mean and standard deviation.**

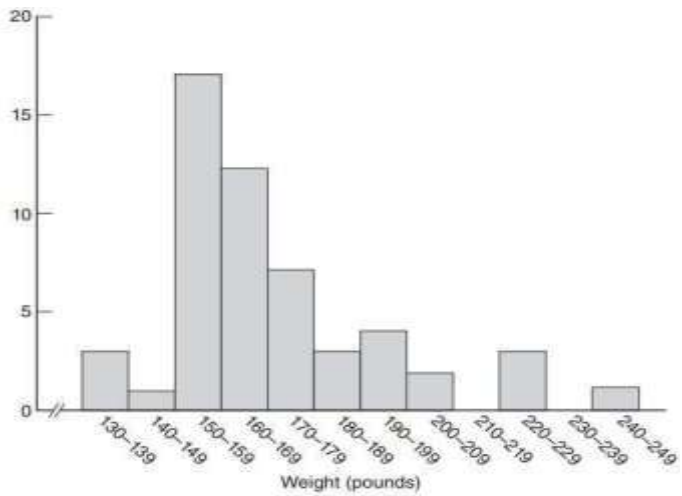



**Figure 2.13.1 Histogram Value of Standard Deviation Cannot Be Negative**

Standard deviation distances always originate from the mean and are expressed as positive deviations above the mean or negative deviations below the mean. Note, however, that although the actual value of the standard deviation can be zero or a positive number, it can never be a negative number because any negative deviation disappears when squared. When a negative sign appears next to the standard deviation, as in the expression X – 1⁄2s, the negative sign indicates that one-half of a standard deviation unit

### 2.13.1. DETAILS: STANDARD DEVIATION

As with the mean, statisticians distinguish between population and sample for both the variance and the standard deviation, depending on whether the data are viewed as a complete set (population) or as a subset (sample). This distinction is introduced here, and it will be very important in inferential statistics.

**Sum of Squares (SS)**

Calculating the standard deviation requires that we obtain first a value for the variance. However, calculating the variance requires, in turn, that we obtain the sum of the squared deviation scores. The sum of squared deviation scores, or more simply the sum of squares, symbolized by SS, merits special attention because it's a major component in calculations for the variance, as well as many other statistical measures.

**CALCULATION OF POPULATION STANDARD DEVIATION $\sigma$ (DEFINITION FORMULA)**

**A. COMPUTATION SEQUENCE**

Assign a value to $N$ **1** representing the number of $X$ scores
Sum all $X$ scores **2**
Obtain the mean of these scores **3**
Subtract the mean from each $X$ score to obtain a deviation score **4**
Square each deviation score **5**
Sum all squared deviation scores to obtain the sum of squares **6**
Substitute numbers into the formula to obtain population variance, $\sigma^2$ **7**
Take the square root of $\sigma^2$ to obtain the population standard deviation, $\sigma$ **8**

**B. DATA AND COMPUTATIONS**

| $X$ | **4** $X-\mu$ | **5** $(X-\mu)^2$ |
|---|---|---|
| 13 | 3 | 9 |
| 10 | 0 | 0 |
| 11 | 1 | 1 |
| 7 | −3 | 9 |
| 9 | −1 | 1 |
| 11 | 1 | 1 |
| 9 | −1 | 1 |

**1** $N = 7$ **2** $\Sigma X = 70$ **6** $SS = \Sigma (X-\mu)^2 = 22$

**3** $\mu = \frac{70}{7} = 10$

**7** $\sigma^2 = \frac{SS}{N} = \frac{22}{7} = 3.14$ **8** $\sigma = \sqrt{\frac{SS}{N}} = \sqrt{\frac{22}{7}} = \sqrt{3.14} = 1.77$

**Table 5.1**

**Table 5.1** shows how to use the definition formula to calculate the sum of squares for distribution C in **Figure**. (Ignore the last two steps in this table until later, when formulas for the variance and standard deviation are introduced.)

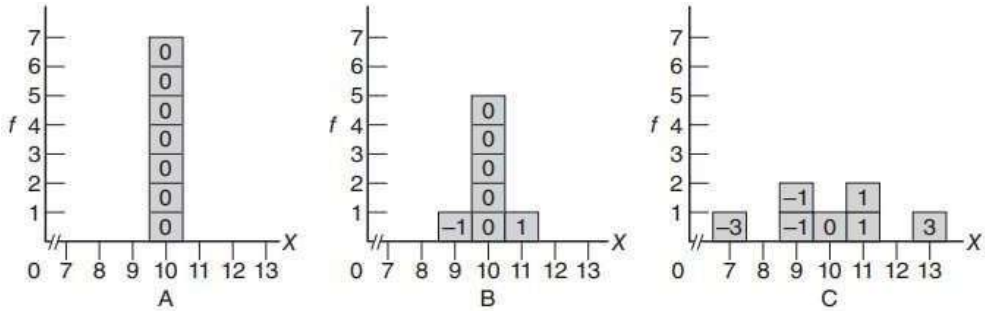



**Figure:** Three distributions with the same mean (10) but different amounts of variability. Numbers in the boxes indicate distances from the mean.

The definition formula is cumbersome when, as often occurs, the mean equals some complex number, such as 169.51, or the number of scores is large. In these cases, use the more efficient computation formula:

**SUM OF SQUARES (*SS*) FOR POPULATION (COMPUTATION FORMULA)**

$$SS = \Sigma X^2 - \frac{(\Sigma X)^2}{N}$$

where $\Sigma X^2$, the sum of the squared X scores, is obtained by first squaring each X score and then summing all squared X scores; $(\Sigma X^2)$ , the square of sum of all X scores, is obtained by first adding all X scores and then squaring the sum of all X scores; and N is the population size.

**CALCULATION OF POPULATION STANDARD DEVIATION (σ) (COMPUTATION FORMULA)**

**A. COMPUTATIONAL SEQUENCE**

Assign a value to *N* representing the number of *X* scores **1**
Sum all *X* scores **2**
Square the sum of all *X* scores **3**
Square each *X* score **4**
Sum all squared *X* scores **5**
Substitute numbers into the formula to obtain the sum of squares, *SS* **6**
Substitute numbers into the formula to obtain the population variance, $\sigma^2$ **7**
Take the square root of $\sigma^2$ to obtain the population standard deviation, $\sigma$ **8**

**B. DATA AND COMPUTATIONS**

| $X$ | 4 $X^2$ |
|---|---|
| 13 | 169 |
| 10 | 100 |
| 11 | 121 |
| 7 | 49 |
| 9 | 81 |
| 11 | 121 |
| 9 | 81 |

**1** $N = 7$ **2** $\sum X = 70$ **5** $\sum X^2 = 722$

**3** $(\sum X)^2 = 4900$

**6** $SS = \sum X^2 - \frac{(\sum X)^2}{N} = 722 - \frac{4900}{7} = 722 - 700 = 22$

**7** $\sigma^2 = \frac{SS}{N} = \frac{22}{7} = 3.14$ **8** $\sigma = \sqrt{\frac{SS}{N}} = \sqrt{\frac{22}{7}} = \sqrt{3.14} = 1.77$

**Table 5.2**

**Sum of Squares Formulas for Sample**

Sample notation can be substituted for population notation in the above two formulas without causing any essential changes:

**SUM OF SQUARES (*SS*) FOR SAMPLE (DEFINITION FORMULA)**

$$SS = \Sigma(X - \bar{X})^2$$

**(COMPUTATION FORMULA)**

$$SS = \Sigma X^2 - \frac{(\Sigma X)^2}{n}$$

where $\bar{X}$, the sample mean, replaces μ, the population mean, and n, the sample size, replaces N, the population size. Notwithstanding these two changes in notation, the numerical result for the sample sum of squares (22) is the same as that for the population sum of squares in **Tables 5.1 and 5.2.** Accordingly, the same symbol, SS, will represent the sum of squared deviation scores for both populations and samples.

**Standard Deviation for Population σ**

***Population Standard Deviation (σ)***
*A rough measure of the average amount by which scores in the population deviate on either side of their population mean.*

Recall that, most generally, a mean is defined as the sum of all scores divided by the number of scores. Since the variance is the mean of all squared deviation scores, it can be defined as the sum of all squared deviation scores divided by the number of scores:

$$variance = \frac{sum\ of\ all\ squared\ deviation\ scores}{number\ of\ scores}$$

or, in symbols:

**VARIANCE FOR POPULATION**

$$\sigma^2 = \frac{SS}{N}$$

Where the squared lowercase Greek letter, $\sigma^2$ (pronounced "sigma squared"), represents the population variance, SS is the sum of squared deviations for the population, and N is the population size.

To rid us of the bizarre squared units of measurement, take the square root of the variance to obtain the standard deviation, that is,

**STANDARD DEVIATION FOR POPULATION**

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{SS}{N}}$$

where σ represents the population standard deviation, $\sqrt{\ }$ instructs us to take the square root of the covered expression, and SS and N are defined above. By referring to the last two steps in either **Table 5.1 or 5.2**, you can verify that the value of the variance, σ 2, equals 3.14 for distribution C because

where σ represents the population standard deviation, instructs us to take the square root of the covered expression, and SS and N are defined above. By referring to the last two steps in either Table 4.1 or 4.2, you can verify that the value of the variance, σ 2, equals 3.14 for distribution C because

$$\sigma^2 = \frac{SS}{N} = \frac{22}{7} = 3.14$$

and that the value of the standard deviation, σ, equals 1.77 for distribution C because

$$\sigma = \sqrt{\frac{SS}{N}} = \sqrt{\frac{22}{7}} = \sqrt{3.14} = 1.77$$

**Standard Deviation for Sample ( s )**

***Sample Standard Deviation (s)***

*A rough measure of the average amount by which scores in the sample deviate on either side of their sample mean.*

Although the sum of squares term remains essentially the same for both populations and samples, there is a small but important change in the formulas for the variance and standard deviation for samples.

This change appears in the denominator of each formula where N, the population size, is replaced not by n, the sample size, but by n − 1, as shown:

**VARIANCE FOR SAMPLE**

$$s^2 = \frac{SS}{n-1}$$

**STANDARD DEVIATION FOR SAMPLE**

$$s = \sqrt{s^2} = \sqrt{\frac{SS}{n-1}}$$

where s2 and s represent the sample variance and sample standard deviation, SS is the sample sum of squares, and n is the sample size.

The reason for using n − 1 will be explained in the next section. But first, spend a few moments studying **Tables 5.3 and 5.4**, which show the calculations for the sample standard deviation, using the definition and computation formulas for sample sums of squares and a new set of five scores.

**CALCULATION OF SAMPLE STANDARD DEVIATION ($S$)**
**(DEFINITION FORMULA)**

**A. COMPUTATION SEQUENCE**

Assign a value to $n$ **1** representing the number of $X$ scores
Sum all $X$ scores **2**
Obtain the mean of these scores **3**
Subtract the mean from each $X$ score to obtain a deviation score **4**
Square each deviation score **5**
Sum all squared deviation scores to obtain the sum of squares **6**
Substitute numbers into the formula to obtain the sample variance, $s^2$ **7**
Take the square root of $s^2$ to obtain the sample standard deviation, $s$ **8**

**B. DATA AND COMPUTATIONS**

| $X$ | **4** $X - \bar{X}$ | **5** $(X - \bar{X})^2$ |
|---|---|---|
| 7 | 4 | 16 |
| 3 | 0 | 0 |
| 1 | −2 | 4 |
| 0 | −3 | 9 |
| 4 | 1 | 1 |

**1** $n = 5$ **2** $\Sigma X = 15$ **6** $SS = \Sigma(X - \bar{X})^2 = 30$

**3** $\bar{X} = \frac{15}{5} = 3$

**7** $s^2 = \frac{SS}{n-1} = \frac{30}{4} = 7.50$ **8** $s = \sqrt{\frac{SS}{n-1}} = \sqrt{\frac{30}{4}} = \sqrt{7.50} = 2.74$

**Table 5.3**

**CALCULATION OF SAMPLE STANDARD DEVIATION ($S$)**
**(COMPUTATION FORMULA)**

**A. COMPUTATIONAL SEQUENCE**

Assign a value to $n$ representing the number of $X$ scores **1**
Sum all $X$ scores **2**
Square the sum of all $X$ scores **3**
Square each $X$ score **4**
Sum all squared $X$ scores **5**
Substitute numbers into the formula to obtain the sum of squares, $SS$ **6**
Substitute numbers into the formula to obtain the sample variance, $s^2$ **7**
Take the square root of $s^2$ to obtain the sample standard deviation, $s$ **8**

**B. DATA AND COMPUTATIONS**

| $X$ | **4** $X^2$ |
|---|---|
| 7 | 49 |
| 3 | 9 |
| 1 | 1 |
| 0 | 0 |
| 4 | 16 |

**1** $n = 5$ **2** $\Sigma X = 15$ **5** $\Sigma X^2 = 75$
**3** $(\Sigma X)^2 = 225$

**6** $SS = \Sigma X^2 - \frac{(\Sigma X)^2}{n} = 75 - \frac{225}{5} = 75 - 45 = 30$

**7** $s^2 = \frac{SS}{n-1} = \frac{30}{4} = 7.50$ **8** $s = \sqrt{\frac{SS}{n-1}} = \sqrt{\frac{30}{4}} = \sqrt{7.50} = 2.74$

**Table 5.4**

**Computational Check**

With rare exceptions, the standard deviation should be less than one-half the size of the range, and in most cases, it will be an even smaller fraction (one-third to one-sixth) the size of the range. Use this rule of thumb to detect sizable computation errors. The only fool proof method for detecting smaller errors—whether you're calculating the standard deviation manually or electronically—is to calculate everything twice and to proceed only if your numerical results agree.

## *2.14* DEGREES OF FREEDOM ( d f)

Technically, we have been discussing a very important notion in inferential statistics known as degrees of freedom.

Degrees of freedom (df) refers to the number of values that are free to vary, given one or more mathematical restrictions, in a sample being used to estimate a population characteristic.

The concept of degrees of freedom is introduced only because we are using scores in a sample to estimate some unknown characteristic of the population. Typically, when used as an estimate, not all observed values in the sample are free to vary because of one or more mathematical restrictions. As has been noted, when n deviations about the sample mean are used to estimate variability in the population, only n − 1 are free to vary. As a result, there are only n − 1 degrees of freedom, that is, df = n − 1. One df is lost because of the zero-sum restriction.

**VARIANCE FOR SAMPLE**

$$s^2 = \frac{SS}{n-1} = \frac{SS}{df}$$

**STANDARD DEVIATION FOR SAMPLE**

$$s = \sqrt{\frac{SS}{n-1}} = \sqrt{\frac{SS}{df}}$$

where s2 and s represent the sample variance and standard deviation, SS is the sum of squares as d and df is the degrees of freedom and equals n − 1.

**Other Mathematical Restrictions**

The notion of degrees of freedom is used extensively in inferential statistics. We'll encounter other mathematical restrictions, and sometimes more than one degree of freedom will be lost. In any event, however, degrees of freedom (df ) always indicate the number of values that are free to vary, given one or more mathematical restrictions, in a set of values used to estimate some unknown population characteristic.

## 2.15 MEASURES OF VARIABILITY FOR QUALITATIVE AND RANKED DATA

**Qualitative Data**

Measures of variability are virtually nonexistent for qualitative or nominal data.

It is probably adequate to note merely whether scores are evenly divided among the various classes (maximum variability), unevenly divided among the various classes (intermediate variability), or concentrated mostly in one class (minimum variability).

At the other extreme, if almost all the residents are concentrated in a single ethnic group, the variability will be minimum; there is little heterogeneity. (According to the previous source, virtually no ethnic variability occurs in sparsely populated rural counties, such as Hooker County in Nebraska and King County in Texas, with an almost exclusively

white population.) If the ethnic composition falls between these two extremes—because of an uneven division among several large ethnic groups—the variability will be intermediate, as is true of many U.S. cities and counties.

**Ordered Qualitative and Ranked Data**

If qualitative data can be ordered because measurement is ordinal (or if the data are ranked), then it's appropriate to describe variability by identifying extreme scores (or ranks). For instance, the active membership of an officers' club might include no one with a rank below first lieutenant or above brigadier general.

## 2.16 NORMAL DISTRIBUTIONS AND STANDARD (Z) SCORES

The relative frequency distribution of heights for the 3091 men shown in **Figure 4.1**.1. To find the proportion of men who are a particular height, merely note the value of the vertical scale that corresponds to the top of any bar in the histogram. For example, .10 of these men, that is, one-tenth of 3091, or about 309 men, are 70 inches tall. When expressed as a proportion, any conclusion based on the 3091 men can be generalized to other comparable sets of men, even sets containing an unspecified number.

Therefore, any generalization will contain inaccuracies due to chance irregularities in the original distribution.

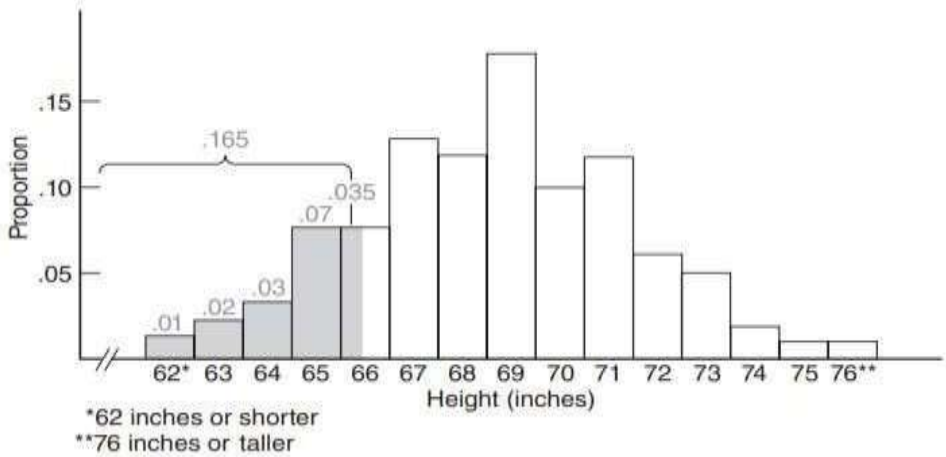



**FIGURE 2.14 Relative frequency distribution for heights of 3091 men. Source: National Center for Health Statistics, 1960–62, Series 11, No.14. Mean updated by authors**

## *2.17 z SCORES*

A z score is a unit-free, standardized score that, regardless of the original units of measurement, indicates how many standard deviations a score is above or below the mean of its distribution.

To obtain a z score, express any original score, whether measured in inches, milliseconds, dollars, IQ points, etc., as a deviation from its mean (by subtracting its mean) and then split this deviation into standard deviation units (by dividing by its standard deviation), that is,

**z SCORE**

$$z = \frac{X - \mu}{\sigma}$$

where X is the original score and μ and σ are the mean and the standard deviation, respectively, for the normal distribution of the original scores. Since identical units of measurement appear in both the numerator and denominator of the ratio for z, the original units of measurement cancel each other and the z score emerges as a unit- free or standardized number, often referred to as a standard score.

**A z score consists of two parts:**

1. a positive or negative sign indicating whether it's above or below the mean; and
2. a number indicating the size of its deviation from the mean in standard deviation units.

## 2.18 STANDARD NORMAL CURVE

If the original distribution approximates a normal curve, then the shift to standard or z scores will always produce a new distribution that approximates the standard normal curve. This is the one normal curve for which a table is actually available. It is a mathematical fact—not proven in this book—that the standard normal curve always has a mean of 0 and a standard deviation of 1. However, to verify (rather than prove) that the mean of a standard normal distribution equals 0, replace X in the z score formula with μ, the mean of any (nonstandard) normal distribution, and then solve for z:

$$\text{Mean of } z = \frac{X - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = \frac{0}{\sigma} = 0$$

Likewise, to verify that the standard deviation of the standard normal distribution equals 1, replace X in the z score formula with μ + 1σ, the value corresponding to one standard deviation above the mean for any (nonstandard) normal distribution, and then solve for z:

$$\text{Standard deviation of } z = \frac{X-\mu}{\sigma} = \frac{\mu + 1\sigma - \mu}{\sigma} = \frac{1\sigma}{\sigma} = 1$$

Although there is an infinite number of different normal curves, each with its own mean and standard deviation, there is only one standard normal curve, with a mean of 0 and a standard deviation of 1.

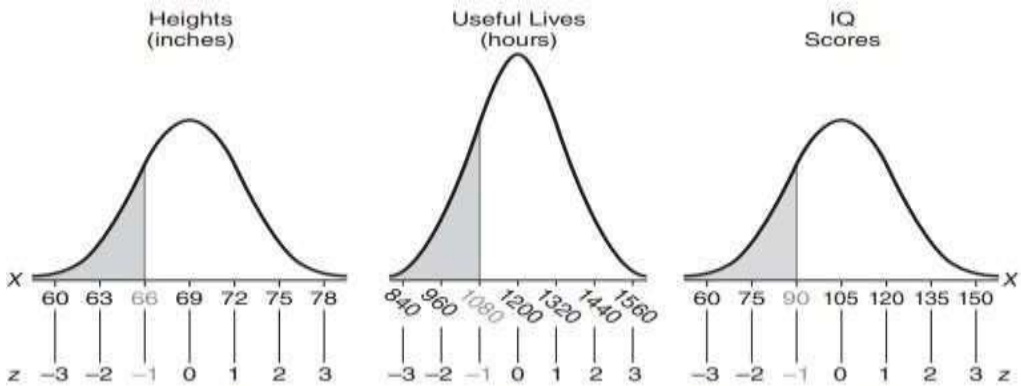



**FIGURE 2.16 converting three normal curves to the standard normal curve**

Accordingly, to find the proportion for the shaded areas in Figure 5.4 (that is, the proportion of applicants who are less than exactly 66 inches tall, or light bulbs that burn for fewer than 1080 hours, or fourth graders whose IQ scores are less than 90), we can use the same z score of –1.00 when referring to the table for the standard normal curve, the one table for all normal curves.

**Standard Normal Table**

Essentially, the standard normal table consists of columns of z scores coordinated with columns of proportions. In a typical problem, access to the table is gained through a z score, such as –1.00, and the answer is read as a proportion, such as the proportion of eligible FBI applicants.

**Using the Bottom Legend of the Table**

Because of the symmetry of the normal curve, the entries in **Table 4.3.2** and Table A of Appendix C also can refer to the lower half of the normal curve. Now the columns are designated as A′, B′, and C′ in the legend at the bottom of the table. When using the bottom legend, all entries refer to the lower half of the standard normal curve.

Imagine that the nonzero entries in column A′ are negative z scores, beginning with –0.01 and ending (in the full-length table of Appendix C) with –4.00. Given a negative z score, columns B′ and C′ indicate how that z score splits the lower half of the normal curve. As suggested by the shading in the bottom legend of the table, column B′ indicates the proportion of area between the mean and the negative z score, and column C′ indicates the proportion of area beyond the negative z score, in the lower tail of the standard normal curve.

## 2.19 SOLVING NORMAL CURVE PROBLEMS

Sections 5 and 6 give examples of two main types of normal curve problems. In the first type of problem, we use a known score (or scores) to find an unknown proportion. For instance, we use the known score of 66 inches to find the unknown proportion of eligible FBI applicants.

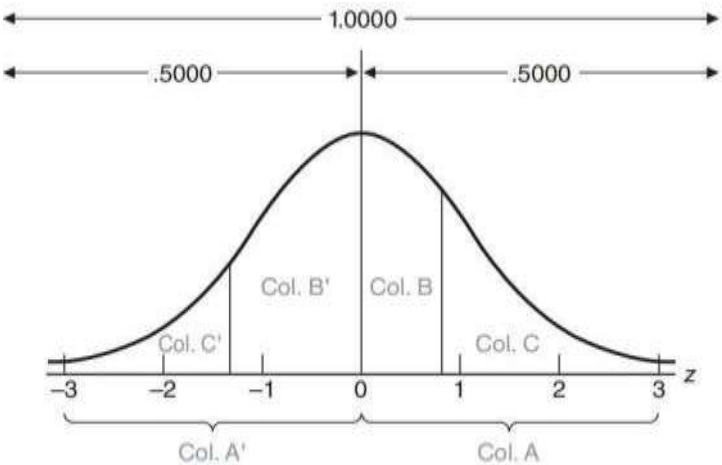



**FIGURE 2.17**

### Key Facts to Remember

When using the standard normal table, it is important to remember that for any z score, the corresponding proportions in columns B and C (or columns B′ and C′) always sum to .5000. Similarly, the total area under the normal curve always equals 1.0000, the sum of the proportions in the lower and upper halves, that is, .5000 + .5000. Finally, although a z score can be either positive or negative, the proportions of area under the curve are always positive or zero but never negative (because an area cannot be negative). **Figure 2.17** summarizes how to interpret the normal curve table in this book.

## 2.20 FINDING PROPORTIONS

Example: Finding Proportions for One Score

Now we'll use a step-by-step procedure, adopted throughout this chapter, to find the proportion of all FBI applicants who are shorter than exactly 66 inches, given that the distribution of heights approximates a normal curve with a mean of 69 inches and a standard deviation of 3 inches.

1. **Sketch a normal curve and shade in the target area**, as in the left part of **Figure 2.17.1** Being less than the mean of 69, 66 is located to the left of the mean. Furthermore, since the unknown proportion represents those applicants who are shorter than 66 inches, the shaded target sector is located to the left of 66.
2. **Plan your solution according to the normal table**. Decide precisely how you will find the value of the target area. In the present case, the answer will be obtained from column C′ of the standard normal table, since the target area coincides with the type of area identified with column C′, that is, the area in the lower tail beyond a negative z.
3. **Convert X to z.** Express 66 as a z score:

$$z = \frac{X - \mu}{\sigma} = \frac{66 - 69}{3} = \frac{-3}{3} = -1$$

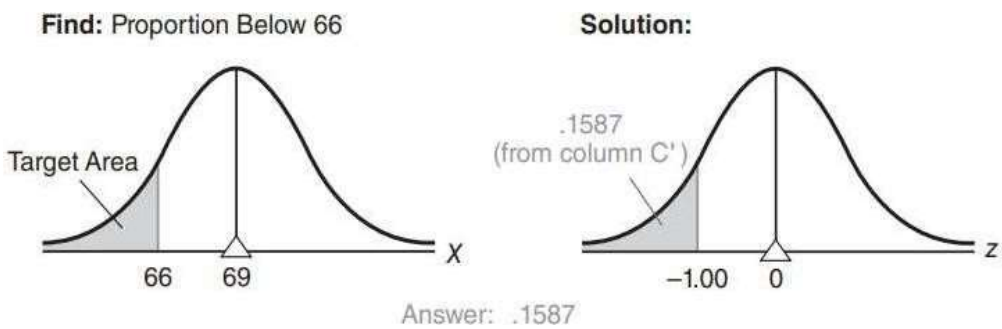



**FIGURE 2.17.1 Finding proportions**

**Find the target area**. Refer to the standard normal table, using the bottom legend, as the z score is negative. The arrows in Table 4.1 show how to read the table. Look up column A' to 1.00 (representing a z score of –1.00), and note the corresponding proportion of .1587 in column C': This is the answer, as suggested in the right part of Figure 4.5.1. It can be concluded that only .1587 (or .16) of all of the FBI applicants will be shorter than 66 inches.

### A Clarification

Because the normal curve is defined for continuous variables, such as height, the same proportion of .1587 would describe not only FBI applicants who are shorter than 66 inches, but also FBI applicants who are shorter than or equal to 66 inches. If you think about it, equal to 66 inches translates into a height of exactly 66 inches— that is, 66.0000 with a string of zeros out to infinity! No measured height can coincide with exactly 66 inches since, in theory, however long the string of zeros for someone's height, measurement always can be carried additional steps until a nonzero appears.

## 2.21 FINDING SCORES

So far, we have concentrated on normal curve problems for which Table A must be consulted to find the unknown proportion (of area) associated with some known score or pair of known scores. For instance, given a GRE score of 650, we found that the unknown proportion of scores larger than 650 equals .07. Now we will concentrate on the opposite type of normal curve problem for which Table A must be consulted to find the unknown score or scores associated with some known proportion.

## 2.22 MORE ABOUT z SCORES

### z Scores for Non-normal Distributions

z scores are not limited to normal distributions. Non-normal distributions also can be transformed into sets of unit-free, standardized z scores. In this case, the standard normal table cannot be consulted, since the shape of the distribution of z scores is the same as that for the original non-normal distribution. For instance, if the original distribution is positively skewed, the distribution of z scores also will be positively skewed. Regardless of the shape of the distribution, the shift to z scores always produces a distribution of standard scores with a mean of 0 and a standard deviation of 1.

# UNIT III
# STATISTICAL FOUNDATIONS OF DATA SCIENCE

**Correlation - scatter plots - correlation coefficient for quantitative data - computational formula for correlation coefficient - regression - regression line - least squares regression line - standard error of estimate - interpretation of $r^2$ - Multiple and Nonparametric Regression -multiple regression equations - regression toward the mean.**

## 3.1 CORRELATION:

- Correlation refers to a process for establishing the relationships between two variables.
- Correlation is used to test relationships between quantitative variables or categorical variables.
- In other words, it's a measure of how things are related.
- The study of how variables are correlated is called correlation analysis

When one measurement is made on each observation, uni-variate analysis is applied. If more than one measurement is made on each observation, multivariate analysis is applied. Here we focus on bivariate analysis, where exactly two measurements are made on each observation.

The two measurements will be called X and Y. Since X and Y are obtained for each observation, the data for one observation is the pair (X, Y).

Some examples:

1. Height (X) and weight (Y) are measured for each individual in a sample.

2. Stock market valuation (X) and quarterly corporate earnings (Y) are recorded for each company in a sample.

3. A cell culture is treated with varying concentrations of a drug and the growth rate (X) and drug concentrations (Y) are recorded for each trial.

4. Temperature (X) and precipitation (Y) are measured on a given day at a set of weather stations.

**Correlation** refers to a relationship between two or more objects. In statistics, the word correlation refers to the relationship between two variables. Correlation exists between two variables when one of them is related to the other in some way.

A positive correlation is where the two variables react in the same way, increasing or decreasing together. Temperature in Celsius and Fahrenheit has a positive correlation.

• The term "correlation" refers to a measure of the strength of association///n between two variables.

• Covariance is the extent to which a change in one variable corresponds systematically to a change in another. Correlation can be thought of as a standardized covariance.

• The correlation coefficient r is a function of the data, so it really should be called the sample correlation coefficient. The (sample) correlation coefficient r estimates the population correlation coefficient p.

• If either the X, or the Y; values are constant (i.e. all have the same value), then one of the sample standard deviations is zero and therefore the correlation coefficient is not defined.

**Types of Correlation**

1. Positive Correlation
2. Zero Correlation
3. Negative Correlation

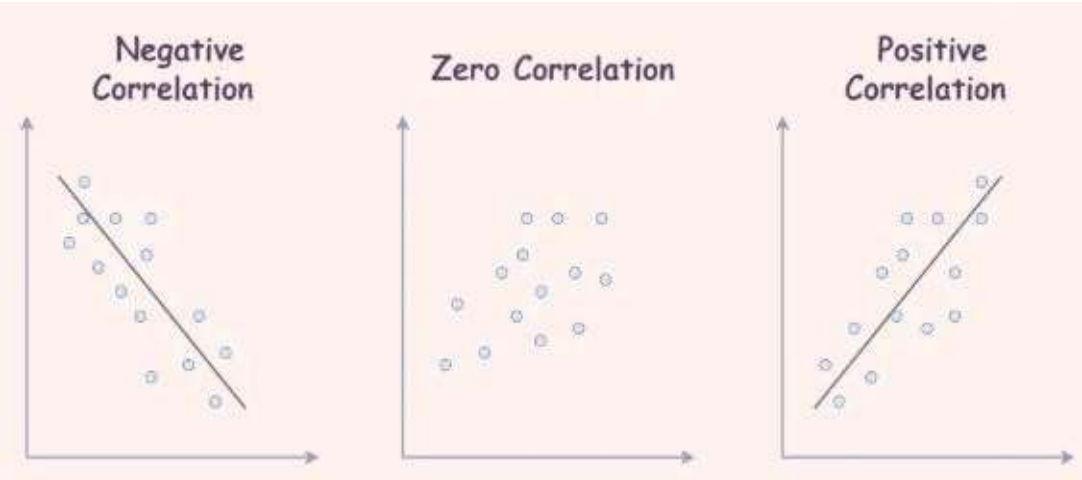



**Fig 3.1 Types of Correlation Classification of correlation**

Two methods are used for finding relationship between variables.

1. Graphic methods

2. Mathematical methods.

Graphic methods contain two sub methods: Scatter diagram and simple graph.

Types of mathematical methods are,

a. Karl 'Pearson's coefficient of correlation

b. Spearman's rank coefficient correlation

c. Coefficient of concurrent deviation

d. Method of least squares.

**Coefficient of Correlation**

Correlation: The degree of relationship between the variables under consideration is measure through the correlation analysis.

• The measure of correlation called the correlation coefficient. The degree of relationship is expressed by coefficient which range from correlation ($-1 \leq r \geq +1$). The direction of change is indicated by a sign.

• If two variables vary in such a way that movement in one are accompanied by movement in other, these variables are called cause and effect relationship.

**Properties of Correlation**

1. Correlation requires that both variables be quantitative.

2. Positive r indicates positive association between the variables and negative r indicates negative association.

3. The correlation coefficient (r) is always a number between - 1 and + 1.

4. The correlation coefficient (r) is a pure number without units.

5. The correlation coefficient measures clustering about a line, but only relative to the SD's.

6. The correlation can be misleading in the presence of outliers or nonlinear association.

7. Correlation measures association. But association does not necessarily show causation.

Example: A sample of 12 fathers and their elder sons gave the following data about their heights in inches. Calculate the coefficient of rank correlation.

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Father | 65 | 63 | 67 | 64 | 68 | 62 | 70 | 66 | 68 | 67 | 69 | 71 |
| Sons | 68 | 66 | 68 | 65 | 69 | 66 | 68 | 65 | 71 | 67 | 68 | 70 |

Solution:

| Fathers heights (X) | Ranks (x) | Sons heights (Y) | Ranks (y) | Rank difference (D = x - y) | $D^2$ |
|---|---|---|---|---|---|
| 65 | 7 | 68 | 4 | 3 | 9 |
| 63 | 9 | 66 | 6 | 3 | 9 |
| 67 | 5 | 68 | 4 | 1 | 1 |
| 64 | 8 | 65 | 7 | 1 | 1 |
| 68 | 4 | 69 | 3 | 1 | 1 |
| 62 | 10 | 66 | 6 | 4 | 16 |
| 70 | 2 | 68 | 4 | -2 | 4 |
| 66 | 6 | 65 | 7 | -1 | 1 |
| 68 | 4 | 71 | 1 | 3 | 9 |
| 67 | 5 | 67 | 5 | 0 | 0 |
| 69 | 3 | 68 | 4 | -1 | 1 |
| 71 | 1 | 70 | 2 | -1 | 1 |

$\sum D^2 = 53$

$$\rho = 1 - \frac{6\sum D^2}{N(N^2-1)}$$

$$\rho = 1 - \frac{6 \times 53}{12(12^2-1)}$$

$$\rho = 1 - \frac{318}{1716}$$

$$\rho = 1 - 0.1853$$

$$\rho = 0.8147$$

## 3.2 SCATTER PLOTS

• When two variables x and y have an association (or relationship), we say there exists a correlation between them. Alternatively, we could say x and y are correlated. To find such an association, we usually look at a scatterplot and try to find a pattern.

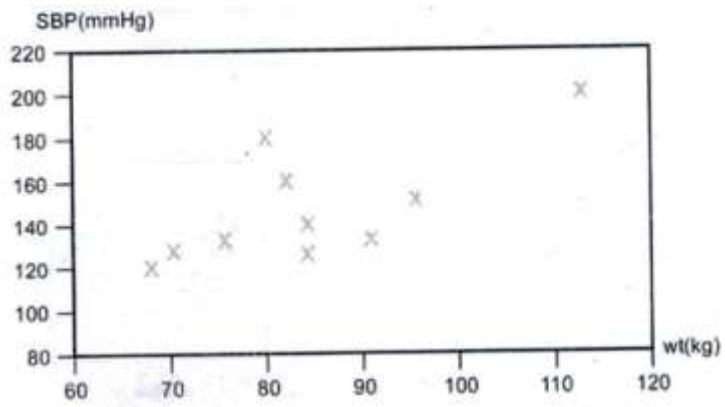



**Fig 3.2 Scatter diagram of weight**

The pattern of data is indicative of the type of relationship between your two variables:

1. Positive relationship
2. Negative relationship
3. No relationship.

The scattergram can indicate a positive relationship, a negative relationship or a zero relationship.

**Advantages of Scatter Diagram**

1. It is a simple to implement and attractive method to find out the nature of correlation.
2. It is easy to understand.
3. User will get rough idea about correlation (positive or negative correlation).
4. Not influenced by the size of extreme item
5. First step in investing the relationship between two variables.

**Disadvantage of scatter diagram**

• Cannot adopt an exact degree of correlation.

## 3.3 CORRELATION COEFFICIENT FOR QUANTITATIVE DATA

The product moment correlation, r, summarizes the strength of association between two metric (interval or ratio scaled) variables, say X and Y. It is an index used to determine whether a linear or straight-line relationship exists between X and Y.

• As it was originally proposed by Karl Pearson, it is also known as the Pearson correlation coefficient. It is also referred to as simple correlation, bivariate correlation or merely the

correlation coefficient.

• The correlation coefficient between two variables will be the same regardless of their underlying units of measurement.

• It measures the nature and strength between two variables of the quantitative type.

• The sign of r denotes the nature of association. While the value of r denotes the strength of association.

• If the sign is positive this means the relation is direct (an increase in one variable is associated with an increase in the other variable and a decrease in one variable is associated with a decrease in the other variable).

• While if the sign is negative this means an inverse or indirect relationship (which means an increase in one variable is associated with a decrease in the other).

• The value of r ranges between (-1) and (+ 1). The value of r denotes the strength of the association as illustrated by the following diagram,

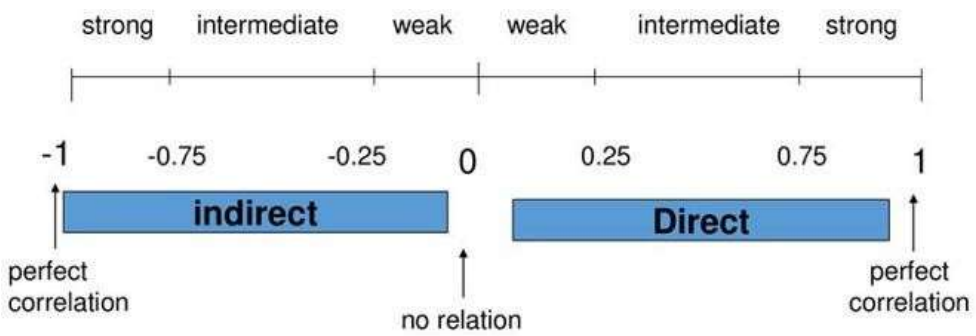
1. If r = Zero this means no association or correlation between the two variables.

2. If $0 < r < 0.25$ = Weak correlation.

3. If $0.25 \leq r < 0.75$ = Intermediate correlation.

4. If $0.75 \leq r < 1$ = Strong correlation.

5. If r=1= Perfect correlation




• Pearson's 'r' is the most common correlation coefficient. Karl Pearson's Coefficient of Correlation denoted by - 'r' The coefficient of correlation 'r' measure the degree of linear relationship between two variables say x and y.

Formula for calculating correlation coefficient (r) :

1. When deviation taken from actual mean :

$$r(X,Y) = \frac{\sum xy}{\sqrt{\sum x^2 \sqrt{\sum y^2}}}$$

2. When deviation taken from an assumed mean:

$$r = \left| \frac{\sum XY - \frac{\sum X \sum Y}{N}}{\sqrt{\left(\sum X^2 - \frac{(\sum X)^2}{N}\right)\left(\sum Y^2 - \frac{(\sum Y)^2}{N}\right)}} \right|$$

| X | x = X – 38 | $x^2$ | Y | y = Y – 34 | $y^2$ | xy |
|---|---|---|---|---|---|---|
| 38 | 0 | 0 | 28 | – 6 | 36 | 0 |
| 45 | 7 | 49 | 34 | 0 | 0 | 0 |
| 46 | 8 | 64 | 38 | 4 | 16 | 32 |
| 38 | 0 | 0 | 34 | 0 | 0 | 0 |
| 35 | – 3 | 9 | 36 | 2 | 4 | – 6 |
| 38 | 0 | 0 | 26 | – 8 | 64 | 0 |
| 46 | 8 | 64 | 28 | – 6 | 36 | – 48 |
| 32 | – 6 | 36 | 29 | – 5 | 25 | 30 |
| 36 | – 2 | 4 | 25 | – 9 | 81 | 18 |
| 38 | 0 | 0 | 36 | 2 | 4 | 0 |
| | $\sum x = 12$ | $\sum X^2 = 226$ | | $\sum y = -26$ | $\sum Y^2 = 266$ | $\sum XY = 26$ |

$$r = \frac{\sum xy - \frac{\sum x \sum y}{N}}{\sqrt{\sqrt{\sum x^2 - \frac{(\sum x)^2}{N}} \cdot \sqrt{\sum y^2 - \frac{(\sum y)^2}{N}}}}$$

$$r = \frac{26 - \frac{(12)(-26)}{10}}{\sqrt{226 - ((12)^2/10)}\sqrt{266 - ((-26)^2/10)}}$$

$$= \frac{26 + 31.2}{\sqrt{226} - 14.4\sqrt{266} - 67.6} = \frac{57.2}{14.546 \times 14.085}$$

$$r = 0.2792$$

## COMPUTATIONAL FORMULA FOR CORRELATION COEFFICIENT

What is Correlation Coefficient Formula?

Correlation coefficient procedure is used to determine how strong a relationship is between the data. The correlation coefficient procedure yields a value between 1 and -1. In which,

- -1 indicates a strong negative relationship
- 1 indicates strong positive relationships
- Zero implies no connection at all

**Understanding Correlation Coefficient**

Correlation coefficient of -1 means there is a negative decrease of a fixed proportion, for every positive increase in one variable. Like, the amount of gas in a tank decreases in a perfect correlation with the speed.

Correlation coefficient of 1 means there is a positive increase of a fixed proportion of others, for every positive increase in one variable. Like, the size of the shoe goes up in perfect correlation with foot length.

Correlation coefficient of 0 means that for every increase, there is neither a positive nor a negative increase. The two just aren't related.

Types of Correlation Coefficient Formula

Various types of Correlation Coeeficient are:

Pearson's Correlation Coefficient Formula

$$R = \frac{n(\sum xy)-(\sum x)(\sum y)}{\sqrt{[n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2}}$$

Sample Correlation Coefficient Formula:

$$r_{xy} = Cov(x,y)/s_x.s_y$$

where,

Sxy is Covariance of Sample

Sx and Sy are Standard Deviations of Sample

Population Correlation Coefficient Formula:

$$\rho_{xy} = \sigma_{xy}/\sigma_x.\sigma_y$$

where,

σx and σy are Populatin Standard Deviation

σxy is Population Covariance

**Pearson's Correlation**

It is the most common correlation in statistics. The full name is Pearson's Product Moment Correlation in short PPMC. It displays the Linear relation between the two sets of data. Two letters are used to represent the Pearson correlation

Greek Letter "rho (ρ)" for a population and the letter "r" for a sample correlation coefficient.

**How to Find Pearson's Correlation Coefficient?**

Follow the steps added below to find the Pearson's Correlation Coefficient of any given data set

Step 1: Firstly make a chart with the given data like subject,x, and y and add three more columns in it xy, $x^2$ and $y^2$.

Step 2: Now multiply the x and y columns to fill the xy column. For example:- in x we have 24 and in y we have 65 so xy will be 24×65=1560.

Step 3: Now, take the square of the numbers in the x column and fill the $x^2$ column. Step

4: Now, take the square of the numbers in the y column and fill the $y^2$ column.

Step 5: Now, add up all the values in the columns and put the result at the bottom. Greek letter sigma (Σ) is the short way of saying summation.

Step 6: Now, use the formula for Pearson's correlation coefficient:

$$R = \frac{n(\sum xy)-(\sum x)(\sum y)}{\sqrt{[n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2}}$$

To know which type of variable we have either positive or negative.

**Linear Correlation Coefficient**

The Pearson's correlation coefficient is the linear correlation coefficient which returns the value between the -1 and +1. In this -1 indicates a strong negative correlation and +1 indicates a strong positive correlation. If it lies 0 then there is no correlation. This is also known as zero correlation.

The “crude estimations” for analyzing the stability of correlations using Pearson’s Correlation:

| r Value | Crude Estimates |
|---|---|
| +.70 or higher | A very strong positive relationship |
| +.40 to +.69 | Strong positive relationship |
| +.30 to +.39. | Moderate positive relationship |
| +.20 to +.29 | Weak positive relationship |
| +.01 to +.19 | No or negligible relationship |
| 0 | No relationship [zero correlation] |
| -.01 to -.19 | No or negligible relationship |
| -.20 to -.29 | Weak negative relationship |
| -.30 to -.39 | Moderate negative relationship |
| -.40 to -.69 | Strong negative relationship |
| -.70 or higher | The very strong negative relationship |

Cramer’s V Correlation

It is as similar as the Pearson correlation coefficient. It is used to calculate the correlation with more than 2×2 rows and columns. Cramer’s V correlation varies between 0 and 1. The value close to zero associates that a very little association is there between the variables and if it’s close to 1 it indicates a very strong association.

The “crude estimates” for interpreting strengths of correlations using Cramer’s V Correlation:

| Cramer’s V | Crude Estimates |
|---|---|
| .25 or higher | Very strong relationship |
| .15 to .25 | Strong relationship |
| .11 to .15 | Moderate relationship |
| .06 to .10 | Weak relationship |
| .01 to .05 | No or negligible relationship |

**Correlation Coefficient Formula Problems**

Problem 1: Calculate the correlation coefficient from the following table:

| SUBJECT | AGE (X) | GLUCOSE LEVEL (Y) |
|---|---|---|
| 1 | 42 | 98 |
| 2 | 23 | 68 |
| 3 | 22 | 73 |
| 4 | 47 | 79 |
| 5 | 50 | 88 |
| 6 | 60 | 82 |

Solution:

Make a table from the given data and add three more columns of XY, $X^2$, and $Y^2$.

| SUBJECT | AGE (X) | GLUCOSE LEVEL (Y) | XY | $X^2$ | $Y^2$ |
|---|---|---|---|---|---|
| 1 | 42 | 98 | 4116 | 1764 | 9604 |
| 2 | 23 | 68 | 1564 | 529 | 4624 |
| 3 | 22 | 73 | 1606 | 484 | 5329 |
| 4 | 47 | 79 | 3713 | 2209 | 6241 |
| 5 | 50 | 88 | 4400 | 2500 | 7744 |
| 6 | 60 | 82 | 4980 | 3600 | 6724 |
| Σ | 244 | 488 | 20379 | 11086 | 40266 |

$\sum xy = 20379$
$\sum x = 244$
$\sum y = 488$
$\sum x^2 = 11086$
$\sum y^2 = 40266$
n = 6.
Put all the values in the Pearson's correlation coefficient formula:

$$R = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2}}$$

$R = 6(20379) – (244)(488) / \sqrt{[6(11086)-(244)^2][6(40266)-(488)^2]}$

$R = 3202 / \sqrt{[6980][3452]}$

R = 3202/4972.238

R = 0.6439

It shows that the relationship between the variables of the data is a strong positive relationship.

Problem 2: Calculate the correlation coefficient from the following table:

| SUBJECT | AGE (X) | Weight (Y) |
|---|---|---|
| 1 | 40 | 99 |
| 2 | 25 | 79 |
| 3 | 22 | 69 |
| 4 | 54 | 89 |

Solution:

Make a table from the given data and add three more columns of XY, $X^2$, and $Y^2$.

| SUBJECT | AGE (X) | Weight (Y) | XY | $X^2$ | $Y^2$ |
|---|---|---|---|---|---|
| 1 | 40 | 99 | 3960 | 1600 | 9801 |
| 2 | 25 | 79 | 1975 | 625 | 6241 |
| 3 | 22 | 69 | 1518 | 484 | 4761 |
| 4 | 54 | 89 | 4806 | 2916 | 7921 |
| ∑ | 151 | 336 | 12259 | 5625 | 28724 |

$\sum xy = 12258$

$\sum x = 151$

$\sum y = 336$

$\sum x^2 = 5625$

$\sum y^2$ 28724

n = 4

Put all the values in the Pearson's correlation coefficient formula:

$$R = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2}}$$

$R = 4(12258) – (151)(336) / \sqrt{[4(5625)-(151)^2][4(28724)-(336)^2]}$

$R = -1704 / \sqrt{[-301][-2000]}$

R=-1704/775.886

R=-2.1961

It shows that the relationship between the variables of the data is a very strong negative relationship.

**3.4 REGRESSION**

• For an input x, if the output is continuous, this is called a regression problem. For example, based on historical information of demand for tooth paste in your supermarket, you are asked to predict the demand for the next month.

• Regression is concerned with the prediction of continuous quantities. Linear regression is the oldest and most widely used predictive model in the field of machine learning. The goal is to minimize the sum of the squared errors to fit a straight line to a set of data points.

• Simple Linear Regression (SLR) is a statistical model in which there is only one independent variable and the functional relationship between the dependent variable and the regression coefficient is linear.

• Regression line is the line which gives the best estimate of one variable from the value of any other given variable.

• Regression line of Y on X: Gives the best estimate for the value of Y for any specific

given values of X:

where

$Y = a + bx$

a = Y - intercept

b = Slope of the line

Y = Dependent variable

X = Independent variable

• By using the least squares method, we are able to construct a best fitting straight line to the scatter diagram points and then formulate a regression equation in the form of:

$\hat{y} = a + bx$

$\hat{y} = \bar{y} + b(x - \bar{x})$

• Regression analysis is the art and science of fitting straight lines to patterns of data. In a linear regression model, the variable of interest ("dependent" variable) is predicted from k other variables ("independent" variables) using a linear equation.

• If Y denotes the dependent variable and X1, ..., Xk are the independent variables, then the assumption is that the value of Y at time t in the data sample is determined by the linear equation:

$Y1 = \beta 0 + \beta 1\ X1t + B2\ X2t + \ldots + \beta k\ Xkt + \varepsilon t$

where the betas are constants and the epsilons are independent and identically distributed normal random variables with mean zero.

**3.5 REGRESSION LINE**

• A way of making a somewhat precise prediction based upon the relationships between two variables. The regression line is placed so that it minimizes the predictive error.

• The regression line does not go through every point; instead it balances the difference between all data points and the straight-line model. The difference between the observed data value and the predicted value (the value on the straight line) is the error or residual. The criterion to determine the line that best describes the relation between two variables is based on the residuals.

Residual = Observed - Predicted

• A negative residual indicates that the model is over-predicting. A positive residual indicates that the model is under-predicting.

**Linear Regression**

• The simplest form of regression to visualize is linear regression with a single predictor. A linear regression technique can be used if the relationship between X and Y can be approximated with a straight line.

• Linear regression with a single predictor can be expressed with the equation:

$$y = \Theta 2x + \Theta 1 + e$$

• Model 'Y', is a linear function of 'X'. The value of 'Y' increases or decreases in linear manner according to which the value of 'X' also changes.

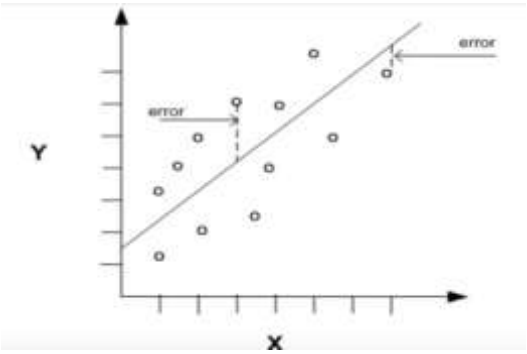



**Fig 3.4 linear Regression**

**Nonlinear Regression:**

• Often the relationship between x and y cannot be approximated with a straight line. In this case, a nonlinear regression technique may be used.

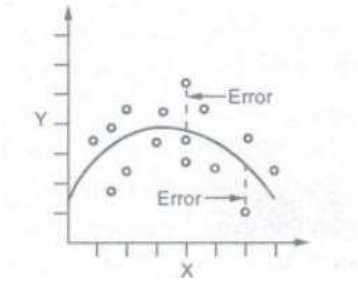



**Fig 3.5 Non linear Regression**

• Alternatively, the data could be preprocessed to make the relationship linear. In Fig. 3.4.2 shows nonlinear regression. (Refer Fig. 3.4.2 on previous page)

• The X and Y have a nonlinear relationship.sp

• If data does not show a linear dependence we can get a more accurate model using a nonlinear regression model.

• For example: y = W0 + W1X + W2 X2 + W3 X3

• Generalized linear model is foundation on which linear regression can be applied to modeling categorical response variables.

**Advantages:**

a. Training a linear regression model is usually much faster than methods such as neural networks.

b. Linear regression models are simple and require minimum memory to implement.

c. By examining the magnitude and sign of the regression coefficients you can infer how predictor variables affect the target outcome.

• There are two important shortcomings of linear regression:

1. Predictive ability: The linear regression fit often has low bias but high variance. Recall that expected test error is a combination of these two quantities. Prediction accuracy can sometimes be improved by sacrificing some small amount of bias in order to decrease the variance.

2. Interpretative ability: Linear regression freely assigns a coefficient to each predictor variable. When the number of variables p is large, we may sometimes seek, for the sake of interpretation, a smaller set of important variables.

## 3.6 LEAST SQUARES REGRESSION LINE

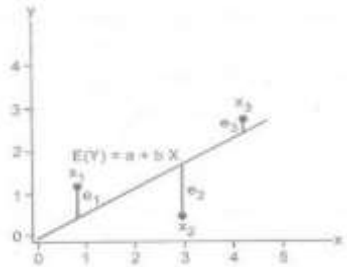
Least square method

• The method of least squares is about estimating parameters by minimizing the squared discrepancies between observed data, on the one hand and their expected values on the other.

• The Least Squares (LS) criterion states that the sum of the squares of errors is minimum. The least-squares solutions yield y(x) whose elements sum to 1, but do not ensure the outputs to be in the range [0, 1].

• How to draw such a line based on data points observed? Suppose a imaginary line of

y = a + bx.




• Imagine a vertical distance between the line and a data point E = Y - E(Y). This error is the deviation of the data point from the imaginary line, regression line. Then what is the best values of a and b? A and b that minimizes the sum of such errors.

**Disadvantages of least square**

1. Lack robustness to outliers.

2. Certain datasets unsuitable for least squares classification.

3. Decision boundary corresponds to ML solution.

**Example: Fit a straight line to the points in the table. Compute m and b by least squares.**

| Points | x | y |
|---|---|---|
| A | 3.00 | 4.50 |
| B | 4.25 | 4.25 |
| C | 5.50 | 5.50 |
| D | 8.00 | 5.50 |

$$\begin{bmatrix} 3.00 & 1 \\ 4.25 & 1 \\ 5.50 & 1 \\ 8.00 & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} 4.50 \\ 4.25 \\ 5.50 \\ 5.50 \end{bmatrix} + \begin{bmatrix} V_A \\ V_B \\ V_C \\ V_D \end{bmatrix}$$

$$X = \begin{bmatrix} m \\ b \end{bmatrix}$$

$$= (A^T A)^{-1} (A^T L)$$

$$= \begin{bmatrix} 121.3125 & 20.7500 \\ 20.7500 & 4.0000 \end{bmatrix}^{-1} \begin{bmatrix} 105.8125 \\ 19.7500 \end{bmatrix}$$

$$= \begin{bmatrix} 0.246 \\ 3.663 \end{bmatrix}$$

$$V = AX - L$$

$$= \begin{bmatrix} 3.00 & 1 \\ 4.25 & 1 \\ 5.50 & 1 \\ 8.00 & 1 \end{bmatrix} \begin{bmatrix} 0.246 \\ 3.663 \end{bmatrix} - \begin{bmatrix} 4.50 \\ 4.25 \\ 5.50 \\ 5.50 \end{bmatrix}$$

$$= \begin{bmatrix} -0.10 \\ 0.46 \\ -0.48 \\ 0.13 \end{bmatrix}$$

**Standard Error of Estimate**

• The standard error of estimate represents a special kind of standard deviation that reflects tells that we the magnitude of predictive error. The standard error of estimate, denoted S approximately how large the prediction errors (residuals) are for our data set in the same units as Y.

Definition formula for standard error of estimate = √Sum of square / √n-2

Definition formula for standard error of estimate = √Y-Y' / √(n-2) Computation

formula for standard error of estimate: Sy/x = √SSy(1-r2) /n-2

**Example: Define linear and nonlinear regression using figures. Calculate the value of Y for X = 100 based on linear regression prediction method.**

| X | Y |
|---|---|
| 4 | 390 |
| 9 | 580 |
| 10 | 650 |
| 14 | 730 |

| | |
|---|---|
| 4 | 410 |
| 7 | 530 |
| 12 | 600 |
| 22 | 790 |
| 1 | 350 |
| 3 | 400 |
| 8 | 590 |
| 11 | 640 |
| 5 | 450 |
| 6 | 520 |
| 10 | 690 |
| 11 | 690 |
| 16 | 770 |
| 13 | 700 |
| 13 | 730 |
| 10 | 640 |

**Solution**

| X | Y | X * Y | $X^2$ |
|---|---|---|---|
| 4 | 390 | 1560 | 16 |
| 9 | 580 | 5220 | 81 |
| 10 | 650 | 6500 | 100 |
| 14 | 730 | 10220 | 196 |
| 4 | 410 | 1640 | 16 |

| | | | |
|---|---|---|---|
| 7 | 530 | 3710 | 49 |
| 12 | 600 | 7200 | 144 |
| 22 | 790 | 17380 | 484 |
| 1 | 350 | 350 | 1 |
| 3 | 400 | 1200 | 9 |
| 8 | 590 | 4720 | 56 |
| TT = 105 | TT = 6660 | TT = 66740 | TT = 1273 |

N = Total number of samples = 12

f =aA + b

but linear equation calculate slope and interception prediction.

$f = a_0 + a_1 N$

$\bar{X} = x / N = 105 / 12 = 8.75$

$\bar{Y} = y / N = 6660 / 12 = 555$

$X_1 = (4 - 8.75)(390 - 555) + (9 - 8.75)(580 - 555) + (10 - 8.75)(650 - 555)$

$+ (14 - 8.75)(730 - 555) + (4 - 8.75)(410 - 555) + (7 - 8.75)(530 - 555)$

$+ (12 - 8.75)(600 - 555) + (22 - 8.75)(790 - 555) + (1 - 8.75)(350 - 555)$

$+ (3 - 8.75)(400 - 555) + (8 - 8.75)(590 - 555) + (11 - 8.75)(640 - 555) (4 - 8.75)^2$

$+ (9 - 8.75)^2 + (10 - 8.75)^2 + (14 - 8.75)^2 + (4 - 8.75)^2$

$+ (7 - 8.75)^2 + (12 - 8.75)^2 + (10 - 8.75)^2 + (14 - 8.75)^2 + (4 - 8.75)^2$

$+ (7 - 8.75)^2 + (11 - 8.75)^2$ $X_1 = 8465 / 36225 = 23.36$

$X_0 = \bar{Y} - X_1$ .

$x = 555 - (8.75)(23.36) = 350.6$

**$F = 350.6 + 23.36 \times 12 = 630.92$**

## 3.7 INTERPRETATION OF $R^2$

The following measures are used to validate the simple linear regression models:

1. Co-efficient of determination (R-square).

2. Hypothesis test for the regression coefficient b1.

3. Analysis of variance for overall model validity (relevant more for multiple linear regression).

4. Residual analysis to validate the regression model assumptions.

5. Outlier analysis.

The primary objective of regression is to explain the variation in Y using the knowledge of X. The coefficient of determination (R-square) measures the percentage of variation in Y explained by the model (ß0 + ß1 X).

**Characteristics of R-square:**

Here are some basic characteristics of the measure:

1. Since $R^2$ is a proportion, it is always a number between 0 and 1.

2. If $R^2 = 1$, all of the data points fall perfectly on the regression line. The predictor x accounts for all of the variation in y!.

3. If$R^2 = 0$, the estimated regression line is perfectly horizontal. The predictor x accounts for none of the variation in y!

• As a result, $R^2$ increases as new predictors are added to a multiple linear regression model, but the adjusted R increases only if the increase in $R^2$ is greater than one would expect from chance alone. In such a model, the adjusted $R^2$ is the most realistic estimate of the proportion of the variation that is predicted by the covariates included in the model.

**$r^2$ INTERPRETATION**

$$r^2 = \frac{SS_{Y'}}{SS_Y} = \frac{SS_Y - SS_{Y|X}}{SS_Y}$$

SSy = 33.2

$SS_{ylx} = \Sigma(Y-Y')^2$

| X(Independent Variable) | Y(Dependent Variable) Actual Value | Y' Predicted Value |
|---|---|---|
| 10 | 12 | 11.8 |
| 20 | 8 | 11.1 |
| 30 | 14 | 10.4 |
| 40 | 11 | 9.7 |
| 50 | 7 | 9 |

$$= [(12-11.8)^2 + (8-11.1)^2 + (14-10.4)^2 + (11-9.7)^2 + (7-9)^2]$$

$$= (0.2)^2 + (-3.1)^2 + (3.6)^2 + (1.3)^2 + (-2)^2$$

$$= 0.04 + 9.61 + 12.96 + 1.69 + 4$$

$$= 28.3$$

$$r^2 = \frac{33.2 - 28.3}{33.2} = 0.15$$

An r-squared of 15% reveals that 15% of the variability observed in the target variable is explained by the regression model.

**Spurious Regression**

• The regression is spurious when we regress one random walk onto another independent random walk. It is spurious because the regression will most likely indicate a non-existing relationship:

1. The coefficient estimate will not converge toward zero (the true value). Instead, in the limit the coefficient estimate will follow a non-degenerate distribution.

2. The t value most often is significant.

3. $R^2$ is typically very high.

**Residual analysis**

Residual (error) analysis is important to check whether the assumptions of regression models have been satisfied. It is performed to check the following:

1. The residuals are normally distributed.

2. The variance of residual is constant (homoscedasticity).

3. The functional form of regression is correctly specified.

4. If there are any outliers.

## 3.8 MULTIPLE REGRESSION EQUATIONS

• Multiple linear regression is an extension of linear regression, which allows a response variable, y to be modelled as a linear function of two or more predictor variables.

• In a multiple regression model, two or more independent variables, i.e. predictors are involved in the model. The simple linear regression model and the multiple regression model assume that the dependent variable is continuous.

What Is Multiple Regression?

Multiple regression is an extension of linear regression models that allow predictions of systems with multiple independent variables. Multiple regression is specifically designed to create regressions on models with a single dependent variable and multiple independent variables.

$y = B_1 * x_1 + B_2 * x_2 + \ldots + B_n * x_n + A$

In this equation, the subscripts denote the different independent variables. For example, $x_1$ is the value of the first independent variable, $x_2$ is the value of the second independent variable, and so on. It keeps going as we add more independent variables until we finally add the last independent variable, $x_n$, to the equation.

## 3.9 REGRESSION TOWARD THE MEAN.

Regression to the mean (RTM) is a statistical phenomenon describing how variables much higher or lower than the mean are often much closer to the mean when measured a second time.

# UNIT IV
# DATA ANALYTICS AND VISUALIZATION WITH PYTHON

**Essential Data Libraries for data analytics: Pandas, NumPy, SciPy. Plotting and visualization with python: Introduction to Matplotlib, Basic Plotting with Matplotlib, Create Histogram, Bar Chart, Pie chart, Box Plot, violin plot using Matplotlib.Introduction to seaborn Library, Multiple Plots, Regression plot, regplot.**

## 4.1 Essential Data Libraries for data analytics

Python is one of the most popular programming languages used across various tech disciplines, especially in data science and machine learning. Python offers an easy-to-code, object-oriented, high-level language with a broad collection of libraries for a multitude of use cases. It has over 137,000 libraries.

### 4.1.1 Pandas

**Pandas Introduction**

Pandas is a powerful and open-source Python library. The Pandas library is used for data manipulation and analysis. Pandas consist of data structures and functions to perform efficient operations on data.

**What is Python Pandas used for?**

The Pandas library is generally used for data science, but have you wondered why? This is because the Pandas library is used in conjunction with other libraries that are used for data science. It is built on top of the NumPy library which means that a lot of the structures of NumPy are used or replicated in Pandas.

**Here is a list of things that we can do using Pandas:**

- Data set cleaning, merging, and joining.
- Easy handling of missing data (represented as NaN) in floating point as well as non-floating point data.
- Columns can be inserted and deleted from DataFrame and higher-dimensional objects.
- Powerful group by functionality for performing split-apply-combine operations on data sets.
- Data Visualization.

**Installing Pandas:**

The first step in working with Pandas is to ensure whether it is installed in the system or not. If not, then we need to install it on our system using the **pip command**.

**Importing Pandas:**

After the Pandas have been installed in the system, we need to import the library. This module is generally imported as follows:

***import pandas as pd***

**Note:** Here, pd is referred to as an alias for the Pandas. However, it is not necessary to import the library using the alias, it just helps in writing less code every time a method or property is called.

**Data Structures in Pandas Library**

Pandas generally provide two data structures for manipulating data. They are:

- Series
- DataFrame
- **Pandas Series**

A Pandas Series is a one-dimensional labeled array capable of holding data of any type (integer, string, float, Python objects, etc.). The axis labels are collectively called **indexes**.

- **Creating a Series**

Pandas Series is created by loading the datasets from existing storage (which can be a SQL database, a CSV file, or an Excel file). It can be created from lists, dictionaries, scalar values, etc.

**Pandas DataFrame:** It is a two-dimensional data structure with labeled axes (rows and columns).

**Creating DataFrame:** Pandas DataFrame is created by loading the datasets from existing storage (which can be a SQL database, a CSV file, or an Excel file). Pandas DataFrame can be created from lists, dictionaries, a list of dictionaries, etc.

### 4.1.2 Numpy

NumPy, short for Numerical Python, is the core library for scientific computing in Python. It has been designed specifically for performing basic and advanced array operations. It primarily supports multi-dimensional arrays and vectors for complex arithmetic operations.

NumPy has risen to become one of the most popular Python science libraries and just secured a round of grant funding.

NumPy's multidimensional array can perform very large calculations much more easily and efficiently than using the Python standard data types.

The library offers many handy features performing operations on a n-arrays and matrices in Python. It helps to process arrays that store values of the same data type and makes performing math operations on arrays easier. In fact, the vectorization of mathematical operations on the NumPy array type increases performance and accelerates the execution time.

**Basics of Numpy Arrays**

Numpy array is a powerful N-dimensional array object which is in the form of rows and columns. We can initialize NumPy arrays from nested Python lists and access it elements. NumPy array is a collection of elements that have the same data type.

A one-dimensional NumPy array can be thought of as a vector, a two-dimensional array as a matrix (i.e., a set of vectors), and a three-dimensional array as a tensor (i.e., a set of matrices).

**Basic array manipulations are as follows :**

**1. Attributes of arrays**: It define the size, shape, memory consumption, and data types of arrays.

**2. Indexing of arrays:** Getting and setting the value of individual array elements.

**3. Slicing of arrays:** Getting and setting smaller subarrays within a larger array.

**4. Reshaping of arrays**: Changing the shape of a given array.

**5. Joining and splitting of arrays:** Combining multiple arrays into one, and splitting one array into many.

**a) Attributes of array**

In Python, arrays from the NumPy library, called N-dimensional arrays or the ndarray, are used as the primary data structure for representing data.

The main data structure in NumPy is the ndarray, which is a shorthand name for N-dimensional array. When working with NumPy, data in an ndarray is simply referred to as an array. It is a fixed-sized array in memory that contains data of the same type, such

as integers or floating point values.

**ndarray.shape:** By using this method in numpy, we can know the array dimensions. It can also be used to resize the array. Each array has attributes ndim (the number of dimensions), shape (the size of each dimension), and size (the total size of the array).

**ndarray.size:** The total number of elements of the array. This is equal to the product of the elements of the array's shape.

**ndarray.dtype:** An object describing the data type of the elements in the array. Recall that NumPy's ND-arrays are homogeneous: they can only posses numbers of a uniform data type.

**b) Indexing of arrays**

Array indexing always refers to the use of square brackets ("[ ]') to index the elements of the array. In order to access a single element of an array we can refer to its index.

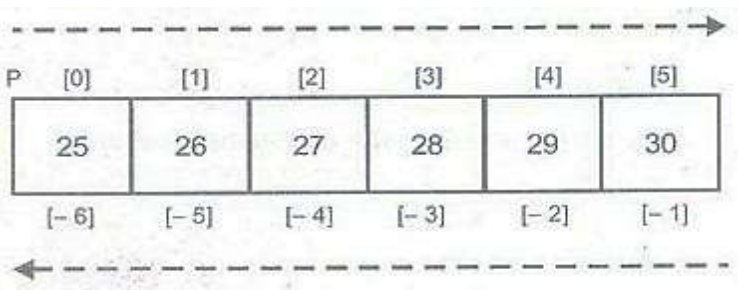



**Fig. 4.1.2.1 The indexing of an ndarray mono-dimensional.**

```
>>> a = np.arange(25, 31)
>>>P
array([25, 26, 27, 28, 29, 30])
>>>P[3]
28
```

The NumPy arrays also accept negative indexes. These indexes have the same incremental sequence from 0 to -1, -2, and so on,

```
>>>P[-1]      30
>>>P[-6]      25
```

In a multidimensional array, we can access items using a comma-separated tuple of indices. To select multiple items at once, we can pass array of indexes within the square brackets.

```
>>>a[[1, 3, 4]]
```

array([26, 28, 29])

Moving on to the two-dimensional case, namely, the matrices, they are represented as rectangular arrays consisting of rows and columns, defined by two axes, where axis 0 is represented by the rows and axis 1 is represented by the columns.

| A | [,0] | [,1] | [,2] |
|---|---|---|---|
| [0,] | 10 | 11 | 12 |
| [1,] | 13 | 14 | 15 |
| [,2] | 16 | 17 | 18 |

**Fig. 4.1.2.2 shows the indexing of a bi-dimensional array**

```
>>> A = np.arange(10, 19).reshape((3, 3))
>>> A
array([[10, 11, 12],[13, 14, 15],[16, 17, 18]])
```

**c) Slicing of arrays**

Slicing is the operation which allows to extract portions of an array to generate new ones. Whereas using the Python lists the arrays obtained by slicing are copies, in NumPy, arrays are views onto the same underlying buffer.

Slicing of array in Python means to access sub-parts of an array. These sub-parts can be stored in other variables and further modified.

**Synatx :** arr[ start stop step], Arr[slice(start, stop, step)]

The start parameter represents the starting index, stop is the ending index, and step is the number of items that are "stepped" over. If any of these are unspecified, they default to the values start=0, stop-size of dimension, step-1.

```
importnumpy as np

arr = np.array([1,2,3,4])

print(arr[1:3:2])

print(arr[:3])

print(arr[::2])
```

**Output:**

[2]

[1 2 3]

[13]

**Multidimensional sub-arrays:**

Multidimensional slices work in the same way, with multiple slices separated by commas. For example:

In[24]: x2

Out[24]: array([[12, 5, 2, 4],[ 7, 6, 8, 8],[ 1, 6, 7, 7]])

In[25]: x2[:2, :3]  # two rows, three columns

Out[25]: array([[12, 5, 2],[ 7, 6, 8]])

In[26]: x2[:3, ::2]  # all rows, every other column

Out[26]: array([[12, 2],[7, 8],[ 1, 7]])

Let us create an array using the package Numpy and access its columns.

# Creating an array

importnumpy as np

a= np.array([[1,2,3],[4,5,6],[7,8,9]])

Now let us access the elements column-wise. In order to access the elements in a column-wise manner colon(:) symbol is used let us see that with an example.

importnumpy as np

a = np.array([[1,2,3],[4,5,6],[7,8,9]])

print(a[:,1])

**Output:**

[258]

**d) Reshaping of array**

The numpy.reshape() function is used to reshape a numpy array without changing the data in the array.

**Syntax:**

numpy.reshape(a, newshape, order='C')

Where order: {'C', 'F', 'A'}, optional Read the elements of a using this index order, and place the elements into the reshaped array using this index order.

**Step 1: Create a numpy array of shape (8,)**

```
num_array = np.array([1,2,3,4,5,6,7,8])

num_array
```

**Output:**

array([1, 2, 3, 4, 5, 6, 7, 8])

**Step 2: Use np.reshape() function with new shape as (4,2)**

```
np.reshape(num_array,(4,2))

array([[1,2],[3,4],[5,6],[7,8]])
```

The shape of the input array has been changed to a (4,2). This is a 2-D array and contains the same data present in the original input 1-D array.

**e) Array concatenation and splitting**

**np.concatenate()** constructor is used to concatenate or join two or more arrays into one. The only required argument is list or tuple of arrays.

```
#first, import numpy
importnumpy as np
# making two arrays to concatenate
arr1 = np.arange(1,4)
arr2 = np.arange(4,7)
print("Arrays to concatenate:")
print(arr1); print(arr2)
print("After concatenation:")
print(np.concatenate([arr1,arr2]))
```

**Output:**

Arrays to concatenate:

[1 2 3]

[4 5 6]

After concatenation:

[1 2 3 4 5 6]

### 4.1.3 SciPy

Scipy is a Python library useful for solving many mathematical equations and algorithms. It is designed on the top of Numpy library that gives more extension of finding scientific mathematical formulae like Matrix Rank, Inverse, polynomial equations, LU Decomposition, etc. Using its high-level functions will significantly reduce the complexity of the code and helps better in analyzing the data.

**Installation of SciPy**

To install SciPy in your system, you can use Python package manager pip. Before proceeding, make sure that you have Python already installed in your system. Here's the step to install Python in your system.

**Step 1:** Firstly, Open terminal and Command Prompt in your system.

**Step 2:** Run the installation Command to install SciPy in your system. *pip*

*install scipy*

**Step 3:** Pip will be download and install SciPy along with dependencies. This process will may take some time depends on internet connection.

**Step 4:** To verify installation, you need to import SciPy in a Python script or interactive shell.

*importscipy*
*print(scipy.__version__)*

Now the installation of SciPy is successfully completed.

**Key Features:**

- An implementation of unconstrained and constrained optimization algorithms.

- Functionality for determining definite integrals and solving differential equations.
- A collection of functions for interpolating data points, such as spline and polynomial interpolation.
- Robust algorithms for matrix operations, eigenvalues, and singular valuedecomposition (SVD).

**Plotting and Visualization with Python**

Introduction to Matplotlib, Basic Plotting with Matplotlib, Create Histogram, Bar Chart, Pie chart, Box Plot, violin plot using Matplotlib.

### 4.2.1 Introduction to Matplotlib

Matplotlib is a popular plotting library in Python used for creating high-quality visualizations and graphs. It offers various tools to generate diverse plots, facilitating data analysis, exploration, and presentation. Matplotlib is flexible, supporting multiple plot types and customization options, making it valuable for scientific research, data analysis, and visual communication. It can create different types of visualization reports like line plots, scatter plots, histograms, bar charts, pie charts, box plots, and many more different plots. This library also supports 3-dimensional plotting.

**Installation of Matplotlib**

Colab Notebooks are similar to Jupyter Notebooks except for running on the cloud. It is also connected to our Google Drive, making it much easier to access our Colab notebooks anytime, anywhere, and on any system. Now install Matplotlib by using the PIP command.

!pip install matplotlib

To verify the installation, you would have to write the following code chunk:

importmatplotlib

```
print(matplotlib.__version__)
```

**Key Features of Matplotlib**

- **Versatile Plotting:** Create a wide variety of visualizations, including line plots, scatter plots, bar charts, and histograms.
- **Extensive Customization:** Control every aspect of your plots, from colors and markers to labels and annotations.
- **Seamless Integration with NumPy:** Effortlessly plot data arrays directly, enhancing data manipulation capabilities.
- **High-Quality Graphics:** Generate publication-ready plots with precise control over aesthetics.
- **Cross-Platform Compatibility:** Use Matplotlib on Windows, macOS, and Linux without issues.
- **Interactive Visualizations:** Engage with your data dynamically through interactive plotting features.

**What is Matplotlib Used For?**

Matplotlib is a Python library for data visualization, primarily used to create static, animated, and interactive plots. It provides a wide range of plotting functions to visualize data effectively.

**Key Uses of Matplotlib:**

- **Basic Plots:** Line plots, bar charts, histograms, scatter plots, etc.
- **Statistical Visualization:** Box plots, error bars, and density plots.
- **Customization:** Control over colors, labels, gridlines, and styles.
- **Subplots & Layouts:** Create multiple plots in a single figure.
- **3D Plotting:** Surface plots and 3D scatter plots using mpl_toolkits.mplot3d.
- **Animations & Interactive Plots:** Dynamic visualizations with FuncAnimation.
- **Integration:** Works well with Pandas, NumPy and Jupyter Notebooks.

**Different Types of Plots in Matplotlib**

Matplotlib offers a wide range of plot types to suit various data visualization needs. Here are some of the most commonly used types of plots in Matplotlib:

- Line plot
- Histogram
- Bar Chart
- Scatter plot
- Pie charts
- Boxplot
- Violin Plot

Most of the time, we have to work with Pyplot as an interface for Matplotlib. So, we import Pyplot like this:

importmatplotlib.pyplot

To make things easier, we can import it like this:

importmatplotlib.

pyplot as plt

**4.2.1.1 Line Plots**

A line plot shows the relationship between the x and y-axis.

The plot() function in the Matplotlib library's Pyplot module creates a 2D hexagonal plot of x and y coordinates. plot() will take various arguments like plot(x, y, scalex, scaley, data, **kwargs).

x, y are the horizontal and vertical axis coordinates where x values are optional, and its default value is range(len(y)).

scalex, scaley parameters are used to autoscale the x-axis or y-axis, and its default value is actual.

**kwargs specify the property like line label, linewidth, marker, color, etc.

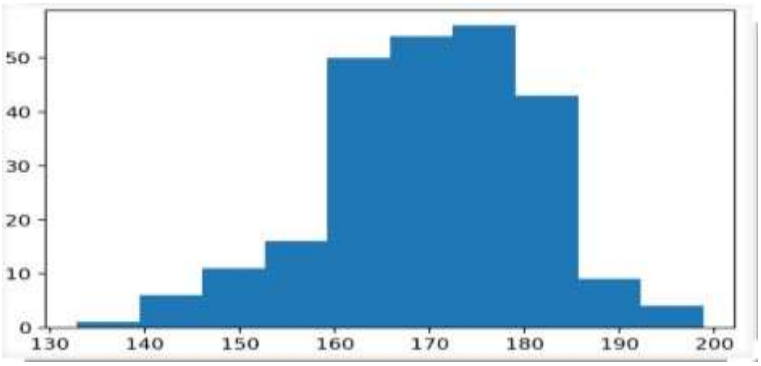
**4.2.1.2 Histogram**

The most common graph for displaying frequency distributions is a histogram. To create a histogram, the first step is to create a bin of ranges, then distribute the whole range of

values into a series of intervals and count the value that will fall in the given interval. We can use plt.Hist () function plots the histograms, taking various arguments like data, bins, color, etc.

**Code:**

```
#draw random samples from random distributions.

x = np.random.normal(170, 10, 250)

#plot histograms plt.hist(x) plt.show()
```



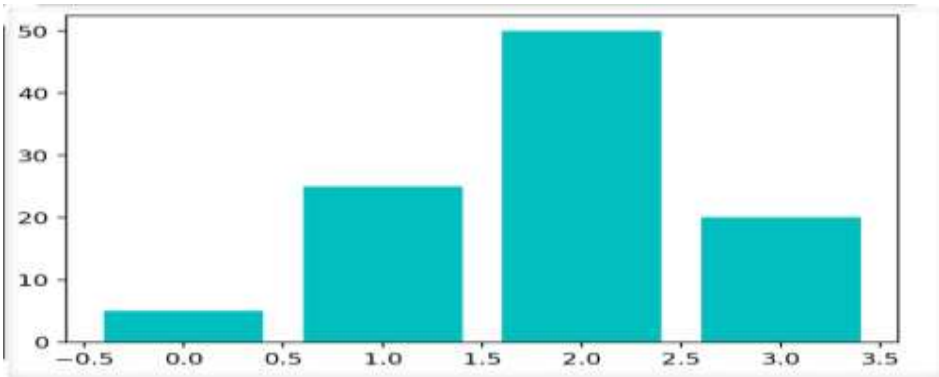

### 4.2.1.3 Bar Plot

Mainly, the barplot shows the relationship between the numeric and categoric values. In a bar chart, we have one axis representing a particular category of the columns and another axis representing the values or counts of the specific category. Barcharts can be plotted both vertically and horizontally using the following line of code:

plt.bar(x,height,width,bottom,align)

**Code:**

```
#define array

data= [5. , 25. , 50. , 20.]

plt.bar(range(len(data)), data,color='c')

plt.show()
```

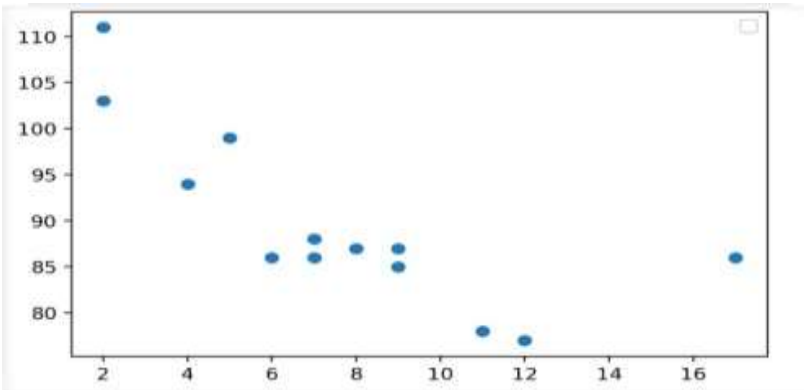



**4.2.1.4 Scatter Plot**

Scatter plots show the relationships between variables by using dots to represent the connection between two numeric variables. The scatter() method in the Matplotlib library is used for plotting.

**Code:**

```
#create the x and y axis coordinates

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])

y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])

plt.scatter(x, y)

plt.legend()

plt.show()
```

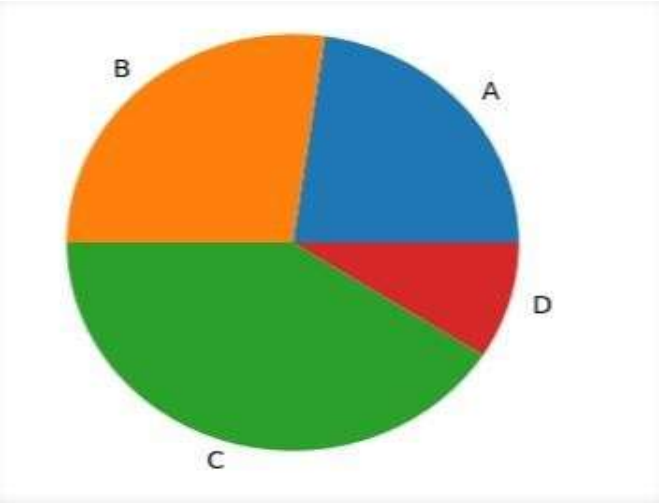



**4.2.1.5 Pie Chart**

A pie chart (or circular chart) is used to show the percentage of the whole. Hence, it is used to compare the individual categories with the whole. Pie() will take the different parameters such as:

**Code:**

```
#define the figure size

plt.figure(figsize=(7,7))

x = [25,30,45,10]

#labels of the pie chart

labels = ['A','B','C','D']

plt.pie(x, labels=labels)

plt.show()
```




### 4.2.1.6 Box Plot

A Box plot in Python Matplotlib showcases the dataset's summary, encompassing all numeric values. It highlights the minimum, first quartile, median, third quartile, and maximum. The median lies between the first and third quartiles. On the x-axis, you'll find the data values, while the y-coordinates represent the frequency distribution.

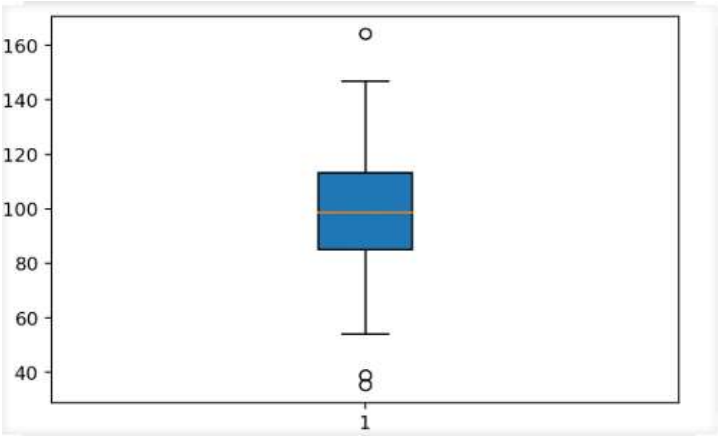
**Code:**

```
#create the random values by using numpy
values= np.random.normal(100, 20, 300)
```

#creating the plot by boxplot() function which is avilable in matplotlib

plt.boxplot(values,patch_artist=True,vert=True)

plt.show()




### 4.2.1.7 Violin Plot in Matplotlib

We can create a violin bin plot in Matplotlib using the violinplot() function. This function creates a graphical representation of the distribution of a dataset, consisting of both box plot elements and kernel density estimation.

- **The violinplot() Function**

The violinplot() function in Matplotlib takes one or more datasets and produces a plot where each dataset is represented by a "violin" shape. The width of the violin corresponds to the data density, and the plot displays the median, quartiles, and probability density function of the data distribution.

Following is the syntax of the violinplot() function in Matplotlib −

plt.violinplot( dataset, positions=None, vert=True, widths=0.5, showmeans=False, showextrema=True, showmedians=False, quantiles=None, points=100, bw_method=None, *, data=None )

- **Basic Violin Plot**

A basic violin plot is a way to show the distribution of data. Imagine your data as a bunch of violins. Each violin represents a set of values, and its shape gives you an idea of how

those values are spread. The wider parts of the violin indicate more data points, and the narrower parts indicate less data points. The line inside the violin is the median value, indicating the center of your data.

- **Horizontal Violin Plot**

To create a horizontal violin plot, the orientation of the usual violins is changed. Instead of extending up and down, they stretch from left to right. Just like regular violins, each one represents a set of data, showing you how the values are spread out. So, a horizontal violin plot helps you understand the distribution of your data, just like the regular one, but in a horizontal direction.

- **Multiple Violins in One Plot**

In Matplotlib, you can use multiple violins in in a single plot to represent various sets of data. By stacking these violins together, you can compare the distributions of multiple datasets, observing how they differ or overlap.

- **Violin Plot with Quantiles**

A violin plot with quantiles displays specific points that divide your data into quarters. Imagine each violin being divided into four equal sections, marking different levels of your data. The width of each section represents the number of data points. Now, you can also observe the plot, marking specific portions where your data lie, such as 25%, 50% (median), and 75%.

## 4.3 Introduction to seaborn Library

Built on top of Matplotlib, Seaborn is a well-known Python library for data visualization that offers a user-friendly interface for producing visually appealing and informative statistical graphics. It is designed to work with Pandas dataframes, making it easy to visualize and explore data quickly and effectively.

### 4.3.1 Seaborn vs. Matplotlib

Python's two most widely used data visualization libraries are Matplotlib and Seaborn. While both libraries are designed to create high-quality graphics and visualizations, they have several key differences that make them better suited for different use cases.

### 4.3.2 Installing Seaborn

Seaborn is supported on Python 3.7+ and has very minimal core dependencies. Installing Seaborn is pretty straightforward. You can either install it with Python’s pip manager or conda package manager.

# install seaborn with pip

pip install seaborn

### 4.3.3 Seaborn Plot types

Seaborn provides a wide range of plot types that can be used for data visualization and exploratory data analysis. Broadly speaking, any visualization can fall into one of the three categories.

- **Univariate** – x only (contains only one axis of information)
- **Bivariate** – x and y (contains two axis of information)
- **Trivariate** – x, y, z (contains three axis of information)

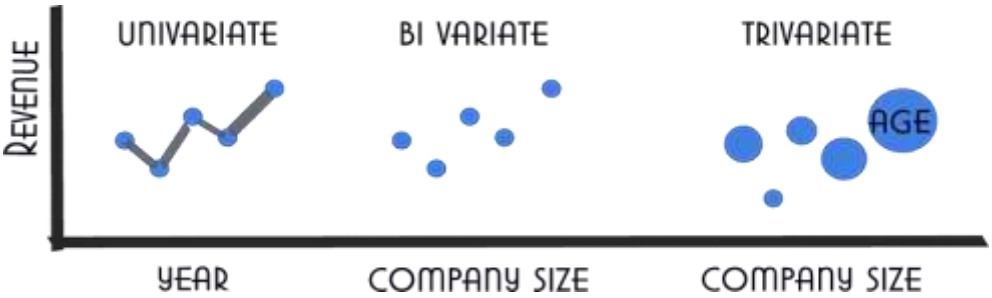



**Figure 4.3.2.1 Seaborn Plot types**

**Here are some of the most commonly used plot types in Seaborn**

- **Scatter Plot**. A scatter plot is used to visualize the relationship between two variables. Seaborn'sscatterplot() function provides a simple way to create scatter plots.
- **Line Plot**. A line plot is used to visualize the trend of a variable over time. Seaborn'slineplot() function provides a simple way to create line plots.
- **Histogram**. A histogram is used to visualize the distribution of a variable. Seaborn'shistplot() function provides a simple way to create histograms.
- **Box Plot**. A box plot is used to visualize the distribution of a variable. Seaborn'sboxplot() function provides a simple way to create box plots.
- **Violin Plot**. A violin plot is similar to a box plot, but provides a more detailed view of the distribution of the data. Seaborn'sviolinplot() function provides a simple way to create violin plots.

- **Heatmap**. A heatmap is used to visualize the correlation between different variables. Seaborn'sheatmap() function provides a simple way to create heatmaps.
- **Pairplot**. A pairplot is used to visualize the relationship between multiple variables. Seaborn'spairplot() function provides a simple way to create pairplots.

#### 4.3.3.1 Seaborn scatter plots

Scatter plots are used to visualize the relationship between two continuous variables. Each point on the plot represents a single data point, and the position of the point on the x and y-axis represents the values of the two variables. The plot can be customized with different colors and markers to help distinguish different groups of data points. In Seaborn, scatter plots can be created using the scatterplot() function.

#### 4.3.3.2 Seaborn line plots

Line plots are used to visualize trends in data over time or other continuous variables. In a line plot, each data point is connected by a line, creating a smooth curve. In Seaborn, line plots can be created using the lineplot() function.

#### 4.3.3.3 Seaborn bar plots

Bar plots are used to visualize the relationship between a categorical variable and a continuous variable. In a bar plot, each bar represents the mean or median (or any aggregation) of the continuous variable for each category. In this, bar plots can be created using the barplot()function.

#### 4.3.3.4 Seaborn histograms

Histograms visualize the distribution of a continuous variable. In a histogram, the data is divided into bins and the height of each bin represents the frequency or count of data points within that bin. In Seaborn, histograms can be created using the histplot() function.

#### 4.3.3.5 Seaborn box plots

Box plots are a type of visualization that shows the distribution of a dataset. They are commonly used to compare the distribution of one or more variables across different categories.

#### 4.3.3.6 Seaborn violin plots

A violin plot is a type of data visualization that combines aspects of both box plots and density plots. It displays a density estimate of the data, usually smoothed by a kernel density estimator, along with the interquartile range (IQR) and median in a box plot-like form.

#### 4.3.3.7 Seabornheatmaps

A heatmap is a graphical representation of data that uses colors to depict the value of a variable in a two-dimensional space. Heatmaps are commonly used to visualize the correlation between different variables in a dataset.

#### 4.3.3.8 Seaborn pair plots

Pair plots are a type of visualization in which multiple pairwise scatter plots are displayed in a matrix format. Each scatter plot shows the relationship between two variables, while the diagonal plots show the distribution of the individual variables.

### 4.4 Multiple Plots

**Introduction to Seaborn Multiple Plots**

Seaborn multiple plots are used to plot multiple graphs in a single window. We can plot the multiple graphs using two ways, first with the help of function, facetgrid, and other with the help of a matplotlib. The facet grid function is a general way of plotting the grids based on a function. It will help visualize the variable and its relationship with multiple variables.

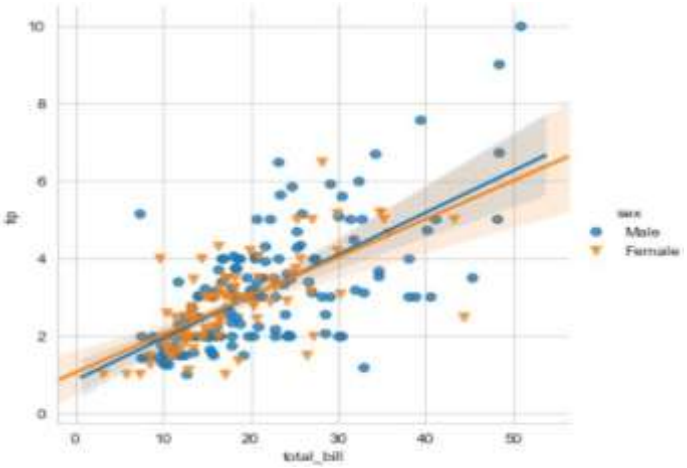
### 4.5 Regression plot

The regression plots in seaborn are primarily intended to add a visual guide that helps to emphasize patterns in a dataset during exploratory data analyses. Regression plots as the name suggests creates a regression line between 2 parameters and helps to visualize their

linear relationships. This article deals with those kinds of plots in seaborn and shows the ways that can be adapted to change the size, aspect, ratio etc. of such plots.

- **Linear plot with additional parameters**

```
sns.set_style('whitegrid')

sns.lmplot(x ='total_bill', y ='tip', data = dataset, hue ='sex', markers =['o', 'v'])
```




**4.6 Regplot**

Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics. Seaborn helps resolve the two major problems faced by Matplotlib; the problems are?

- Default Matplotlib parameters
- Working with data frames

As Seaborn compliments and extends Matplotlib, the learning curve is quite gradual. If you know Matplotlib, you are already half-way through Seaborn.

**seaborn.regplot()**

This method is used to plot data and a linear regression model fit. There are a number of mutually exclusive options for estimating the regression model.

**Syntax :**

seaborn.regplot( x, y, data=None, x_estimator=None, x_bins=None, x_ci='ci', scatter=True, fit_reg=True, ci=95, n_boot=1000, units=None, order=1, logistic=False, lowess=False, robust=False, logx=False, x_partial=None, y_partial=None, truncate=False, dropna=True, x_jitter=None, y_jitter=None, label=None, color=None, marker='o', scatter_kws=None, line_kws=None, ax=None)

**Parameters:** The description of some main parameters are given below:

- **x, y:** These are Input variables. If strings, these should correspond with

column names in “data”. When pandas objects are used, axes will be labeled with the series name.

- **data:** This is dataframe where each column is a variable and each row is an observation.
- **lowess:** (optional) This parameter take boolean value. If “True”, use “statsmodels” to estimate a nonparametric lowess model (locally weighted linear regression).
- **color:** (optional) Color to apply to all plot elements.

- **marker:** (optional) Marker to use for the scatterplot glyphs.

**Return:** The Axes object containing the plot.

# UNIT V
# APPLICATIONS AND CASE STUDIES

**Crawling the Web: Sentiment Analysis with Twitter Data - Unsupervised Learning: Customer Segmentation in Retail Industry-Breast cancer prediction - Dynamic Plots: Visualizing Piketty's Capital - Fraud Detection via Graph Analysis: Enron Email Dataset - Character Recognition from Images - Analyzing Marketing Campaigns with pandas.**

## 5.1 Crawling the Web:Sentiment Analysis with Twitter Data

### Introduction

Sentiment Analysis is the process of analyzing the emotions behind textual data. When applied to Twitter data, it helps in understanding public opinions on various topics such as politics, products, or events. This process involves:

- Web Crawling – Collecting tweets from Twitter
- Data Preprocessing – Cleaning and filtering the tweets
- Sentiment Classification – Categorizing tweets into positive, negative, or neutral
- Visualization – Graphical representation of sentiment trends

**1. Web Crawling: Collecting Twitter Data**

**To extract data from Twitter, we use:**

- Tweepy API (Official Twitter API)
- Snscrape (Non-API alternative)
- BeautifulSoup/Selenium (For scraping web-based tweets)

**2. Data Preprocessing (Cleaning Tweets)**

**Tweets often contain noise, so we remove:**

- URLs, hashtags, mentions (@user)
- Special characters & emojis
- Stopwords (e.g., is, are, the, etc.)

**3. Sentiment Analysis on Tweets For sentiment classification, we use:**

- Lexicon-based Models (VADER, TextBlob)
- Machine Learning Models (Naïve Bayes, LSTM, Transformer-based models like BERT)

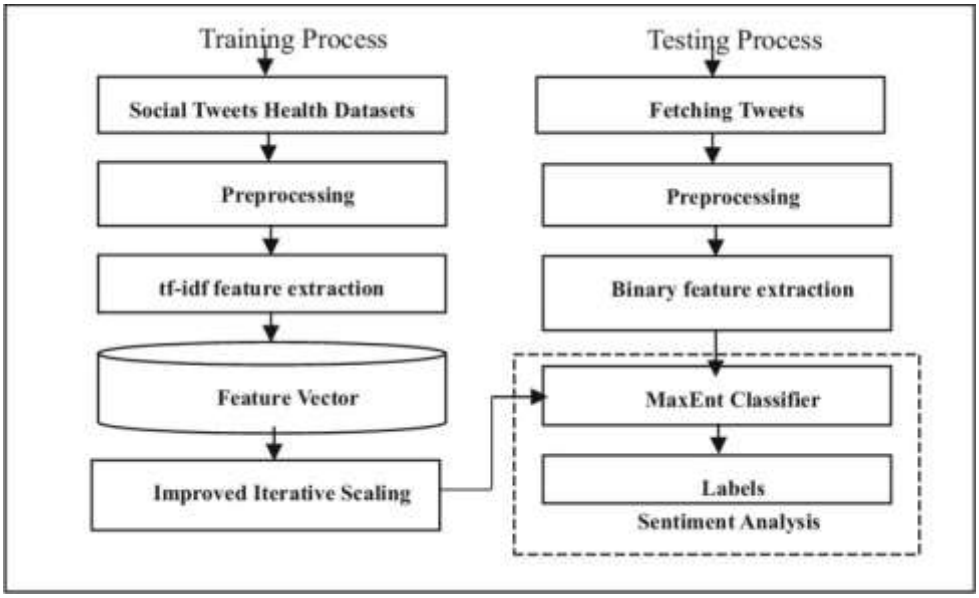



**Fig 5.1 Schematic block of tweets sentiment analyzer**

Sentiment analysis is a machine learning approach that determines the polarity of textual data, such as tweets, reviews, or comments. This workflow consists of two main processes: Training and Testing.

1. Training Process

The training phase is crucial for building an effective sentiment classification model. It involves the following steps:

- Social Tweets Health Datasets:
  - A dataset containing tweets related to health topics is collected.
  - These tweets include different sentiments such as positive, negative, or neutral.
- Preprocessing:
  - Cleaning the text by removing special characters, stop words, and punctuations.
  - Tokenization and stemming/lemmatization are performed to normalize words.
- TF-IDF Feature Extraction:

    - Term Frequency-Inverse Document Frequency (TF-IDF) is used to convert textual data into numerical form.
    - This technique assigns weights to words based on their importance in the dataset.
- Feature Vector:
    - The extracted features are stored in a vector representation, which is used as input for the classifier.
- Improved Iterative Scaling:
    - A statistical optimization technique is applied to improve the model's performance.
    - This helps in adjusting feature weights for better classification accuracy.

**2. Testing Process**

The testing phase involves using a trained model to classify new tweets. The following steps are involved:

- Fetching Tweets:
    - Real-time or stored tweets are collected from social media platforms (e.g., Twitter API).
- Preprocessing:
    - The same preprocessing steps as in the training phase are applied to the new data.
- Binary Feature Extraction:
    - Features are converted into a binary representation (presence or absence of words).
- MaxEnt Classifier (Maximum Entropy Model):
    - A probabilistic classifier that predicts sentiment labels (positive, negative, or neutral).
    - It uses the learned feature weights from the training phase.
- Labels & Sentiment Analysis:
    - The classifier assigns a sentiment label to each tweet based on extracted features.
    - The final result represents the sentiment polarity of the text.

**4. Visualization of Sentiments**

Once the sentiment scores are obtained, we can visualize them using Matplotlib or Seaborn.

**5. Applications of Sentiment Analysis**

- Business & Marketing→ Brand reputation monitoring
- Politics→ Understanding public opinion on policies
- Stock Market→ Predicting trends based on sentiment
- Crisis Management→ Identifying negative sentiment for quick responses

**5.2 Unsupervised Learning:**

Unsupervised learning is a type of machine learning where the model is trained on unlabeled data without predefined outputs. The goal is to find patterns, structures, and relationships within the data. It is widely used in clustering, anomaly detection, and

dimensionality reduction.

**Types of Unsupervised Learning**

1. Clustering – Grouping similar data points together
2. Dimensionality Reduction – Reducing the number of features while preserving information
3. Anomaly Detection – Identifying rare or unusual data points

Applications of Unsupervised Learning

✓ Customer segmentation

✓ Anomaly detection in fraud detection

✓ Image compression

✓ Market basket analysis

✓ Recommendation systems

**5.2.1 Customer Segmentation in Retail Industry**

Unsupervised Learning Customer Segmentation in Retail Industry

Customer segmentation is a vital strategy in the retail industry that enables businesses to group customers based on shared characteristics, such as purchasing behavior, demographics, and preferences. Unsupervised learning techniques, such as K-Means clustering, Hierarchical clustering, and DBSCAN, help retailers identify customer segments without prior labels, allowing them to optimize marketing strategies, personalize promotions, and enhance customer satisfaction.

**5.2.1.1 Types of Unsupervised Learning Techniques for Customer Segmentation**

1. K-Means Clustering – Groups customers based on numerical attributes (e.g., spending patterns).
2. Hierarchical Clustering – Forms a tree-like cluster structure for better interpretability.
3. DBSCAN (Density-Based Clustering) – Identifies clusters based on data density, suitable for non-uniform data.

**5.2.1.2 Implementing Customer Segmentation Using K-Means in Python**

Step 1: Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as snsfrom sklearn.cluster
import KMeansfrom sklearn.preprocessing
import StandardScaler
```

Step 2: Load and Explore Retail Customer Data

```
# Sample Data (Replace with actual dataset)
data = {'CustomerID': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
   'Annual Income (k$)': [15, 16, 17, 18, 19, 20, 30, 40, 50, 60],
   'Spending Score (1-100)': [39, 81, 6, 77, 40, 76, 60, 30, 50, 75]}
df = pd.DataFrame(data)
df.drop(columns=['CustomerID'], inplace=True)
df.head()
```

Step 3: Data Preprocessing

```
scaler = StandardScaler()
df_scaled = scaler.fit_transform(df)
```

Step 4: Determine Optimal Clusters Using the Elbow Method

```
wcss = []
for i in range(1, 11):kmeans = KMeans(n_clusters=i, init='k-means++', random_state=42)
kmeans.fit(df_scaled)
wcss.append(kmeans.inertia_)
# Plot Elbow Curve
plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), wcss, marker='o', linestyle='--')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.title('Elbow Method for Optimal K')
plt.show()
```

Step 5: Apply K-Means Clustering

```
# Apply K-Means with optimal clusters (e.g., K=3)
kmeans = KMeans(n_clusters=3, init='k-means++', random_state=42)
df['Cluster'] = kmeans.fit_predict(df_scaled)
# Visualize Clusters
plt.figure(figsize=(8, 5))
sns.scatterplot(x=df['Annual Income (k$)'], y=df['Spending Score (1-100)'],
hue=df['Cluster'], palette='viridis', s=100)
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score')
plt.title('Customer Segmentation')
plt.legend(title="Cluster")
plt.show()
```

**5.2.1.3 Insights and Business Impact**

✓High-Income, High-Spending Customers → VIP customers, target them with loyalty programs

✓Low-Income, Low-Spending Customers → Budget-friendly promotions and discounts

✓Mid-Income, Moderate-Spending Customers → Encourage more spending with personalized offers

**5.2.2 Breast cancer prediction**

**5.2.2.1 Early Detection and Diagnosis of Breast Cancer Using Machine Learning**

Breast cancer is a leading cause of mortality among women worldwide. Early detection significantly improves treatment outcomes and survival rates. Machine learning (ML) and artificial intelligence (AI) play a crucial role in assisting radiologists and healthcare professionals in diagnosing breast cancer with greater accuracy.

**Understanding the Image**

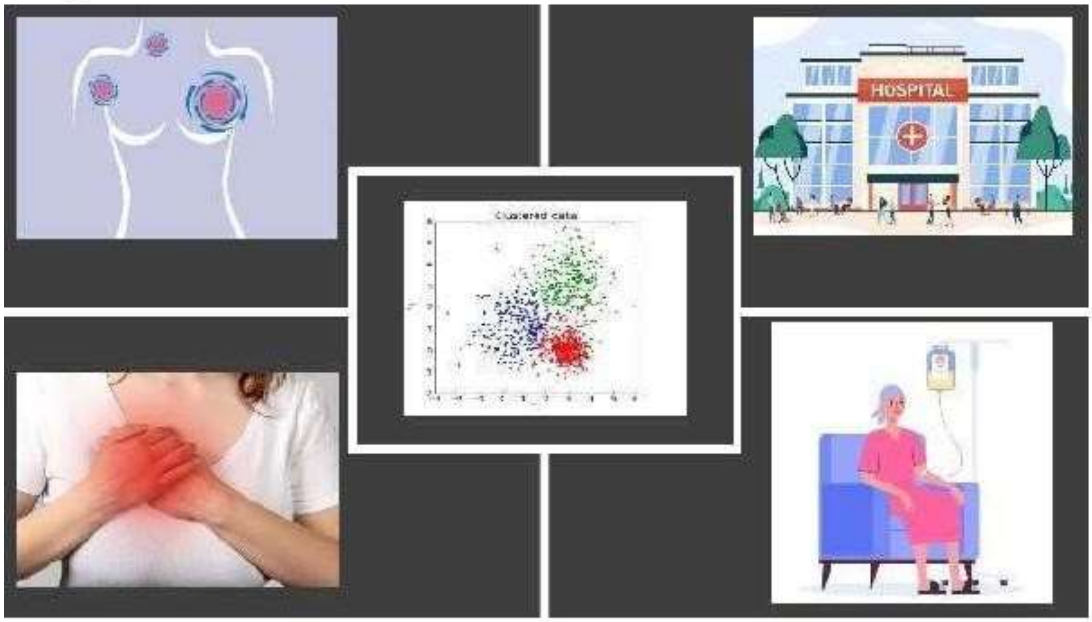



**Figure 5.2 Scenario of early detection and diagnosis of breast cancer by ML** The provided image illustrates a scenario of early detection and diagnosis of breast cancer, featuring:

1. Breast cancer illustration – Representation of tumor detection.
2. Hospital facility – Where patients undergo medical screening.
3. Patient with chest pain – A common symptom of breast cancer.
4. Chemotherapy patient – Undergoing cancer treatment.
5. Clustered data graph – Depicting machine learning-based classification.

These elements emphasize the importance of early detection, clinical diagnosis, and ML-driven analysis.

**5.2.2.2 Role of Machine Learning in Breast Cancer Detection**

1. Data Collection and Preprocessing

- Medical imaging datasets (e.g., Mammograms, Ultrasound, MRI)
- Feature extraction (e.g., tumor size, shape, density)
- Data cleaning and normalization

**2. Machine Learning Techniques for Diagnosis**

- Unsupervised Learning (K-Means Clustering, PCA):
  - Groups breast tumor data into benign and malignant clusters.
  - Helps radiologists analyze patterns in unlabeled data.
- **Supervised Learning (SVM, Random Forest, Deep Learning - CNNs):**
  - Trained on labeled datasets for precise tumor classification.
  - CNN models detect abnormalities in mammograms with high accuracy.

**3. Image Processing Techniques for Early Detection**

- Edge detection (Sobel, Canny operators) for tumor boundary detection
- Histogram equalization for enhancing contrast in medical images

- Feature extraction (texture, shape, intensity) for classification

**5.2.2.3 Clinical Impact of ML-Based Early Detection**

- **Reduces false positives and false negatives** in mammogram screening.
- **Speeds up the diagnostic process**, assisting radiologists.
- **Enhances precision in staging cancer** (benign vs. malignant tumors).
- **Provides personalized treatment plans** through predictive analytics.

**5.2.2.4 Breast Cancer Prediction Using K-Means Clustering**

Breast cancer is one of the most common cancers affecting women globally. Early detection plays a crucial role in improving survival rates. **Machine learning techniques** have proven effective in assisting medical professionals by automating the detection of cancerous tumors. This study focuses on using **unsupervised learning**, specifically **K-Means Clustering**, to classify breast cancer tumors as **benign or malignant** based on extracted features.

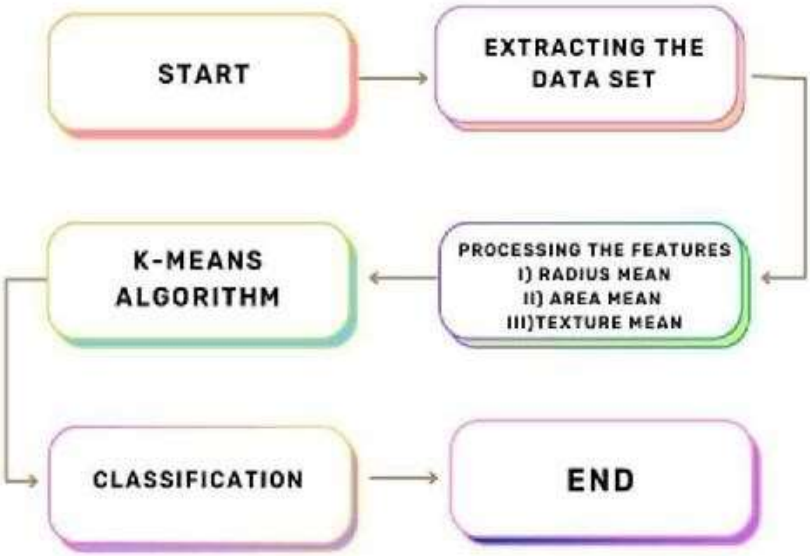



**Figure 5.3 Breast Cancer Prediction using Unsupervised Learning**

Workflow for Breast Cancer Prediction Using K-Means Clustering

**1. Start**

The process begins by defining the objective—predicting breast cancer using clustering techniques.

**2. Extracting the Dataset**

A well-known dataset such as the **Wisconsin Breast Cancer Dataset (WBCD)** is used. The dataset consists of various features extracted from cell nuclei of breast tissue samples.

**3. Processing the Features**

Feature selection and preprocessing are performed to enhance accuracy. Key features used in this model include:

- **Radius Mean** – Average distance from the center to points on the perimeter.
- **Area Mean** – Average area of the tumor cells.
- **Texture Mean** – Standard deviation of gray-scale values.

Data is normalized using **MinMaxScaler** to ensure that all features are within a similar range.

**4. Applying K-Means Algorithm**

- **K-Means Clustering** groups data points into **two clusters** (Benign and Malignant).
- The **Elbow Method** is used to determine the optimal number of clusters.
- The algorithm minimizes intra-cluster distances, ensuring that similar data points are grouped together.

**5. Classification**

Once the clustering is complete, each tumor is classified into one of the two groups:

- **Cluster 1:** Benign Tumors
- **Cluster 2:** Malignant Tumors

The classification is validated by comparing cluster labels with actual diagnostic results.

**6. End**

The final classification results are analyzed to determine model effectiveness. If necessary, further optimizations such as feature selection or dimensionality reduction (PCA) can be applied.

**5.2.2.5 Advantages of Using K-Means for Breast Cancer Prediction**

- **Unsupervised Learning** – Does not require labeled data for training.
- **Fast and Scalable** – Efficient for large datasets.
- **Easy Interpretation** – Simple clustering mechanism to differentiate benign and malignant tumors.
- **Feature Extraction Based on Medical Parameters** – Uses medically relevant features for classification.

**5.3 Dynamic Plots:**

**5.3.1 Visualizing Piketty's Capital**

Thomas Piketty's Capital in the Twenty-First Century (2013) provides an in-depth analysis of wealth inequality across time, supported by extensive economic data. While Piketty's research is largely quantitative, traditional static plots can limit our ability to fully engage with and explore the nuances of wealth distribution over centuries. Dynamic visualizations—such as interactive graphs, animated time series, and user-controlled dashboards—offer a powerful way to make Piketty's findings more accessible and engaging.

**Methods for Creating Dynamic Plots**

1. Time-Series Animations

Animated line charts can illustrate the evolution of wealth and income inequality over time. Tools like Plotly, D3.js, and Matplotlib (with animation modules) allow for smooth

transitions between different years, making it easier to see trends and shifts in inequality.

2. **Interactive Scatter and Bubble Charts**

These visualizations can represent wealth accumulation across different countries, showing relationships between income distribution, economic policies, and capital accumulation. Users can interact with data points to explore specific country trends or zoom in on particular time periods.

3. **Geospatial Maps of Wealth Distribution**

Using tools like Leaflet.js or Tableau, we can create maps showing regional variations in wealth inequality. Heatmaps can illustrate which areas have experienced the most significant shifts in capital accumulation.

4. **Dynamic Dashboards**

Platforms like Tableau, Power BI, and Shiny (R) allow for the creation of dashboards where users can filter data by region, time period, or demographic group. These dashboards provide real-time data exploration capabilities.

Example: Visualizing Capital-to-Income Ratios

A core argument in Piketty's book is that the ratio of capital to income has risen dramatically since the 19th century. A dynamic visualization can effectively depict:

- The increase in capital ownership by the top 1%
- The shifting role of labor vs. capital income
- The impact of taxation policies on wealth inequality
- Using Python and Plotly, we can create an animated line chart that highlights these trends over time, providing users with an intuitive understanding of economic disparities.

**Real-Time Example: Wealth Inequality in the United States**

A real-time example of dynamic visualization in action is the use of the World Inequality Database (WID) to track wealth inequality in the United States. An interactive dashboard can show:

- The share of total U.S. wealth held by the top 1% over time.
- How income distribution has shifted since the 1970s with policy changes.
- Real-time updates when new tax policies are introduced and their impact on capital accumulation.

## 5.4 FRAUD DETECTION Via GRAPH ANALYSIS:

Fraud detection graph analysis is an approach leveraging the power of graph theory and analytics to safeguard digital assets against such threats. By mapping transactional data and user interactions in the form of nodes and edges, graph analysis uncovers hidden patterns and anomalies indicative of fraudulent activities. This post discusses how fraud detection graph analysis plays a pivotal role in protecting digital assets, ensuring their integrity, and maintaining user trust in an increasingly digital world.

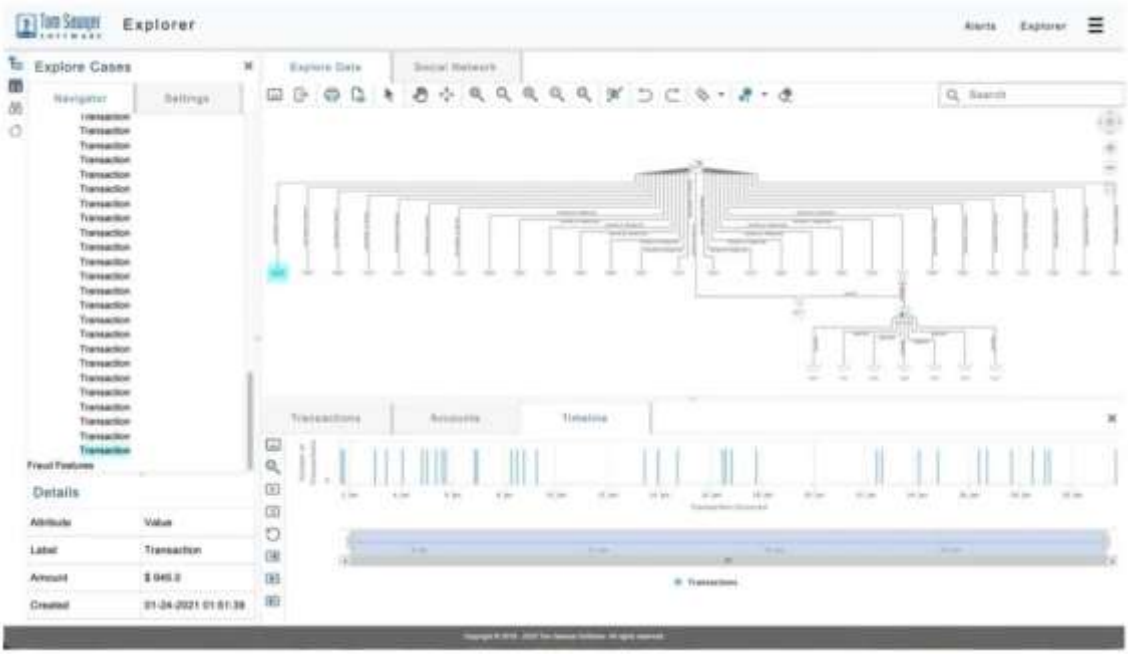



**Figure 5.3 A fraud analysis application showing a graph and chart of financial transactions.**

**Understanding Digital Assets and the Need for Protection:**

Digital assets encompass a wide range of online valuables, including digital currencies, intellectual property, digital rights, and more. The allure of these assets lies not just in their novelty but in their potential to redefine ownership, investment, and value exchange in the digital ecosystem. However, their digital nature makes them prime targets for cybercriminals. Fraudulent activities ranging from phishing scams to sophisticated hacking attempts pose significant risks, threatening the security and trust in digital platforms.

**Fundamentals of Fraud Detection Graph Analysis:**

At its core, fraud detection graph analysis is a method that utilizes graph theory to model and analyze relationships and interactions between entities within a network. This approach is particularly effective in the context of digital transactions and interactions, where entities (such as users, accounts, and transactions) and their connections can be represented as nodes and edges in a graph.

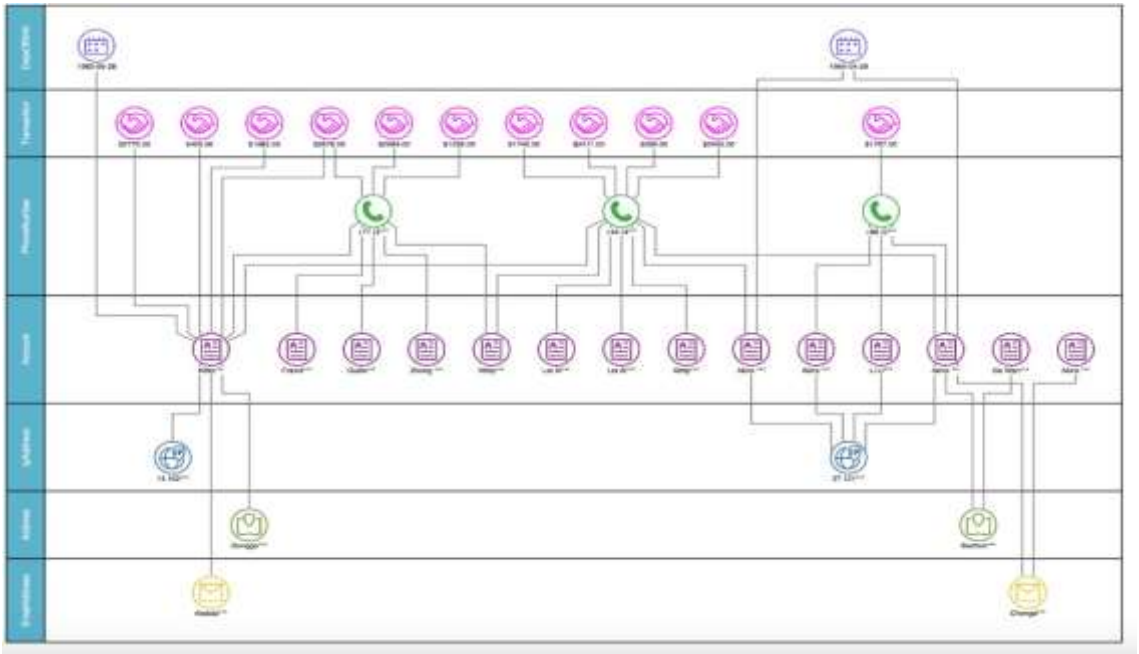


**Figure 5.4 A graph of financial transactions and related accounts, phone numbers, and more**

**Key concepts in fraud detection graph analysis include:**

Nodes: Represent entities such as accounts, users, or transactions.

Edges: Signify the relationships or interactions between entities, such as money transfers or communication links.

Patterns and Anomalies: Recognizable formations or irregularities in the graph that indicate potential fraud.

Understanding these concepts is crucial for developing effective fraud detection strategies that can adapt to the evolving tactics of cybercriminals.

**Methodologies in Fraud Detection Graph Analysis:**

The methodologies employed in fraud detection graph analysis are diverse, each tailored to uncover and combat specific types of fraud. By leveraging the interconnected nature of transactional and social data, these methods provide a holistic view of user behavior and interactions, making it possible to detect anomalies that would otherwise go unnoticed.

**Machine Learning Models and Algorithms:**

Machine learning plays a crucial role in modern fraud detection strategies. Algorithms capable of pattern recognition are trained on vast datasets to identify the subtle signs of fraudulent activity within a network. Supervised learning models, for example, learn from historical fraud cases to recognize similar patterns in new transactions. On the other hand, unsupervised learning models detect outliers or anomalies in data without prior labeling, making them ideal for identifying novel fraud tactics.

**Pattern Recognition and Anomaly Detection Techniques:**

Graph analysis relies heavily on pattern recognition to identify common fraud schemes, such as unusual transaction volumes or abnormal account connections. Anomaly

detection techniques are employed to spot deviations from normal behavior, flagging potential fraud for further investigation. These techniques can be as simple as threshold-based alerts or as complex as clustering algorithms that group similar entities and highlight outliers.

**Network Analysis and Community Detection:**

Network analysis provides insights into the overall structure of the graph, identifying key players, central nodes, and vulnerable points within the network. Clustering algorithms segment the graph into groups of closely related nodes, which can be particularly useful in identifying fraud rings or collusion among users.

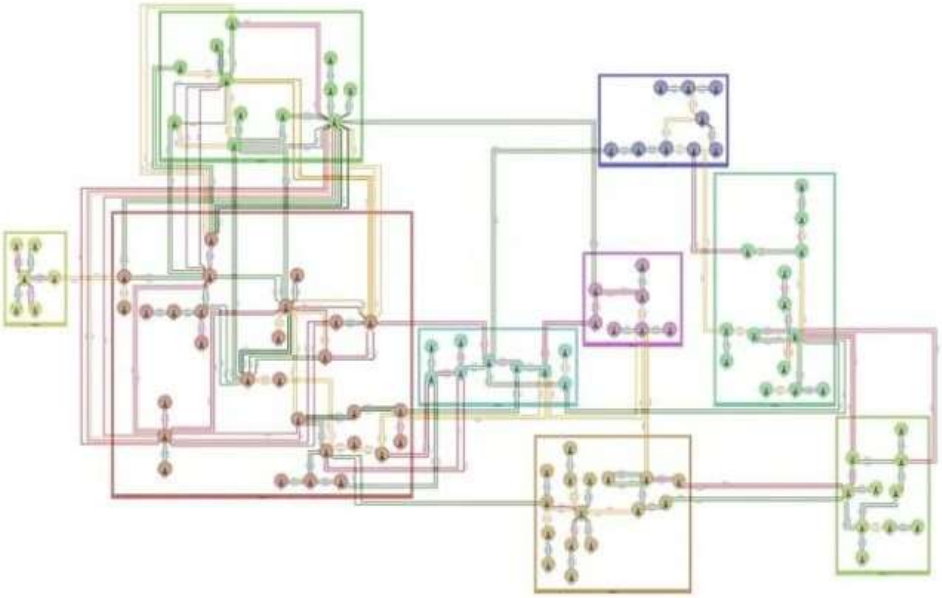


**Figure 5.5 A graph showing clusters of related elements.**

**Enhancing Fraud Detection Graph Analysis with Advanced Software Solutions:**

Advanced software solutions that are equipped with sophisticated graph visualization and analysis capabilities like Tom Sawyer Perspectives, are increasingly crucial when it comes to fraud detection. These tools can fundamentally transform the approach organizations take toward detecting and analyzing fraudulent activities, offering more dynamic, insightful, and efficient methodologies.

Below, we explore how advanced software solutions can aid in enhancing Fraud Detection

**Visualization of Complex Relationships:**

Dynamic Graph Visualization: intuitive and interactive visualization of complex networks of transactions and user relationships aids in identifying unusual patterns, such as dense clusters or strange connections, which can be indicative of fraudulent behavior.

Real-time Interaction: real-time interaction provides analysts with the visual data in real time, enabling a deeper investigation into suspicious patterns or anomalies to understand the context of potentially fraudulent activities better.

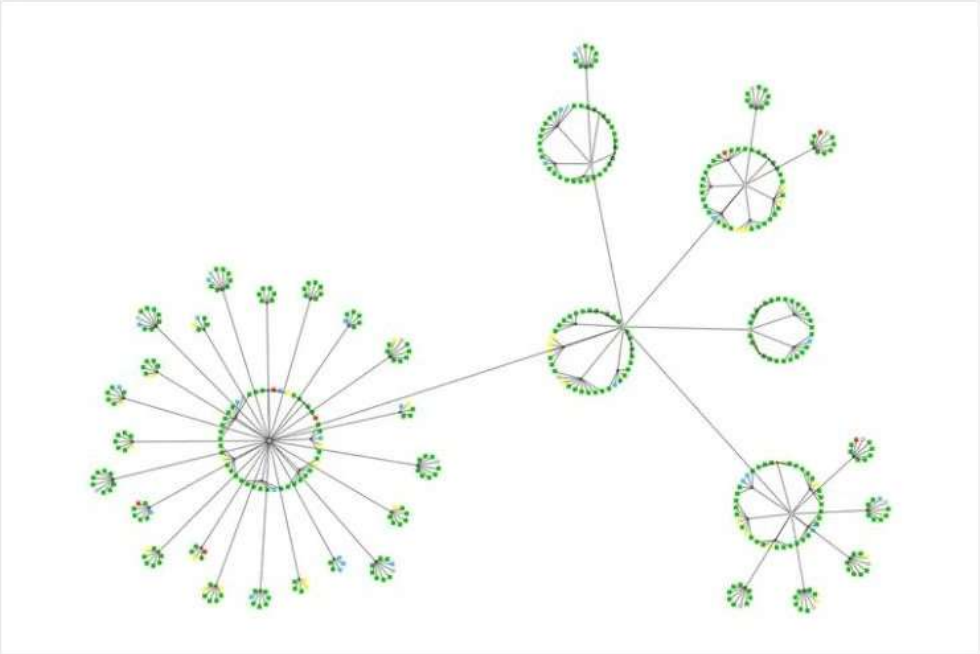


**Figure 5.6 A graph visualization of a network of transactions produced with Tom Sawyer Perspectives.**

**Advanced Analytics for Anomaly Detection:**

Pattern Recognition: Advanced algorithms assist in recognizing patterns and behaviors typical of fraud within graph data. By identifying standard patterns, the software can spotlight deviations that suggest fraudulent activities.

Anomaly Detection: Sophisticated analytics tools can automatically detect anomalies within large and complex datasets. This is vital for identifying new and emerging fraud tactics that haven't been previously encountered.

**Scalability and Performance:**

Handling Big Data: Solutions designed to process and analyze large volumes of data efficiently are essential for organizations dealing with vast amounts of transactional information. Scalability ensures that fraud detection efforts can keep pace with the growth of digital transactions.

Performance Optimization: optimized performance of graph analysis operations ensure that fraud detection processes are accurate and timely. This is crucial for the real-time detection and mitigation of fraudulent activities.

**Integration and Customization:**

Seamless Integration: seamless integration with existing IT infrastructure and data sources, including transaction monitoring systems, customer databases, and external intelligence feeds is critical for a holistic fraud detection approach.

Customization: extensive customization options allows organizations to tailor the analysis

and visualization tools to their specific fraud detection needs. This flexibility ensures the software remains effective as fraud strategies evolve and new types of digital assets emerge.

**Collaboration and Information Sharing:**

analyzing fraud patterns fosters collaboration among different teams and departments. This collaborative approach is essential for a comprehensive fraud detection strategy.

Information Sharing: facilitating the sharing of insights and data within the organization and with external partners, enhances the collective ability to combat fraud.

**Challenges in Implementing Fraud Detection Graph Analysis:**

Despite its effectiveness, fraud detection graph analysis faces several challenges:

- Data Privacy and Security Concerns

The use of sensitive personal and transactional data raises privacy and security concerns. Ensuring that data is handled in compliance with regulations like GDPR and CCPA is paramount.

- The Complexity of Real-Time Analysis

Fraud detection often requires analyzing data in real-time to prevent fraud before it occurs. Handling the volume and velocity of data in large-scale networks poses technical and computational challenges.

- Evolving Fraud Tactics

As detection methods become more sophisticated, so too do the tactics of fraudsters. Keeping detection models up-to-date and capable of identifying new fraud schemes is a constant challenge.

**The Impact of Fraud Detection Graph Analysis on Digital Asset Security:**

Graph analysis significantly enhances the security framework around digital assets. Its ability to quickly identify and isolate fraudulent activities not only minimizes financial losses but also helps in maintaining the integrity and trustworthiness of digital platforms.

Real-world examples are common where fraud detection graph analysis has been instrumental in thwarting major fraud attempts, showcasing its value in protecting digital assets across industries.

**Future Directions and Innovations in Fraud Detection Graph Analysis:**

The future of fraud detection graph analysis involves ongoing advancements in technology and provides new opportunities for innovation.

Key areas of focus include:

Integration of Artificial Intelligence (AI) and Machine Learning:

Enhanced AI capabilities will allow for more sophisticated analysis of complex networks, improving the detection of fraudulent activities.

Cross-industry collaboration:

Sharing insights and data across industries can help in creating a more comprehensive approach to fraud detection, making it harder for fraudsters to exploit system vulnerabilities.

Adoption of Blockchain Technology:

Blockchain's inherent transparency and traceability features could complement graph analysis, providing an additional layer of security to digital transactions.

The continuous evolution of fraud detection strategies, fueled by advancements in graph analysis, is essential for staying ahead of fraudsters in the digital age.

**How a typical graph data model in graph analytics works (for fraud detection)**

When used in fraud detection, a graph data model acts as a virtual detective that finds evidence of fraud. It links individuals, transactions, and financial institutions involved in crime. Then, it compares these connections with previously identified fraud patterns to find similarities or anomalies.

Let's consider a money laundering pattern as an example. One person wants to transfer large amounts of money without being detected by banking systems. The recipient also wants to get the funds without detection. The sender usually divides the funds into smaller portions. They also open several accounts with different banks to set up the transfers. Similarly, the recipient prepares several accounts to appear as though they are receiving payments from different clients.

Graph analytics can also determine if such fraudulent activities are expanding rapidly, e.g., new criminal rings emerging in unexpected locations. It reveals shared data points, e.g., the same devices or email addresses accessing multiple banks or websites.

In terms of e-commerce fraud, graph analytics can alert businesses about frequent chargebacks or transaction reversals from specific customers. These patterns indicate that the customers intend to keep purchased goods without paying for them, which is a criminal offense.

**The key approaches that graph analytics employ to identify fraud**

There are four approaches that data scientists use to identify fraud using graph analytics, as we'll explain below:

**1. Querying**

Graph databases require a graph query language to extract and modify data. An example of a query language is NebulaGraph'snGQL. These are specialized programming languages that enable graph analysts to ask questions and derive valuable answers from their databases, such as:

- How many individuals are transacting with a known fraudster?
- How far apart (in miles) are transactions of $10,000?
- How many of those transactions were conducted at 3AM in the morning?

Using such questions, graph analysts can identify any data points of interest and determine how they are related. The queries can be as detailed as possible, e.g., differences between a shipping address and a billing address on specific orders. They can also analyze the contents of a checkout basket to find anomalies in purchasing patterns. These patterns may be difficult to see without graph technology.

**2. Machine learning**

The specific form of machine learning that is utilized in this context is what we call supervised machine learning. So what is supervised machine learning? This is a process that classifies the data used to train a graph analytics model. This means assigning labels to the training datasets and allowing the machine learning model to learn and adapt.

A common example of supervised machine learning is image recognition. This is where data scientists label the training images, e.g., photos of a bus. Then, the model is shown photos of all kinds of vehicles (planes, cars, trains, etc.) and learns how to identify a bus.
In fraud detection, graph analysts can label training data as fraud or non-fraud. For example, a fraudulent transaction can be tagged and fed into the ML model. A tag can be a particular tax haven location which signals fraud. Using this tag, the model can analyze all transaction data and pick out all matching incidents.
This approach is particularly helpful to track money laundering activities. Data scientists can tag the training datasets with parameters such as:

- User identities
- Frequency of purchases
- Transfer amounts
- Internet Protocol (IP) addresses, etc.

**3. Graph Algorithms**

A graph algorithm is a mathematical tool used to detect patterns in graph databases automatically. It can identify specific qualities and connections between data points in ways that graph queries might miss. Some examples of graph algorithms include:
Centrality algorithms: These detect the most influential individuals or data points in a network, e.g., accounts with the most frequent chargeback activities on a payment platform.
Community detection algorithms: These identify data points that have the most connections, e.g., transaction patterns that show signs of a fraud ring.
Similarity algorithms: These identify data points that have matching qualities, e.g., clusters of accounts that are linked to the same email addresses.
Link prediction algorithms: These predict connections between two data points, e.g.,how likely a suspicious account activity will turn out to be fraud, based on established patterns.
There are numerous other algorithms that apply to graph analytics. Data scientists can customize these algorithms to match specific use cases and derive detailed insights. This means that they can keep up with changing fraud tactics before they impact an organization.

**4. Real-time analytics**

Some types of fraud are extremely time-sensitive, such as phone-based scams targeting senior citizens. With graph analytics, such types of fraud can be detected in real time. For example, phone communications can be scanned in real-time based on the scam callers' patterns.
The calls are automatically redirected to a fraud detection team. The call recipient can also see an alert on their devices saying that the caller is a potential scammer.

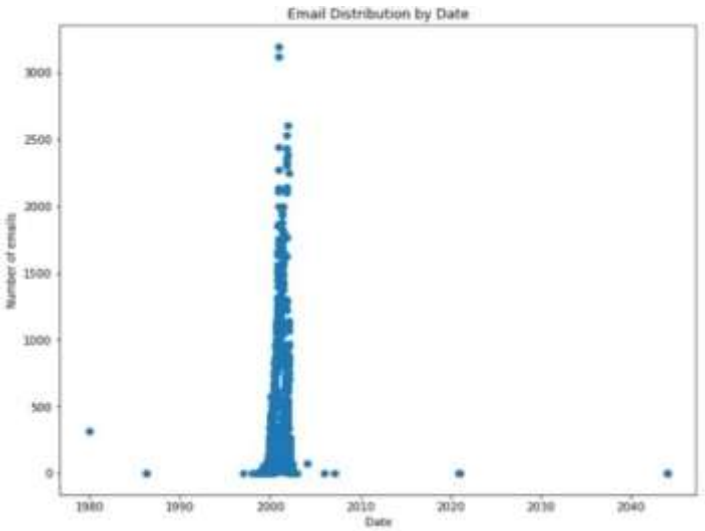
**5.5 FRAUD DETECTION IN THE ENRON EMAILS**

So we all remember, or have read about, the massive risk management failure at Enron. It had all the classic red flags, but blew up to such a massive scale because somehow the Board of Directors remained ignorant of what was really going on. They instead relied on Lay and then Skilling and their supposed ‘expertise’ as well as the opinions of the now dismantled Arthur Anderson. Could this have been avoided? Certainly. As I mentioned

many of the red flags from other risk management failures were there. It's easy to ignore qualitative issues — they're explainable and you can talk your way around ignoring them. But quantitative…that is a much harder reality to ignore. Numbers don't really lie the same way people do.

I set out to build a machine learning model that would allow senior managers or board members to see people who may warrant additional scrutiny. I used the email traffic to classify the text and with fairly accurate results could put people on a 'investigate further' list. Perhaps had the board been required to review these findings, the scale of the scandal would not have caused nearly 20 years of repercussions for both the energy and financial markets.

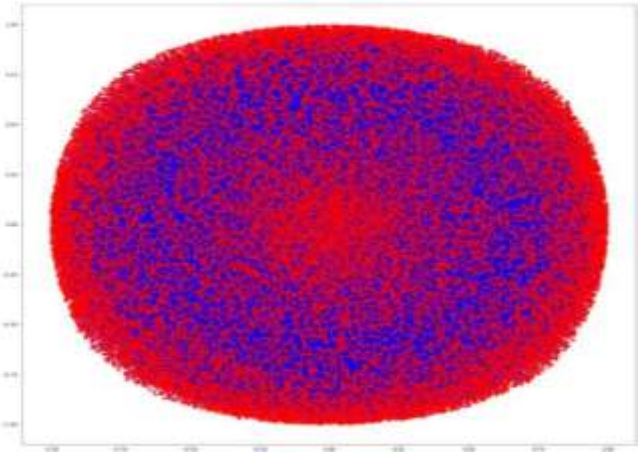
I wanted to see if I would be able to review the dates of the scandal by plotting the number of emails sent in any one day. This took some serious scrubbing but I was able to arrive at the following. In the real world, where you have a full body of emails instead of those just pertaining to the scandal, one could fathom even an escalating number of emails could be worth investigating, particularly if it did not flow with previous patterns.




I also wanted to see what a network map would look like of the dataset. I used NXVIZ which is a visualization toolkit for Network X. I used three different types of mapping to show the interplay between individuals. Certain people at the centers of these maps could be seen as high-risk given the scope of their interactions. I included the network map before but in my code you'll also find an arc plot and a circos plot.

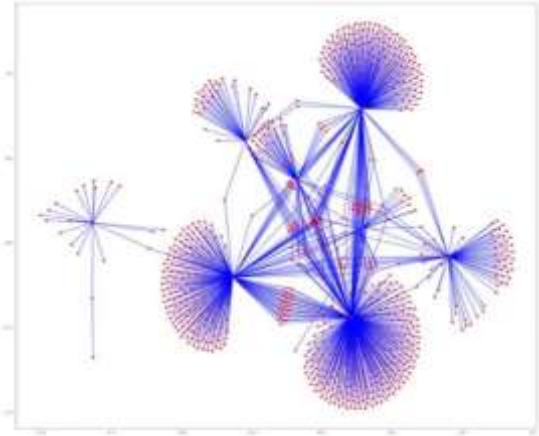
*#drawing the network map*
*plt.figure(figsize=(20,20))*

```
position = nx.spring_layout(G, k=.1)
nx.draw_networkx(G, position, node_size=25, node_color='red', with_labels=False, edge_color='blue')
plt.show()
```



Obviously, this is ridiculous and is what happens when you look at something so large. Here's a slice of the top 1,000 senders and their interactions in the form of recipients. These are sorted based on node weight; those with higher weights should be more 'important' to the investigation we are conducting.

```
top_edges = sorted(G.edges(data=True), key=lambda t: t[2].get('weight', 1))
top_edges = top_edges[:1000]
```



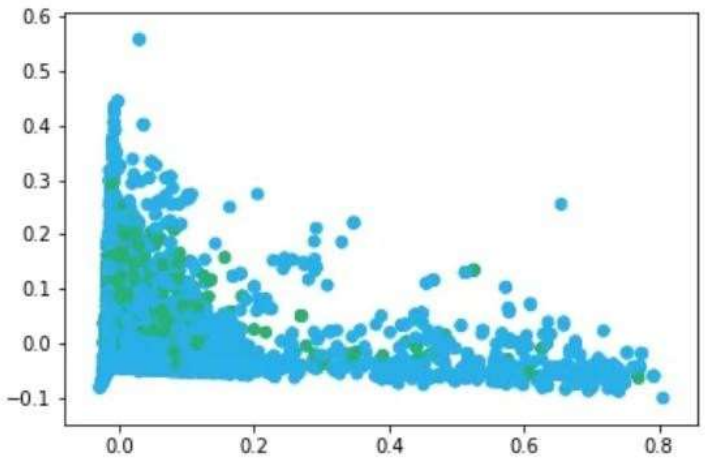
Given the size of the data set, I started some additional exploring using emails from only one individual, Kay Mann. We wanted to look at word frequency but tokenizing the text is processor intensive and I chose to test my method on a smaller dataset before using the whole thing. This allowed us to use the frequency distribution from NLTK to get top words and their use counts. With such a simple metric, displaying the top words as a wordcloud was a natural step. The library WordCloud makes this incredibly easy and

allows us to use Matplot and seaborn for formatting and display options.

```
top_keys =list(top_words.keys())
most_frequent={}
for key in top_keys:
new = key.lower()
most_frequent[new]=top_words[key]
#creating WordCloud object
comment_words2 = str(most_frequent)
wordcloud2 = WordCloud(width = 800, height = 800,
background_color ='white',
stopwords = stopwords_list,
min_font_size = 10).generate(comment_words2)
# plot the WordCloud image
plt.figure(figsize = (15,15), facecolor = None)
plt.imshow(wordcloud2)
plt.axis("off")
plt.tight_layout(pad = 0)
plt.show()
```

For those interested, some of the people included in this word cloud had very fascinating parts to play in the Enron scandal. I included a few in my code descriptions available on my github repo.

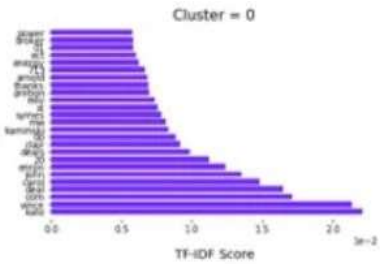
Now that we have some of the fun stuff down, including some statistics I chose not to include here, the next step in the analysis is, well, the analysis. What we are going to do is take an unsupervised machine learning model and cluster the dataset to create the labels that will be used for a supervised classification model. This should give a 'to-be-investigated-further' list. Which in terms of resources, allows fraud analysts to use their time for additional tasks or for the business to allocate those resources elsewhere.

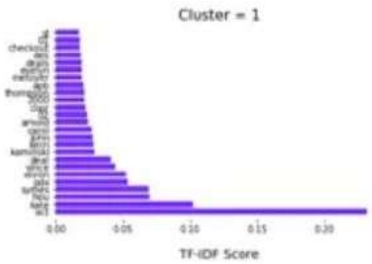



Two — Cluster Kmeans Analysis

Now with two clusters, each saved in its own dataframe we can make the below plot. It

is the weighted top 25 words for each of the two clusters we used. While the plot isn't perfectly segregated, if we were to use the whole dataset (somehow!) we could extrapolate the accuracy here would be higher. Regardless, I was satisfied enough to use these as the 'true' labels for the supervised model.




With available labels, the attention turns to a supervised model. I chose K-nearest neighbours for its classification and lazy computing properties. My hope was to create a model that would output the appropriate cluster so in turn an analyst would be able to know who might need additional scrutiny. We implemented the KNN model with the nearest 3 neighbours and came up with a greater than 90% accuracy score on the test data. That sounds pretty good to me.

If I had an infinite amount of time to continue plugging away at this we could do some additional things. First, we could find the necessary resources to be able to run the algorithm on the full dataset. Second, we could break our clustering and classifying into two steps: find the people of interest and then going on to look at the language they used in their emails to see if that could be a red flag for fraud in itself. Finally, we could try a number of different clusters or neighbors to see if we could further improve model performance. Since my time is not infinite, these will have to be shelved ideas for a rainy day.

I know for a fact, as I worked in corporate finance and risk management for over a decade,

that these sorts of algorithms are on the cutting edge of Fraud detection and financial forensics. Hopefully, this application carries some additional insight for the community looking at how to prevent these sorts of risk management failures in the future.

**5.6 ENRON EMAIL DATASET:**

The Enron Email Dataset consists of emails from employees of the Enron Corporation, a major energy company that went bankrupt in 2001 due to corporate fraud. The dataset was released by the Federal Energy Regulatory Commission (FERC) and has since become a valuable resource for researchers.

**Key Features of the Dataset:**

Size: Around 500,000 emails from 150 employees.

Format: Stored in plain text files, structured as a mailbox dataset.

Usage:

- Sentiment analysis
- Fraud detection
- Spam filtering
- Social network analysis
- Email classification

**Structure of the Dataset**

The dataset is organized into folders representing employees' mailboxes, with subfolders like:

inbox/

sent/

deleted_items/

drafts/

discussion_threads/

**Applications in Data Science**

Natural Language Processing (NLP)

Sentiment Analysis: Extracting positive or negative sentiment from emails.

Named Entity Recognition (NER): Identifying people, organizations, and financial terms in emails.

Machine Learning

Spam Detection: Training classifiers to differentiate spam from legitimate emails.

Topic Modeling: Using techniques like LDA (Latent Dirichlet Allocation) to discover discussion topics.

Social Network Analysis

- Mapping communication networks among employees.
- Detecting communities and key influencers.

Fraud Detection

- Identifying unusual communication patterns.
- Extracting insider trading information.

**1. Email Network Graph**

(Example visualization of employee email exchanges in a network format)

This type of graph shows who communicated with whom, highlighting key people in the network.
An email network graph visually represents email communication patterns, with nodes representing email addresses and edges signifying email exchanges, helping analyze communication flows and identify key individuals or groups.
What is an Email Network Graph?
Nodes: Represent email addresses or individuals within an organization.
Edges: Represent email exchanges between those addresses (or individuals).
Purpose: To visualize and analyze communication patterns within a network, such as an organization or research institution.
Benefits of Email Network Analysis:

- Understanding Communication Dynamics: Helps identify information gatekeepers, influential individuals, and communication bottlenecks.
- Identifying Communities: Reveals how employees or individuals are clustered and communicate within an organization.
- Assessing Communication Effectiveness: Can help determine if email is an effective communication method and identify areas for improvement.
- Tracing Information Flow: Shows how information propagates through an organization or network.
- Analyzing temporal dynamics: Can track how communication changes over time

Examples of Email Network Datasets:
Enron Email Network:
A well-known dataset containing email communication data from Enron employees.
email-Eu-core:
A dataset representing email communication within a large European research institution.
email-dnc:
A dataset representing email communication within a Democratic National Committee.
Tools for Creating and Analyzing Email Network Graphs:
Network Data Repository: A website that provides various network datasets, including email datasets.
Graph-tool: A Python library for creating and analyzing graphs**.**
Microsoft Graph: A Microsoft API that allows developers to access and manage data in Microsoft products, including email data.
SNAP (Stanford Network Analysis Package): A software package for network analysis.
Kaggle: A platform for data science and machine learning that hosts various datasets, including email datasets.
Netzschleuder: A website for exploring and visualizing network data.
Highcharts: A JavaScript charting library that supports network graph visualization.
GitHub: A platform for hosting code repositories, including tools for email network analysis.

**2. Word Cloud from Enron Emails**

(Word cloud representation of frequently occurring words in emails)
This helps identify major themes and topics discussed.

A word cloud generated from the Enron email dataset, a collection of emails from Enron Corporation employees, can visually represent the frequency of words used in the emails, highlighting keywords and patterns in their communication.
Here's a more detailed breakdown:
What is the Enron Email Dataset?
The Enron email dataset is a collection of emails sent and received by employees of the Enron Corporation, a now-defunct energy trading company.
The corpus was generated from Enron email servers by the Federal Energy Regulatory Commission (FERC) during its investigation following the company's collapse in 2001.
The dataset contains over 600,000 emails from 158 Enron employees.
What is a Word Cloud?
A word cloud is a visual representation of text data, where the size of each word corresponds to its frequency in the text.
It's a way to quickly identify the most common or important words in a dataset.
How can a word cloud be used with the Enron Email Dataset?
Researchers and analysts can use word clouds to explore the Enron email dataset and gain insights into the company's communication patterns, identify key topics, and potentially uncover patterns related to the company's downfall.
For example, a word cloud can highlight the frequency of certain keywords like "Lay", "Skilling", "energy", "market", "accounting", or "fraud".
A word cloud can also be used to analyze the emails of specific individuals, such as Ken Lay or Jeff Skilling, to understand their communication style and the topics they discussed.
**Example:**
One can create a word cloud using the words in Ken Lay's emails and even use a picture of Ken Lay as a mask to create a word cloud in the shape of his face.
Where to find the dataset:
The CMU CALO Project dataset is available for download at http://www.cs.cmu.edu/~enron/.
**3. Email Timeline Analysis**

(Chart displaying email activity over time, showing peak periods of communication)
Such a visualization can reveal critical moments leading up to the scandal.
Email timeline analysis involves examining email communications chronologically, often to understand events, identify patterns, or trace the flow of information, which can be useful in various contexts like incident investigation or project management.
**Here's a more detailed explanation:**
What is Email Timeline Analysis?
Chronological Sequencing:
It's the process of arranging emails in the order they were sent or received, creating a timeline of communication.
**Purpose:**
**This analysis helps to**:

- Understand the sequence of events: By seeing the emails in order, you can

reconstruct the flow of information and identify key decisions or actions.

- Identify trends or patterns: Analyzing the timeline can reveal recurring themes, communication styles, or delays in responses.
- Pinpoint responsibilities: By tracing the emails, you can determine who was involved in a particular communication and when.
- Investigate incidents: In situations like cybersecurity breaches or workplace disputes, email timelines can help to reconstruct events and identify potential culprits.

**Tools:**

You can use email clients, dedicated timeline analysis software, or even spreadsheets to create and analyze email timelines.

**Examples:**

Project Management: Tracking email exchanges related to a project to see how progress is made and identify potential roadblocks.

Incident Investigation: Analyzing emails related to a security incident to understand the timeline of events and identify the source of the problem.

Legal Cases: Examining email exchanges to build a case or understand the context of a situation.

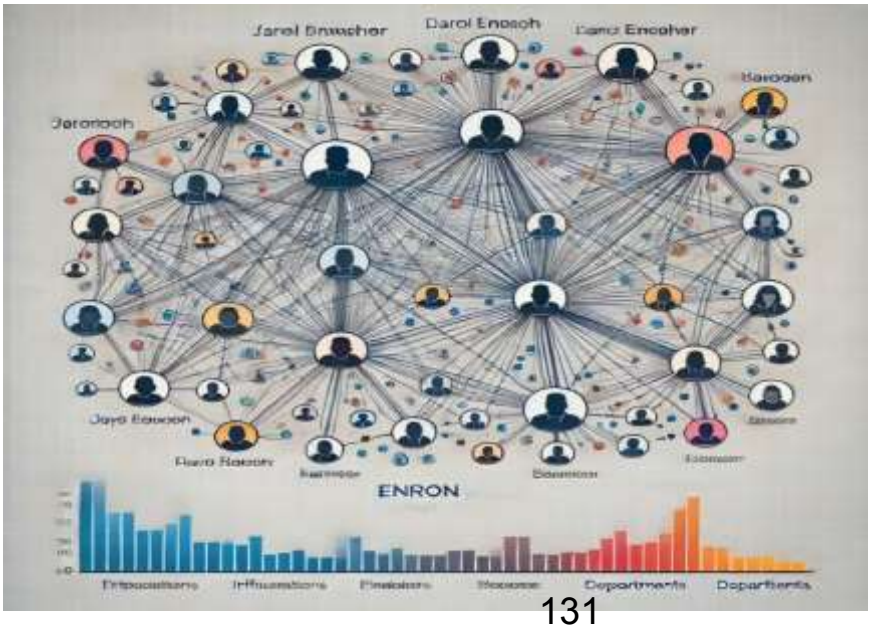
**How to Conduct Email Timeline Analysis:**

Gather Emails: Collect all relevant emails related to the specific event or project. Organize Emails: Sort the emails by date and time to create a chronological sequence.

Identify Key Information: Highlight important dates, senders, recipients, subjects, and content.

Analyze the Timeline: Look for patterns, trends, and connections between the emails.

Draw Conclusions: Based on the analysis, formulate conclusions and identify potential actions or next steps.



Here's the network graph visualization of the Enron Email Dataset. It highlights key influencers with larger nodes and differentiates departments using color coding. Let me know if you need any modifications or additional visualizations!

## 5.7 CHARACTER RECOGNITION FROM IMAGES

Character recognition from images, or Optical Character Recognition (OCR), is a data science technique that converts images of text (printed or handwritten) into machine-readable text, enabling digital access and manipulation of scanned documents or images containing text.

**What is OCR?**

**OCR is a technology that uses computer vision and machine learning to recognize and extract text from images.**

Optical Character Recognition (OCR) is the process that converts an image of text into a machine-readable text format. For example, if you scan a form or a receipt, your computer saves the scan as an image file. You cannot use a text editor to edit, search, or count the words in the image file.

**Purpose:**

It aims to convert images of text (like scanned documents, photos of text, or even text superimposed on an image) into a digital, editable, and searchable format.

**How it works:**

OCR systems analyze the image to identify characters, words, and sentences, then translate them into a text format.

**Applications:**

- Digitizing documents: Converting physical documents into digital archives.
- Data extraction: Extracting data from forms, invoices, or other documents.
- Text-based analysis: Enabling text analysis, translation, or search within images.
- Accessibility: Assisting visually impaired individuals by converting images of text into audio format.

**The history of OCR**

- In 1974, Ray Kurzweil started Kurzweil Computer Products, Inc., whose omni-font OCR product might recognize text printed in virtually any font. He decided that the best application of this technology would be a machine learning (ML) device for the vision-impaired, so he created a reading machine that might read text aloud in a text-to-speech format. In 1980, Kurzweil sold his company to Xerox, which was interested in further commercializing paper-to-computer text conversion.
- OCR technology became popular in the early 1990s while digitizing historical newspapers. Since then, the technology has undergone several improvements. Today, products can deliver near-to-perfect OCR accuracy. Advanced methods can automate complex document-processing workflows.
- Before OCR technology became available, the only option to digitally format documents was manually reentering the text. Not only is the redundant input time-consuming, but it also comes with inevitable inaccuracies and typing errors. Today, OCR services are widely available to the public. For example, Google

Cloud Vision OCR can be used to scan and store documents on your smartphone.

**OCR in Data Science**

Machine Learning:

OCR systems often rely on machine learning algorithms to recognize patterns in images and accurately identify characters.

Data Preprocessing:

Before applying OCR, image preprocessing techniques (like noise reduction, binarization, and skew correction) are used to improve the accuracy of the recognition process.

Feature Extraction:

OCR algorithms use feature extraction techniques to identify relevant features in the image, such as edges, corners, and textures, to aid in character recognition.

Classification:

Once features are extracted, machine learning models classify the features to identify the corresponding characters.

**Tools and Technologies:**

- Tesseract OCR: A popular open-source OCR engine.
- OpenCV: A library for image processing and computer vision tasks.
- Python: A programming language often used for implementing OCR pipelines.
- Cloud Vision API (Google): A cloud-based API for text detection and OCR.
- Azure AI services: Microsoft's Azure AI services offer OCR capabilities.
- IBM Watson: IBM offers OCR solutions as part of its Watson AI services

**Why is OCR important?**

Most business workflows involve receiving information from print media. Paper forms, invoices, scanned legal documents, and printed contracts are all part of business processes. These large volumes of paperwork take a lot of time and space to store and manage. Though paperless document management is the way to go, scanning the document into an image creates challenges. The process requires manual intervention and can be tedious and slow.

**How does OCR work?**

The OCR engine or OCR software works by using the following steps:

Image acquisition

A scanner reads documents and converts them to binary data. The OCR software analyzes the scanned image and classifies the light areas as background and the dark areas as text.

Preprocessing

The OCR software first cleans the image and removes errors to prepare it for reading. These are some of its cleaning techniques:

- Deskewing or tilting the scanned document slightly to fix alignment issues during the scan.
- Despeckling or removing any digital image spots or smoothing the edges of text images.
- Cleaning up boxes and lines in the image.
- Script recognition for multi-language OCR technology

**Text recognition**

The two main types of OCR algorithms or software processes that an OCR software uses for text recognition are called pattern matching and feature extraction.

The dark portions are processed to find alphabetic letters, numeric digits or symbols. This stage typically involves targeting one character, word or block of text at a time. Characters are then identified by using one of two algorithms, either pattern recognition or feature recognition.

**Pattern recognition (or pattern matching):** The OCR program has previously been trained on examples of text in various fonts and formats to recognize characters by comparison to a template in the scanned document or image file. Each unique combination of shape, scale and font is called a glyph. For this to work, the characters must be in a font that the OCR program has already been trained on. Given the number of fonts worldwide and languages that use different characters, such as Arabic, Chinese, English, French, German, Greek, Japanese, Korean or Spanish, training on every combination of font and language would be an enormous system drain.

**Feature recognition (detection or extraction):** This is used when the OCR program is analyzing a font that it has not been trained on. OCR applies rules regarding the features of a specific letter or number to recognize characters in the scanned document. Features include the number of angled lines, line intersections, loops or curves in a character. For example, the capital letter "A" is stored as two diagonal lines that meet with a horizontal line across the middle. When a character is identified, it is converted into an American Standard Code for Information Interchange (ASCII) code that computer systems use to handle further manipulations.

**Pattern matching**

Pattern matching works by isolating a character image, called a glyph, and comparing it with a similarly stored glyph. Pattern recognition works only if the stored glyph has a similar font and scale to the input glyph. This method works well with scanned images of documents that have been typed in a known font.

**Layout recognition:**

A more complete OCR program will also analyze the structure of a document image. It divides the page into elements, such as blocks of text, tables or images. The lines are divided into words and then into characters. After the characters have been singled out, the program compares them with a set of pattern images. After processing all likely matches, the program returns the recognized text.

**Feature extraction:**

Feature extraction breaks down or decomposes the glyphs into features such as lines, closed loops, line direction, and line intersections. It then uses these features to find the best match or the nearest neighbor among its various stored glyphs.

**Postprocessing:**

After analysis, the system converts the extracted text data into a computerized file. Some OCR systems can create annotated PDF files that include both the before and after versions of the scanned document.

**What are the types of OCR?**

Data scientists classify different types of OCR technologies based on their use and

application. The following are a few examples:

**Simple optical character recognition software**

A simple OCR engine works by storing many different font and text image patterns as templates. The OCR software uses pattern-matching algorithms to compare text images, character by character, to its internal database. If the system matches the text word by word, it is called optical word recognition. This solution has limitations because there are virtually unlimited font and handwriting styles, and every single type cannot be captured and stored in the database.

**Intelligent character recognition software**

Modern OCR systems use intelligent character recognition (ICR) technology to read the text in the same way humans do. They use advanced methods that train machines to behave like humans by using machine learning software. A machine learning system called a neural network analyzes the text over many levels, processing the image repeatedly. It looks for different image attributes, such as curves, lines, intersections, and loops, and combines the results of all these different levels of analysis to get the final result. Even though ICR typically processes the images one character at a time, the process is fast, with results obtained in seconds.

**Intelligent word recognition**

Intelligent word recognition systems work on the same principles as ICR, but process whole word images instead of preprocessing the images into characters.

**Intelligent character recognition (ICR)**

ICR brings in the power of AI. By using ML or deep learning, the OCR program learns to read just as humans do: through continual practice and training. A neural network reviews text repeatedly looking for distinctive attributes: the locations of curves, intersections, lines and loops.

**Optical mark recognition**

For identifying checked boxes and other marks, such as bubbles in surveys or a signature on a form, plus logos, symbols and watermarks. All can be identified by matching to stored images, as with simple OCR.

**What are the benefits of OCR?**

The following are major benefits of OCR technology:

Searchable text

Businesses can convert their existing and new documents into a fully searchable knowledge archive. They can also process the text database automatically by using data analytics software for further knowledge processing.

Operational efficiency

You can improve efficiency by using OCR software to automatically integrate document workflows and digital workflows within your business. Here are some examples of what OCR software can do:

- Scan hand-filled forms for automated verification, reviews, editing, and analysis. This saves the time required for manual document processing and data entry.
- Find the required documents by quickly searching for a term in the database so that you don't have to manually sort through files in a box.

- Convert handwritten notes to editable texts and documents.

**Artificial intelligence solutions**

OCR is often part of other artificial intelligence solutions that businesses might implement. For example, it scans and reads number plates and road signs in self-driving cars, detects brand logos in social media posts, or identifies product packaging in advertising images. Such artificial intelligence technology helps businesses make better marketing and operational decisions that reduce expenses and improve the customer experience.

The benefits of employing OCR technology include the ability to:

- Cut costs by reducing or eliminating redundant manual input.
- Streamline workflows with the input of preprinted documents or written forms and speed research with searchable digital data.
- Automate document routing, content processing and preparation for text mining.
- Save the cost of storing yet more paper records.
- Centralize and secure data sets for protection against fires, break-ins and documents lost in the bank vaults.
- Enable greater access to data for visually impaired staff and customers.
- Improve service by giving employees the most up-to-date and accurate information.

**What is OCR used for?**

The following are some common OCR use cases in various industries:

**Banking**

The banking industry uses OCR to process and verify paperwork for loan documents, deposit checks, and other financial transactions. This verification has improved fraud prevention and enhanced transaction security. For example, BlueVine is a financial technology company that provides financing to small and medium-sized businesses. It used Amazon Textract, a cloud-based OCR service, to develop a product for small businesses in the US to quickly access Paycheck Protection Program (PPP) loans as part of the COVID-19 relief stimulus package. Amazon Textract automatically processed and analyzed tens of thousands of PPP forms per day so that BlueVine could help several thousand businesses get funds, saving over 400,000 jobs in the process.

**Healthcare**

The healthcare industry uses OCR to process patient records, including treatments, tests, hospital records, and insurance payments. OCR helps to streamline workflow and reduce manual work at hospitals while keeping records up to date. For example, the nib Group provides health and medical insurance to over 1 million Australians and receives thousands of medical claims per day. Its customers can take photos of their medical invoice and submit them through the nib mobile app. Amazon Textract processes these images automatically so that the company can approve claims much faster.

**Logistics**

Logistics companies use OCR to track package labels, invoices, receipts, and other documents more efficiently. For example, the Foresight Group uses Amazon Textract to automate invoice processing in SAP. Manual entry of these business documents was time-

consuming and error-prone because Foresight employees had to enter the data in multiple accounting systems. With Amazon Textract, Foresight software can read characters more accurately across many different layouts, which increases business efficiency.

**What types of OCR does Google Cloud offer?**

Google Cloud offers two types of OCR: OCR for documents and OCR for images and videos.

While they share a foundational technology, Document AI is a document understanding platform optimized for document processing. Its Custom Extractor is powered by GenAI that processes both generic and domain-specific documents with higher accuracy and faster, without the need to choose a specialized processor.

Cloud Vision, is commonly used to detect text, handwriting and a wide range of objects from images and videos.

**How does OCR work at Google Cloud?**

Google Cloud powers OCR with best-in-class AI. It goes beyond traditional text recognition by understanding, organizing and enriching data, ultimately generating business-ready insights.

It gives you the flexibility to either use the OCR tools as a unified suite for streamlined efficiency (e.g. Document AI), or simply call the relevant APIs directly available in Google Cloud console to integrate OCR functionalities into your applications.

**How Google Cloud AI and OCR work together?**

All the OCR solutions mentioned above give you access to pre-trained ML models that you can deploy right away through an API, or uptrain to improve accuracy for your specific needs.

You can also train your own custom models with AutoML - no machine learning expertise needed.

Check out AutoML documentation on building custom ML models.

**Latest advances in OCR**

OCR has advanced significantly past the first business systems in 1974 and progress continues. Superior OCR programs can provide extraction of key insights from documents in suboptimal conditions, such as irregular fonts, insufficient resolution, bad lighting from mobile capture and various colors and backgrounds.

By incorporating computer vision and natural language processing, improved information representation and model optimization, businesses can now enjoy state-of-the-art document understanding. Improvements can include analysis of layout and reading order in complex documents understanding visuals and representing them as charts and diagrams. Some OCR programs now are driven by generative AI to help structure document data even faster. An “old” technology continues to learn new tricks.